# Coefficient criteria for finite nonnegative linearization

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**Abstract.** We give two sufficient criteria stated directly in the coefficients of a finite stochastic three-term recurrence. The first assumes that the sum of its diagonal and downward coefficients has positive geometric increments with ratio $q \geq 1$, and that the ratio of those coefficients is constant and nonnegative. A finite algebra construction gives nonnegative linearization, including products beyond the terminating degree. This includes the full positive Krawtchouk range and the dual $q$-Krawtchouk family with nonconstant increments. The underlying quadratic algebra occurs in Koelink's work on type B Hecke algebras. The second criterion gives explicit perturbation inequalities admitting a relatively open neighborhood of recurrence arrays around every Krawtchouk system in the interior of the positive linearization range. For symmetric finite systems, a quadratic decomposition reduces positivity to two nonsymmetric finite systems. This yields a parity-split criterion covering Szwarc's quadratic system for $p_0 \leq 1/2$ and a $q$-analogue, and explicit symmetric neighborhoods. For arrays with constant ratio of diagonal to downward coefficients, we prove that nonnegative log-concave Newton coefficients of the profile imply nonnegative linearization, for every size and every product; this inequality criterion contains the Krawtchouk and geometric arrays. The ratio may also increase: it suffices that the Newton coefficients of the downward coefficients be log-concave and that those of the diagonal coefficients be obtained from them by nonincreasing nonnegative weights, or that both coefficient sequences be quadratic with convex downward coefficients and profile and nondecreasing diagonal coefficients. For log-concave Newton profiles the ratio may also increase linearly, or as a convex quadratic in the index, as long as the diagonal coefficients are nondecreasing. Log-concavity of Newton coefficients is preserved under pointwise products. Using this, we show that on every log-concave Newton profile, ratios of arbitrary degree with nonnegative, log-concave Newton coefficients without internal zeros and positive constant coefficient give nonnegative linearization whenever the diagonal coefficients are nondecreasing. The proof of the key inequality for this result is computer-assisted; on profiles of Newton degree at most two the proof is direct. For cubic ratios $r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with nonnegative coefficients, the single inequality $r\sigma \leq t\tau$ can replace log-concavity; this proof is computer-assisted as well. Nonnegative Newton coefficients of the ratio alone do not suffice. None of the criteria requires computing general linearization coefficients.

**Keywords.** Finite orthogonal polynomials; nonnegative linearization; Krawtchouk polynomials; dual $q$-Krawtchouk polynomials; quadratic transformation; polynomial hypergroups.

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## 1. Introduction

Nonnegative linearization (property (P)) asks whether the product of two orthogonal polynomials is a nonnegative linear combination of polynomials in the same basis. For a finite system, the identity is understood on the orthogonality support. This distinction is essential when the formal product degree exceeds the terminal degree. In a normalization with $R_i(1) = 1$, nonnegative coefficients have sum one and define the product probabilities of a finite polynomial hypergroup.

The Krawtchouk case is classical. Eagleson [2] studied positive definite sequences for these polynomials in 1969; Szwarc [7, §5, Problem 2] records the nonnegative-linearization range as $0 < p_0 \le 1/2$ for binomial weights $\binom{N}{j} p_0^{N-j}(1-p_0)^j$ when normalized at $j = 0$.[1] Our proof below fixes that convention directly from a two-point model. It does not treat Krawtchouk positivity itself as a new result.

Recurrence methods also have a long history. In the second note of his 1969 report, Askey [1, Part II, Theorem 1 and Corollary 1] gives finite recurrence conditions leading to a hyperbolic difference-equation maximum principle. The stated polynomial conclusion concerns products with degree sum at most $N$. Szwarc's later characterization [8, Theorem 1] treats finite as well as infinite systems through an auxiliary discrete hyperbolic problem. His 2005 survey nevertheless asks for criteria for finite systems, including Krawtchouk and its quadratic system $Q_n$ (Section 7) [7, §5, Problem 2]. Here we seek structural assumptions on the recurrence coefficients that imply positivity for every finite product.

Koelink [3, Theorems 3.1(ii) and 7.8] constructed a triangular quadratic algebra associated with the type B Hecke algebra, and identified the corresponding $q$-Krawtchouk spherical functions. The dual recurrence in that construction is the geometric family used below. Our algebraic proof extracts its positive multiplication mechanism and expresses the resulting sufficient condition as two scalar patterns in a stochastic recurrence: a constant ratio and geometric increments. No priority is claimed for the algebra or for the resulting named family. For comparison, Miclo [4, Proposition 3.3] gives a spectral characterization of the hypergroup property at a boundary point for irreducible finite birth–death chains by entrywise nonnegativity of products of transition matrices shifted by eigenvalues of their leading principal submatrices.

The second criterion is quantitative stability. Around each Krawtchouk array in the interior of the positive linearization range ($0 < r < 1$, equivalently $0 < p_0 < 1/2$), we exhibit an explicit radius in the space of recurrence coefficients within which all finite products remain nonnegative. Its proof uses one positive term to obtain a reference margin, an error recurrence to control perturbations, and orthogonality to preserve exact zeros outside the triangle support. The resulting neighborhoods contain independently varying coefficient arrays, rather than only a reparametrized classical family. The radius is deliberately conservative and depends on $N$; no uniform stability radius at the symmetric endpoint is

[1] The linked author version, §5, Problem 2, printed p. 125, uses inconsistent parameter conventions: its displayed weights have mean $N(1-p)$, whereas its degree-zero recurrence gives $K_1(x) = (x - pN)/((1-p)N)$. Thus the displayed weights give $\int K_1\, d\mu = (1-2p)/(1-p)$. Corollary 6.1 derives our weights and normalization directly.

asserted.

Section 2 states the geometric criterion. Sections 3 and 4 give its algebraic proof. Section 5 proves the perturbation inequalities, including explicit nongeometric examples. Section 6 fixes the Krawtchouk normalization and compares the geometric recurrence with Koelink's construction. Section 7 treats Szwarc's quadratic system and inequality hypotheses away from the deepest products. Section 8 proves an inequality criterion for constant-ratio arrays, including all deep products: nonnegative log-concave Newton coefficients of the profile suffice. Section 9 proves that uniformly sampled polynomial deformation rules, and analytic rules whose defining functions extend holomorphically to an open neighborhood of the region $\Omega$ specified there, must satisfy two first-order equalities if they keep property (P) at the symmetric Krawtchouk systems, shows that the stability radius of Theorem 5.3 has the optimal order in $1-r$, and records counterexamples to inequality shape conditions at smaller $r$. The final section lists questions left by these results.

## 2. Statement and conventions

For integers $j \geq 0$ put $[j]_q = 1 + q + \cdots + q^{j-1}$, with $[0]_q = 0$; in particular $[j]_1 = j$. Consider real coefficients indexed by $0 \leq i \leq N$, where $N \geq 1$, satisfying

$$\begin{gathered} a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \\ a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0). \end{gathered} \tag{2.1}$$

Set $R_{-1} = 0$, $R_0 = 1$, and determine $R_1, \ldots, R_N$ by

$$xR_i = a_i R_{i+1} + b_i R_i + c_i R_{i-1} \qquad (0 \leq i < N). \tag{2.2}$$

At $i = 0$, (2.2) gives $R_1 = x$; induction at $x = 1$, using the row sums and $a_i > 0$ for $i < N$, gives $R_i(1) = 1$ for $0 \leq i \leq N$. The terminal relation is $(x - b_N)R_N - c_N R_{N-1} = 0$ on the orthogonality support. Linearization means equality of functions on this support, rather than an unrestricted polynomial identity when the degree sum exceeds $N$.

**Theorem 2.1** *(Geometric increment criterion).* *Assume (2.1). Set $d_i = b_i + c_i$. Suppose*

1. *$b_i/c_i = \beta$ for every $1 \leq i \leq N$, where $\beta \geq 0$;*
2. *$\delta_i = d_i - d_{i-1} > 0$ for $1 \leq i \leq N$;*
3. *for some $q \geq 1$, $\delta_{i+1} = q\delta_i$ whenever $1 \leq i < N$.*

*Then the recurrence has a positive orthogonality measure on exactly $N+1$ distinct real points. Its normalized polynomials satisfy*

$$R_m R_n = \sum_{k=0}^{N} g(m,n;k) R_k, \qquad g(m,n;k) \geq 0, \qquad \sum_{k=0}^{N} g(m,n;k) = 1 \tag{2.3}$$

*on that support for all $0 \leq m, n \leq N$.*

**Lemma 2.2** *(Coefficient parametrization).* *The hypotheses of Theorem 2.1 are equivalent to*

$$a_i = \frac{q^i[N-i]_q}{[N]_q}, \qquad b_i = (1-r)\frac{[i]_q}{[N]_q}, \qquad c_i = r\frac{[i]_q}{[N]_q} \tag{2.4}$$

*for $q \geq 1$, $0 < r \leq 1$. When $N = 1$, one may choose $q = 1$.*

*Proof.* The terminal sum gives $d_N = 1$, and $d_0 = 0$. Geometric increments give $d_i = \delta_1[i]_q$, hence $\delta_1 = 1/[N]_q$. The constant ratio gives $c_i = d_i/(1+\beta)$ and $b_i = \beta d_i/(1+\beta)$. Put $r = (1+\beta)^{-1}$. Finally $[N]_q - [i]_q = q^i[N-i]_q$. These calculations reverse to give the converse, including $q = 1$ and $N = 1$. ∎

The parameter conventions used below are

| | |
|---|---|
| $q$ | $q \geq 1$ (geometric increment ratio) |
| $\beta$ | $b_i/c_i \geq 0$ |
| $r$ | $1/(1+\beta) \in (0,1]$ |
| $p$ | $1/r = 1+\beta \geq 1$ |
| $p_0$ | $r/(1+r) \in (0,1/2]$ |
| symmetric Krawtchouk case | $r = 1, \quad p_0 = 1/2$ |

## 3. A finite algebra with positive multiplication

Fix $p \geq 1$ and $q \geq 1$. Starting with $A_{N+1} = \mathbb{R}$, form recursively the commutative real algebras

$$A_j = A_{j+1}[z_j]/\left(z_j^2 - \left((q-1)\sum_{l=j+1}^{N} z_l + p - 1\right) z_j - pq^{N-j}\right). \tag{3.1}$$

Write $A = A_1$. For $S \subseteq \{1,\dots,N\}$ let $z_S = \prod_{j\in S} z_j$, with $z_\varnothing = 1$.

**Lemma 3.1** *(Basis and positivity).* *The elements $z_S$ form a basis of $A$, and all their multiplication coefficients are nonnegative.*

*Proof.* Division by a monic quadratic over any commutative coefficient ring gives a unique remainder of degree at most one. Applying this successively in (3.1) gives the asserted basis and dimension $2^N$. For positivity, replace a square by

$$z_j^2 = (q-1)\sum_{l>j} z_j z_l + (p-1)z_j + pq^{N-j}.$$

Every displayed coefficient is nonnegative. This rewriting terminates: give $z_j$ weight $2^{N-j}$ and give a monomial the sum of the weights of its factors, counted with multiplicity. Each replacement strictly decreases this nonnegative integer weight, because $l > j$ implies $2^{N-l} < 2^{N-j}$. The final expression is in the unique squarefree basis, proving the claim. ∎

**Lemma 3.2** *(Positive associative inner product).* *Let $\tau$ extract the coefficient of $z_\varnothing$. Then*

$$\tau(z_S z_T) = \mathbf{1}_{S=T}\, p^{|S|} q^{\sum_{j\in S}(N-j)}. \tag{3.2}$$

*Consequently $\langle u, v\rangle = \tau(uv)$ is positive definite, and multiplication by every element of $A$ is self-adjoint.*

*Proof.* Induct downwards through the tower. If $j$ belongs to exactly one of $S, T$, their product has the form $z_j v$ with $v \in A_{j+1}$ and has zero constant coefficient in $z_j$. If it belongs to both, the constant coefficient in $z_j$ is $pq^{N-j} z_{S\setminus\{j\}} z_{T\setminus\{j\}}$ by (3.1). If it belongs to neither, pass to $A_{j+1}$. Iteration proves (3.2). All diagonal factors are positive. Associativity and commutativity give $\tau((hu)v) = \tau(u(hv))$. ∎

Put $E_k = \sum_{|S|=k} z_S$ for $0 \le k \le N$, and set $E_{-1} = E_{N+1} = 0$.

**Lemma 3.3** *(Three-term multiplication).* *For $0 \le k \le N$,*

$$E_1 E_k = [k+1]_q E_{k+1} + (p-1)[k]_q E_k + pq^{k-1}[N-k+1]_q E_{k-1}. \tag{3.3}$$

*The last term is interpreted as zero for $k = 0$.*

*Proof.* For $k = 0$ the identity reads $E_1 = E_1$. For $N = 1, k = 1$ it is exactly $z_1^2 = (p-1)z_1 + p$. For the induction on $N$, write $F_j$ for the elementary squarefree sums in $z_2, \dots, z_N$, with $F_j = 0$ outside $0 \le j \le N-1$. Then $E_k = F_k + z_1 F_{k-1}$ and $E_1 = F_1 + z_1$. For $1 \le k \le N-1$, use (3.1) to obtain

$$E_1 E_k = F_1 F_k + z_1 F_k + q z_1 F_1 F_{k-1} + (p-1) z_1 F_{k-1} + pq^{N-1} F_{k-1}.$$

Apply the induction hypothesis to $F_1 F_k$ and $F_1 F_{k-1}$. The coefficients of $F_{k+1}$ and $z_1 F_k$ are respectively $[k+1]_q$ and $1 + q[k]_q = [k+1]_q$. The coefficients of $F_k$ and $z_1 F_{k-1}$ are both $(p-1)[k]_q$, since $1 + q[k-1]_q = [k]_q$. The coefficient of $F_{k-1}$ is

$$pq^{k-1}[N-k]_q + pq^{N-1} = pq^{k-1}[N-k+1]_q,$$

which is also the coefficient of $z_1 F_{k-2}$. Regrouping gives (3.3). For the endpoint $k = N$, $E_N = z_1 F_{N-1}$, so the same square relation gives

$$E_1 E_N = q z_1 F_1 F_{N-1} + (p-1) z_1 F_{N-1} + pq^{N-1} F_{N-1}.$$

The tail endpoint identity is

$$F_1 F_{N-1} = (p-1)[N-1]_q F_{N-1} + pq^{N-2} F_{N-2}.$$

Substitution gives $(p-1)[N]_q E_N + pq^{N-1} E_{N-1}$, as required. This separately checks the terminating boundary. ∎

**Proposition 3.4** *(Positive subalgebra).* *The space $B = \operatorname{span}\{E_0, \dots, E_N\}$ is the subalgebra generated by $E_1$. Every product $E_m E_n$ has nonnegative coefficients in this basis.*

*Proof.* All $E_k$ are independent, since their squarefree supports are nonempty and disjoint. Equation (3.3) shows that $B$ is invariant under multiplication by $E_1$. Since $[k+1]_q > 0$, its interior recurrence also expresses each $E_{k+1}$ as a polynomial in $E_1$. Thus $B = \mathbb{R}[E_1]$,

proving closure under multiplication, including products of total index greater than $N$. Expand such a product as $\sum_k h_k E_k$. Its coefficient at each squarefree monomial $z_S$ is $h_{|S|}$. Lemma 3.1 makes that coefficient nonnegative. There is a subset of each cardinality $0, \dots, N$, so every $h_k$ is nonnegative. ∎

## 4. Normalization and the finite orthogonal system

**Lemma 4.1** *(Positive normalization).* *Evaluation $z_j \mapsto pq^{N-j}$ defines an algebra character $\epsilon$ of $A$. Put $t_k = \epsilon(E_k) > 0$. Then*

$$t_0 = 1, \quad t_1 = p[N]_q, \qquad \frac{t_{k+1}}{t_k} = pq^k \frac{[N-k]_q}{[k+1]_q} \quad (0 \le k < N). \tag{4.1}$$

*Proof.* For $h = N - j$, substitution in the right-hand side of (3.1) gives

$$(q-1)p^2q^h[h]_q + (p-1)pq^h + pq^h = p^2q^h\big((q-1)[h]_q + 1\big) = p^2q^{2h}.$$

Thus all defining relations hold. The generating polynomial of the $t_k$ is $T(u) = \prod_{j=0}^{N-1}(1 + pq^j u)$, whose coefficients are strictly positive. When $q > 1$, the identity $(1+pu)T(qu) = (1+pq^N u)T(u)$ gives $(q^k - 1)t_k = p(q^N - q^{k-1})t_{k-1}$ for $1 \le k \le N$, which is (4.1). When $q = 1$, $t_k = \binom{N}{k}p^k$ and the same ratio follows directly. ∎

*Proof of Theorem 2.1.* Use Lemma 2.2 and put $p = 1/r \ge 1$. Construct $A, B$ as above and set $X = E_1/t_1$, $U_k = E_k/t_k$. Divide (3.3) by $t_1 t_k$ and use (4.1). The three coefficients are

$$\begin{aligned} \frac{[k+1]_q t_{k+1}}{p[N]_q t_k} &= \frac{q^k[N-k]_q}{[N]_q}, \\ \frac{(p-1)[k]_q}{p[N]_q} &= (1-r)\frac{[k]_q}{[N]_q}, \\ \frac{pq^{k-1}[N-k+1]_q t_{k-1}}{p[N]_q t_k} &= r\frac{[k]_q}{[N]_q}. \end{aligned}$$

At $k = 0$ omit the third expression; at $k = N$ omit the first. Consequently $U_k = R_k(X)$ for every $k \le N$, with the stated terminal relation.

The orthogonality support follows from the real spectral theorem in finite dimensions. On $B$ use the positive inner product from Lemma 3.2. Multiplication by $X$ is self-adjoint. The vector $1 = E_0$ is cyclic because the $E_k$ are polynomials of successive degrees in $X$ and form a basis. In an orthogonal eigenspace decomposition of a self-adjoint operator, the cyclic span of one vector has dimension at most the number of distinct eigenvalues. Here the cyclic span has dimension $N+1$, so there are exactly $N+1$ distinct eigenvalues, and the projection of 1 on every eigenspace is nonzero. Give each eigenvalue the squared norm of that projection as its mass. All masses are positive, sum to $\langle 1, 1\rangle = 1$, and yield

$$\int f(x)h(x)\,d\mu(x) = \langle f(X), h(X)\rangle$$

for real polynomials $f, h$, identifying an algebra element with its action on 1. The $E_k$ are orthogonal by (3.2), so the $R_k$ are orthogonal for this measure.

Proposition 3.4 and positive $t_k$ give nonnegative product coefficients for the $U_k$. Applying the character $\epsilon$ gives their sum equal to one because $\epsilon(U_k) = 1$. Polynomial identities in the algebra hold on every spectral point; conversely the $R_k$, having successive degrees at most $N$, form a basis of functions on the $N+1$ distinct points. This proves uniqueness and (2.3) for all products. ■

## 5. A criterion for small coefficient perturbations

Write the tridiagonal recurrence matrix with the row convention

$$J_{i,i+1} = a_i, \qquad J_{i,i} = b_i, \qquad J_{i,i-1} = c_i,$$

with all other entries zero and indices outside $0, \dots, N$ omitted.

**Lemma 5.1** *(Finite recurrence realization and triangle support).* *Every recurrence satisfying (2.1) has a positive orthogonality measure on exactly $N+1$ real points. If $J$ is its tridiagonal recurrence matrix, then its finite product coefficients satisfy*

$$g(m,n;k) = [R_m(J)]_{n,k}. \tag{5.1}$$

*They vanish unless $|m-n| \le k \le \min(m+n, N)$, regardless of their signs inside this interval.*

*Proof.* Choose positive $\pi_i$ with $\pi_0 = 1$ and $\pi_{i+1} = \pi_i a_i / c_{i+1}$. For $D = \operatorname{diag}(\sqrt{\pi_i})$, $H = DJD^{-1}$ is real symmetric. Induction in (2.2) gives the row-vector identity $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$, hence $e_0^{\mathsf{T}} R_i(H) = \pi_i^{-1/2} e_i^{\mathsf{T}}$. In particular $e_0$ is cyclic for $H$. The same finite spectral argument as in Theorem 2.1 gives $N+1$ distinct eigenvalues and positive spectral masses at $e_0$, and

$$\int R_i R_j \, d\mu = e_0^{\mathsf{T}} R_i(H) R_j(H) e_0 = \mathbf{1}_{i=j} \pi_i^{-1}.$$

The terminal relation holds on this support: its polynomial $Q(x) = (x - b_N) R_N - c_N R_{N-1}$ satisfies $e_0^{\mathsf{T}} Q(J) = 0$. Since $e_0$ is cyclic and $Q(J)$ commutes with $J$, $Q(J) = 0$. Multiplying a basis polynomial by $x$ acts by the corresponding row of $J$, including the terminal row. Induction on powers of $x$ therefore gives (5.1). Orthogonality also gives

$$g(m,n;k) = \frac{\int R_m R_n R_k \, d\mu}{\int R_k^2 \, d\mu}.$$

If one of $m, n, k$ exceeds the sum of the other two, its polynomial is orthogonal to their product, whose degree is smaller. This proves the asserted support. ■

**Lemma 5.2** *(A uniform positive margin in the asymmetric case).* *For the Krawtchouk recurrence (2.4) with $q = 1$ and $0 < r < 1$, put $s = \min(r, 1-r)$. Every coefficient in its triangle support obeys*

$$g(m,n;k) \ge (s/4)^N \quad \text{if } |m-n| \le k \le \min(m+n, N). \tag{5.2}$$

*Proof.* Specialize the algebra above to $q = 1$, and put $y_i = z_i/p$, where $p = 1/r$. Then $y_i^2 = (1-r)y_i + r$, and $U_m = \binom{N}{m}^{-1} \sum_{|S|=m} y_S$. The reduction of a product $y_S y_T$ has nonnegative coefficients, each chosen term being a product of at most $N$ factors from $\{r, 1-r\}$. Its coefficient is therefore at least $s^N$. For each admissible $k$, at least one such term has degree $k$. Assume $m \geq n$. If $k \leq m$, take $T \subseteq S$, with sizes $n, m$, and retain the linear term for exactly $k-(m-n)$ overlap coordinates. This number lies between 0 and $n$. If $k \geq m$, choose sets with overlap $m+n-k$ and union of size $k \leq N$, and retain the linear term at every overlap coordinate. Permutation symmetry makes the coefficient of $y_U$ depend only on $|U| = k$. Hence the coefficient in the normalized elementary-symmetric basis is at least

$$\frac{s^N \binom{N}{k}}{\binom{N}{m}\binom{N}{n}} \geq \frac{s^N}{4^N}.$$

This uses a single positive term and a denominator bound, not an explicit formula for the product coefficients. ∎

**Theorem 5.3** *(Explicit perturbation inequalities).* *Let $N \geq 1$, $0 < r < 1$, $s = \min(r, 1-r)$, and*

$$\eta_N(r) = \frac{1}{2}\left(\frac{s}{32N}\right)^N. \tag{5.3}$$

*Suppose coefficients $\widetilde{a}_i, \widetilde{b}_i, \widetilde{c}_i$ satisfy (2.1), with $\widetilde{b}_i \geq 0$ for every $i$, and*

$$\max_{0 \leq i \leq N} \left\{ \left| \widetilde{a}_i - \frac{N-i}{N} \right|, \left| \widetilde{b}_i - \frac{(1-r)i}{N} \right|, \left| \widetilde{c}_i - \frac{ri}{N} \right| \right\} \leq \eta_N(r). \tag{5.4}$$

*Then the finite system has nonnegative linearization. Throughout its triangle support,*

$$\widetilde{g}(m, n; k) \geq \tfrac{1}{2}(s/4)^N.$$

*The product coefficients sum to one.*

*Proof.* Use tildes for the perturbed system and untilded letters for its $q = 1$ reference at $r$. Let $\eta = \eta_N(r)$ and use the maximum absolute row-sum matrix norm. Both $J$ and $\widetilde{J}$ are stochastic matrices, so $\|J\|_\infty = \|\widetilde{J}\|_\infty = 1$ and $\|\widetilde{J} - J\|_\infty \leq 3\eta$. Put $B_m = R_m(J)$, $\widetilde{B}_m = \widetilde{R}_m(\widetilde{J})$ and $e_m = \|\widetilde{B}_m - B_m\|_\infty$. Theorem 2.1 at $q = 1$ and (5.1) give $\|B_m\|_\infty = 1$ for all $m \leq N$. Also $e_0 = 0$ and $e_1 \leq 3\eta$.

Subtract the two matrix recurrences for $1 \leq m < N$. Writing $\Delta_m = \widetilde{B}_m - B_m$ gives exactly

$$\begin{aligned} \widetilde{a}_m \Delta_{m+1} = {} & (\widetilde{J} - \widetilde{b}_m I)\Delta_m - \widetilde{c}_m \Delta_{m-1} \\ & + \big(\widetilde{J} - J - (\widetilde{b}_m - b_m) I\big) B_m \\ & - (\widetilde{c}_m - c_m) B_{m-1} - (\widetilde{a}_m - a_m) B_{m+1}. \end{aligned}$$

Since $\eta \leq 1/(2N)$, $\widetilde{a}_m \geq 1/(2N)$. The terms not containing a $\Delta$ have combined norm at most $6\eta$. Also $0 \leq \widetilde{b}_m, \widetilde{c}_m \leq 1$, so

$$e_{m+1} \leq 4Ne_m + 2Ne_{m-1} + 12N\eta. \tag{5.5}$$

Set $L = 8N$. Induction gives $e_m \le \eta L^m$ for $1 \le m \le N$: the base case follows from $3 \le L$, and, for $m \ge 1$,

$$4N + \frac{2N}{L} + \frac{12N}{L^m} \le 4N + \frac{1}{4} + \frac{3}{2} \le 8N = L.$$

Thus every matrix entry differs from its reference by at most

$$\eta(8N)^N = \frac{1}{2}(s/4)^N.$$

Lemma 5.2 gives the positive lower bound on triangle entries, while Lemma 5.1 makes all entries outside the triangle exactly zero. Finally $\widetilde{R}_m(1) = 1$ follows by recurrence from the stochastic coefficient sums, so evaluating each product at 1 gives coefficient sum one. Here 1 belongs to the support because $\widetilde{J}\mathbf{1} = \mathbf{1}$ and Lemma 5.1 identifies that support with its spectrum. No product-positivity inequality for the perturbed system has been assumed. ■

**Corollary 5.4** *(Finite systems outside the geometric family).* *For every $N \ge 2$, take the Krawtchouk coefficients at $r = 1/2$, and replace only*

$$a_1 \longmapsto a_1 - h, \qquad b_1 \longmapsto b_1 + h, \qquad 0 < h \le \frac{1}{4}(64N)^{-N}.$$

*These recurrences have nonnegative linearization and do not satisfy the constant-ratio hypothesis of Theorem 2.1. More generally, Theorem 5.3 admits a relatively open neighborhood of the asymmetric Krawtchouk array in the affine space defined by the row-sum and boundary equalities in (2.1).*

*Proof.* All initial and terminal conventions and stochastic sums are preserved. The bound on $h$ is half the radius (5.3) at $r = 1/2$, and is smaller than $a_1 = (N-1)/N$, so all required signs hold. The perturbation theorem applies. At index 1, $b_1/c_1 = 1 + 2Nh > 1$, whereas at index $N$ the ratio remains 1; these indices differ because $N \ge 2$. Thus this array lies outside the geometric family. At the asymmetric reference, all free coefficients constrained to be positive are strictly positive; the strict version of (5.4) therefore contains a relatively open neighborhood in that affine space; the sign conditions define the admissible region within it. ■

## 6. Applications and scope

**Corollary 6.1** *(Full positive Krawtchouk range).* *For every $N \ge 1$ and $0 < p_0 \le 1/2$, Krawtchouk polynomials orthogonal for binomial weights $\binom{N}{j}p_0^{N-j}(1-p_0)^j$ have nonnegative linearization when normalized at $j = 0$.*

*Proof.* To fix the convention without relying on a named recurrence, put $r = p_0/(1-p_0) \in (0, 1]$, and let $Y_1, \dots, Y_N$ be independent variables with values $1, -r$ of probabilities $p_0, 1 - p_0$. Their mean is zero and $Y_i^2 = (1-r)Y_i + r$. The normalized elementary symmetric sums

$$K_i(j) = \binom{N}{i}^{-1} \sum_{|S|=i} \prod_{l \in S} Y_l$$

depend only on the number $j$ of negative coordinates. They are orthogonal, because the expectation of two distinct squarefree monomials is zero: some independent mean-zero variable occurs once. Their weights in $j$ are exactly the stated binomial weights. With $x = N^{-1}\sum_l Y_l = 1 - j/(N(1-p_0))$, multiplication by $x$ gives

$$xK_i = \frac{N-i}{N}K_{i+1} + (1-r)\frac{i}{N}K_i + r\frac{i}{N}K_{i-1}.$$

Indeed, in multiplying the elementary sum by $\sum_l Y_l$, indices outside a subset give $(i+1)$ copies of each subset of size $i+1$; the linear part of $Y_l^2$ gives $(1-r)i$ copies of each subset of size $i$; and the constant part gives $r(N-i+1)$ copies of each subset of size $i-1$. Division by the binomial normalizers gives the displayed recurrence, including $i = N$. Its leading coefficient in $x$ is positive, its degree is $i$, and its value at $j = 0$ is one. These properties identify the normalized orthogonal polynomials for the binomial measure. Apply Theorem 2.1 with $q = 1$. The range is sharp in this endpoint normalization: if $1/2 < p_0 < 1$, then $r = p_0/(1-p_0) > 1$ and $g(1,1;1) = b_1 = (1-r)/N < 0$. For $N = 1$, this coefficient follows from the terminal relation. ■

**Corollary 6.2** *(The dual q-Krawtchouk case).* *For every $N \geq 2$, $q > 1$, and $0 < r \leq 1$, the recurrence (2.4) has nonnegative linearization. In the normalization $R_0 = 1$, $R_1 = x$, it is outside the Krawtchouk recurrence class.*

*Proof.* Its increments are $d_i - d_{i-1} = q^{i-1}/[N]_q$, so all hypotheses hold. For a Krawtchouk recurrence in this normalization the increments of $b_i + c_i$ are constant. Here their ratio is $q > 1$ and $N \geq 2$, so they are not constant. An affine change of the support transporting the same endpoint-normalized basis leaves the multiplication coefficients in the basis unchanged; it cannot remove this distinction. ■

**Proposition 6.3** *(Relation to the existing algebraic construction).* *The quadratic relations (3.1) are those of Koelink [3, Theorem 3.1(ii), equation (3.3)], with $z_j = u(x^j)$ and the same parameters $p, q$. The recurrence in this paper is the normalized recurrence of his Theorem 7.1, with $r = 1/p$; its polynomial-in-the-spectral-variable interpretation is the dual q-Krawtchouk interpretation described after his Theorem 7.8.*

*Proof.* The source square relation is $u(x^j)^2 = (q-1)u(x^j)\sum_{l>j} u(x^l) + (p-1)u(x^j) + pq^{N-j}$, which is (3.1). In Theorem 7.1 the coefficients of the next, current and previous indices are respectively $pq^i(1-q^{N-i})$, $(p-1)(1-q^i)$ and $1-q^i$. Division by $p(1-q^N)$ gives (2.4) for $q > 1$. For $0 \leq f \leq N$, the corresponding spectral coordinate is

$$x_f = \frac{p(1-q^{N-f}) - (1-q^f)}{p(1-q^N)} = \frac{q^{N-f} - rq^f + r - 1}{q^N - 1}, \qquad r = 1/p.$$

Writing $\phi_f(w_d)$ for Koelink's spherical-function values, we have $R_d(x_f) = \phi_f(w_d)$ for $0 \leq d \leq N$: the initial value is one and the normalized recurrences agree, so induction in $d$ identifies them. As $q \to 1$, dividing numerator and denominator by $q-1$ and using the finite geometric sums gives $x_f \to 1 - (1+r)f/N$. The $q = 1$ case follows directly from the

construction above, without division by zero. His discussion after Theorem 7.8 explicitly interchanges polynomial degree and spectral index to obtain the dual family. ■

The geometric criterion covers the full positive Krawtchouk range, including the symmetric endpoint. The perturbation criterion allows freely varying coefficients around each Krawtchouk array in the interior of the positive linearization range; its stated radius is conservative and decreases with $N$. Both hypotheses can be checked using a number of arithmetic comparisons proportional to $N$, without forming any general product coefficients or testing polynomial matrices. The proofs use positive algebra multiplication, a lower bound from a single positive term, and a norm estimate. They provide sufficient conditions, not a characterization of all finite systems with property (P). The geometric parameter family and underlying algebra predate Szwarc's survey; this paper makes no claim of historical priority for these constructions or criteria.

## 7. Szwarc's quadratic system

Problem 2 of [7, §5] does not stop at the Krawtchouk recurrence. With his parameter written as $p_0$, Szwarc introduces the symmetric recurrence of length $2N+1$

$$yQ_{2n} = (N-n)Q_{2n+1} + nQ_{2n-1}, \qquad yQ_{2n-1} = (1-p_0)Q_{2n} + p_0Q_{2n-2}, \tag{7.1}$$

records that $Q_{2n}(y)$ is the Krawtchouk polynomial $K_n(y^2)$ of his recurrence, states that the $Q_n$ have nonnegative linearization for $p_0 \le 1/2$, and asks for a general criterion that would handle this case. His Theorem 3.3 for infinite symmetric systems does not apply. In the normalization (7.1) its monotonicity hypotheses hold, and its hypothesis $\alpha_n \le \gamma_n$ holds at odd indices exactly when $p_0 \le 1/2$; at an even index $2n$ it reads $n \le N-n$ and fails in the upper half. Some failure of this kind is forced in every finite system, whose terminal upward coefficient vanishes. This section proves the positivity of (7.1) by a graded version of the algebra of Section 3, and then gives an explicit coefficient criterion for symmetric finite systems near (7.1).

Put $M = 2N$, $R_{2n}(t) = Q_{2n}(\sqrt{N}t)$ and $R_{2n-1}(t) = \sqrt{N}\,Q_{2n-1}(\sqrt{N}t)$. Substitution in (7.1) shows that the $R_k$ satisfy (2.2), with $N$ replaced by $M$, for the array

$$b_k = 0, \qquad (a_{2n}, c_{2n}) = \Big(\frac{N-n}{N}, \frac{n}{N}\Big), \qquad (a_{2n-1}, c_{2n-1}) = (1-p_0,\, p_0). \tag{7.2}$$

It satisfies (2.1) with $M$ in place of $N$ whenever $0 < p_0 < 1$. The two normalizations differ by positive factors and a positive dilation, so they have the same product coefficients up to positive factors.

**Theorem 7.1** *(Szwarc's quadratic system).* *Let $N \ge 1$ and $0 < p_0 \le 1/2$. The array (7.2) has a symmetric positive orthogonality measure on $2N+1$ points, and for all $0 \le m, n \le 2N$*

$$R_mR_n = \sum_{k=0}^{2N} g(m,n;k)R_k, \qquad g(m,n;k) \ge 0, \qquad \sum_k g(m,n;k) = 1$$

*on that support. If $1/2 < p_0 < 1$, then $g(2,2;2) < 0$.*

Throughout this section put $r = p_0/(1-p_0)$, so that $0 < r \le 1$ exactly when $p_0 \le 1/2$. Let $A$ be the commutative algebra generated by $y_1, \dots, y_N$ subject to $y_j^2 = (1-r)y_j + r$. This is the algebra of Section 3 with $q = 1$ after the rescaling $y_j = z_j/p$ used in Lemma 5.2. Its squarefree monomials $y_S$ form a basis, and for $r \le 1$ all multiplication coefficients are nonnegative. Call an element of $A$ nonnegative if its coefficients in this basis are nonnegative.

**Lemma 7.2** *(Edge idempotents).* *For $1 \le l \le N$ put $f_l = (r + y_l)/(1+r)$. Then $y_l f_l = f_l$ and $f_l^2 = f_l$. The ideal $f_l A$ has basis $f_l y_S$, $S \subseteq \{1, \dots, N\} \setminus \{l\}$, and $f_l u$ has nonnegative coefficients in this basis whenever $u \in A$ is nonnegative.*

*Proof.* First $(1+r)y_l f_l = r y_l + (1-r)y_l + r = (1+r)f_l$. Then $(1+r)f_l^2 = r f_l + y_l f_l = (1+r)f_l$. Let $A'$ be generated by the $y_j$ with $j \ne l$, so that $A = A' \oplus y_l A'$. Since $f_l y_l = f_l$, $f_l A = f_l A'$. The map $u \mapsto f_l u = (ru + y_l u)/(1+r)$ is injective on $A'$ by the direct sum, which gives the basis. For the last claim, $f_l y_S = f_l y_{S \setminus \{l\}}$ for every $S$. ∎

Let $V = A \oplus \bigoplus_{l=1}^N f_l A$, on which $A$ acts by multiplication in every summand, and call a vector nonnegative if all of its coordinates are nonnegative in the bases above. Define the $A$-linear operator

$$Y(u; v_1, \dots, v_N) = \Big( \frac{1}{N} \sum_{l=1}^N v_l \,;\, f_1 u, \dots, f_N u \Big).$$

Put $U_{2n} = \binom{N}{n}^{-1} \sum_{|S|=n} y_S \in A$ for $0 \le n \le N$. For $0 \le n \le N-1$, let $U_{2n+1}$ be the vector in $\bigoplus_l f_l A$ whose $l$th coordinate is $\binom{N-1}{n}^{-1} \sum f_l y_S$, summed over $S \subseteq \{1, \dots, N\} \setminus \{l\}$ with $|S| = n$.

**Lemma 7.3** *(Three-term action).* *With $U_{-1} = U_{2N+1} = 0$ and the coefficients (7.2), $Y U_k = a_k U_{k+1} + c_k U_{k-1}$ for $0 \le k \le 2N$.*

*Proof.* The $l$th coordinate of $Y \sum_{|S|=n} y_S$ is $\sum_{S \not\ni l} f_l y_S + \sum_{S' \not\ni l} f_l y_{S'}$, with $|S| = n$ and $|S'| = n-1$, because $f_l y_S = f_l y_{S \setminus \{l\}}$. Dividing by $\binom{N}{n}$ and using $\binom{N-1}{n}/\binom{N}{n} = (N-n)/N$ and $\binom{N-1}{n-1}/\binom{N}{n} = n/N$ gives the even case. For the odd case, $(1+r)f_l y_S = r y_S + y_{S \cup \{l\}}$ when $l \notin S$. Each $n$-set occurs for $N-n$ choices of $l$ and each $(n+1)$-set for $n+1$ choices, so

$$Y U_{2n+1} = \frac{r(N-n)\binom{N}{n} U_{2n} + (n+1)\binom{N}{n+1} U_{2n+2}}{N(1+r)\binom{N-1}{n}} = \frac{r U_{2n} + U_{2n+2}}{1+r}.$$

Finally $r/(1+r) = p_0$ and $1/(1+r) = 1 - p_0$. ∎

*Proof of Theorem 7.1.* Lemma 5.1 gives a positive measure on the $2N+1$ zeros of the terminal polynomial $T(t) = t R_{2N}(t) - R_{2N-1}(t)$. Since $b \equiv 0$, the matrix $J$ is conjugate to $-J$ by $\mathrm{diag}((-1)^i)$, which fixes $e_0$; hence the measure is symmetric and $R_k(-t) = (-1)^k R_k(t)$. Put $\Phi(f) = f(Y)U_0$, where $U_0 = (1; 0, \dots, 0)$. Lemma 7.3 gives $\Phi(R_k) = U_k$ for $k \le 2N$ and $\Phi(T) = Y U_{2N} - U_{2N-1} = 0$. Every polynomial vanishing on the support is a multiple

of $T$, so $\Phi$ maps it to zero. Hence the product identity on the support becomes

$$R_m(Y)R_n(Y)U_0 \;=\; \sum_k g(m,n;k)U_k.$$

The supports of the $U_k$ in the bases of $V$ are nonempty and disjoint: $U_{2n}$ uses the $y_S$ with $|S| = n$, and $U_{2n+1}$ uses the $f_l y_S$ with $l \notin S$, $|S| = n$. Each basis vector lies in exactly one of them, with positive coefficient. Consequently $g(m,n;k) \geq 0$ for every $k$ whenever the left side is nonnegative.

Suppose $m$ is even. Since polynomials in $Y$ commute and $Y$ is $A$-linear,

$$R_m(Y)R_n(Y)U_0 \;=\; R_n(Y)(U_mU_0) \;=\; U_m\,R_n(Y)U_0 \;=\; U_mU_n.$$

Here $U_m \in A$ and $U_n$ are nonnegative, so their product is nonnegative by the positivity of $A$ and Lemma 7.2. The same holds when $n$ is even, since $g(m,n;k) = g(n,m;k)$. If $m,n$ are odd and $k$ is odd, then $g(m,n;k) = 0$ by symmetry of the measure. If $m,n$ are odd and $k$ is even, then $g(m,n;k)\int R_k^2\,d\mu = \int R_mR_nR_k\,d\mu = g(k,m;n)\int R_n^2\,d\mu \geq 0$. The point $t = 1$ belongs to the support because $J\mathbf{1} = \mathbf{1}$, and $R_k(1) = 1$, so the coefficients sum to one.

For $1/2 < p_0 < 1$ we have $r > 1$. The identities above are purely algebraic and remain valid, but $A$ is no longer positive. In particular $U_2^2 = N^{-2}\big((1-r)\sum_j y_j + Nr + 2\sum_{i<j} y_iy_j\big)$, whose $U_2$ coefficient $g(2,2;2) = (1-r)/N$ is negative. ■

Since $Y^2$ acts on $A$ by multiplication by $N^{-1}\sum_l f_l = (r+X)/(1+r)$, with $X = N^{-1}\sum_l y_l$, the even part of the theorem is Corollary 6.1: $R_{2n}(t) = K_n\big((1+r)t^2 - r\big)$ in the normalization of that corollary. Thus Theorem 7.1 recovers the full positive Krawtchouk range by Szwarc's quadratic route. At $p_0 = 1/2$ the operator $Y$ is the normalized adjacency operator of the subdivided $N$-cube seen from a vertex, and the theorem reduces to counting in that graph.

**Lemma 7.4** *(Margin for the quadratic system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1-p_0)$ and $s = \min(r, 1-r)$. For the array (7.2),*

$$g(m,n;k) \;\geq\; (rs/8)^N \qquad \textit{whenever } |m-n| \leq k \leq m+n \textit{ and } m+n+k \textit{ is even.}$$

*All other coefficients vanish.*

*Proof.* The vanishing statements are Lemma 5.1 and the symmetry used above. We use $\binom{N}{k} \leq 2^N$ throughout. Up to the order of $m,n$, there are three cases.

If $m,n,k$ are even, the coefficient is a Krawtchouk coefficient computed in $A$, so Lemma 5.2 gives the lower bound $(s/4)^N \geq (rs/8)^N$.

Let $m = 2a$, $n = 2b+1$ and $k = 2c+1$, so that

$$\max(a-b-1,\, b-a) \;\leq\; c \;\leq\; \min(a+b,\, N-1).$$

Fix $l$ and a set $U \not\ni l$ with $|U| = c$. The $l$th coordinate of $U_mU_n$ is

$$\binom{N}{a}^{-1}\binom{N-1}{b}^{-1}\sum_{|T|=a}\;\sum_{S\not\ni l,\ |S|=b} f_l\,y_{T\setminus\{l\}}\,y_S.$$

Every nonzero term arising from a chosen reduction of $y_{T\setminus\{l\}}y_S$ is a product of at most $N-1$ factors from $\{r, 1-r\}$, and therefore has coefficient at least $s^{N-1}$. Put $a' = |T \setminus \{l\}|$, which can equal $a-1$ when $a \geq 1$ and $a$ when $a \leq N-1$. In the $N-1$ coordinates other than $l$, sets of sizes $a'$ and $b$ with overlap $o$, retaining the linear term on $t \leq o$ overlap coordinates, give monomials of size $a'+b-2o+t$. As $o$ ranges over its admissible values these sizes fill the interval $\big[|a'-b|, \min(a'+b, N-1)\big]$. The two admissible values of $a'$ cover every $c$ above. Hence $f_l y_U$ has coefficient at least $s^{N-1}\binom{N}{a}^{-1}\binom{N-1}{b}^{-1}$, and comparison with $U_k$ gives

$$g(m,n;k) \geq \binom{N-1}{c}\frac{s^{N-1}}{\binom{N}{a}\binom{N-1}{b}} \geq \frac{2s^{N-1}}{4^N} \geq \Big(\frac{s}{4}\Big)^N.$$

Finally let $m = 2b+1$, $n = 2c+1$ and $k = 2a$. With $\pi_i$ as in the proof of Lemma 5.1, (7.2) gives $\pi_{2a} = \binom{N}{a}r^{-a}$ and $\pi_{2c+1} = \binom{N-1}{c}r^{-c}/p_0$. Orthogonality gives

$$g(2b+1,2c+1;2a) = g(2a,2b+1;2c+1)\frac{\pi_{2a}}{\pi_{2c+1}} \geq \frac{2s^{N-1}}{4^N}\cdot\frac{p_0 r^N}{2^{N-1}} \geq \frac{2s^{N-1}r^{N+1}}{8^N},$$

using $r^{c-a} \geq r^N$ and $p_0 \geq r/2$. Since $s \leq r$, the last quantity is at least $(rs/8)^N$. ∎

The proof of Theorem 5.3 uses only three properties of its reference: nonnegative linearization with sums one, a lower bound for the upward coefficients, and a margin on the known support. We record it in that generality.

**Lemma 7.5** *(Stability of a reference with a margin).* *Let $a, b, c$ satisfy (2.1) on the indices $0, \dots, M$, have nonnegative linearization, and satisfy $a_i \geq \alpha$ for $i < M$, where $0 < \alpha \leq 1$. Suppose $g(m,n;k) \geq \delta > 0$ on a set $\Sigma$ of index triples. Let $\widetilde{a}, \widetilde{b}, \widetilde{c}$ satisfy (2.1), with $\widetilde{b}_i \geq 0$, and differ from $a, b, c$ by at most $\eta$ in every entry. If $\eta \leq \alpha/2$ and $\eta(8/\alpha)^M \leq \delta/2$, then $|\widetilde{g}(m,n;k) - g(m,n;k)| \leq \delta/2$ for all $m, n, k$; in particular $\widetilde{g} \geq \delta/2$ on $\Sigma$.*

*Proof.* Repeat the proof of Theorem 5.3 with $N$ replaced by $M$. Now $\widetilde{a}_m \geq \alpha/2$, so the subtracted recurrences give $e_{m+1} \leq (4e_m + 2e_{m-1} + 12\eta)/\alpha$. With $L = 8/\alpha \geq 8$, induction gives $e_m \leq \eta L^m$: the base case is $e_1 \leq 3\eta$, and the step needs $4L^m + 2L^{m-1} + 12 \leq 8L^m$, which holds for $m \geq 1$. Every entry of $\widetilde{B}_m - B_m$ is therefore at most $\eta L^M \leq \delta/2$, and (5.1) identifies these entries with the product coefficients. ∎

Theorem 5.3 is the case $M = N$, $\alpha = 1/N$ and $\delta = (s/4)^N$ of this lemma.

**Theorem 7.6** *(Symmetric criterion near Szwarc's system).* *Let $N \geq 1$, $0 < p_0 < 1/2$, $r = p_0/(1-p_0)$, $s = \min(r, 1-r)$, and*

$$\eta = \frac{1}{2}\Big(\frac{rs}{8}\Big)^N (16N)^{-2N}.$$

*Let $c_k$ be the downward coefficients in (7.2). Take real numbers $\widetilde{c}_1, \dots, \widetilde{c}_{2N-1}$ with $|\widetilde{c}_k - c_k| \leq \eta$, and put $\widetilde{c}_0 = 0$, $\widetilde{c}_{2N} = 1$, $\widetilde{b}_k = 0$ and $\widetilde{a}_k = 1 - \widetilde{c}_k$. The resulting symmetric finite system has nonnegative linearization with sums one. Moreover $\widetilde{g}(m,n;k) \geq \frac{1}{2}(rs/8)^N$ whenever $|m-n| \leq k \leq m+n$ and $m+n+k$ is even, and $\widetilde{g}(m,n;k) = 0$ otherwise.*

*Proof.* The perturbed array satisfies (2.1) with $M = 2N$: $\widetilde{a}_k \geq 1/N - \eta > 0$ for even $k < 2N$, $\widetilde{a}_k \geq 1/2 - \eta > 0$ for odd $k$, and $\widetilde{c}_k \geq \min(p_0, 1/N) - \eta > 0$ for $k > 0$. Theorem 7.1 shows that the reference has nonnegative linearization, and $a_k \geq 1/(2N)$ for $k < 2N$. Lemma 7.4 gives the margin $\delta = (rs/8)^N$ on the stated set. Apply Lemma 7.5 with $\alpha = 1/(2N)$: then $\eta \leq \alpha/2$ and $\eta(16N)^{2N} = \delta/2$. Off the stated set, $\widetilde{g}$ vanishes by Lemma 5.1 and by the symmetry of the perturbed measure, which holds because $\widetilde{b} \equiv 0$. The sums equal one because $\widetilde{R}_m(1) = 1$ and 1 lies in the support. ∎

**Corollary 7.7** *(Symmetric systems outside both families).* *For $N \geq 2$, $0 < p_0 < 1/2$ and $0 < h \leq \eta$, the array obtained from (7.2) by $c_1 \mapsto p_0 + h$, $a_1 \mapsto 1 - p_0 - h$ has nonnegative linearization. It is neither of the form (7.2) nor of the form (2.4). More generally, Theorem 7.6 admits a relatively open set of symmetric stochastic arrays, with $2N - 1$ free coefficients, around each array (7.2) with $0 < p_0 < 1/2$.*

*Proof.* Theorem 7.6 applies. The odd downward coefficients $c_1 = p_0 + h$ and $c_3 = p_0$ differ, so the array is not of the form (7.2); here $N \geq 2$ ensures $3 < 2N$. In (2.4), on $2N + 1$ indices, the downward coefficients $c_i = r[i]_q/[2N]_q$ are strictly increasing, whereas $c_3 < c_1$. ∎

**A criterion for all symmetric finite systems.** Szwarc uses the quadratic transformation [7, §3.5] to pass from a symmetric system to its even part. For finite systems the converse direction also works: positivity of a symmetric system follows from positivity of two nonsymmetric finite systems, both read off from its coefficients. Any criterion for nonsymmetric finite systems therefore becomes a criterion for symmetric ones.

**Theorem 7.8** *(Quadratic decomposition).* *Let $N \geq 2$, and let $a_k, c_k$ $(0 \leq k \leq 2N)$ with $b \equiv 0$ satisfy (2.1) with $2N$ in place of $N$. Put $a_k = c_k = 0$ for $k \notin [0, 2N]$, and define the even component*

$$\hat{a}_n = a_{2n}a_{2n+1}, \quad \hat{b}_n = c_{2n}a_{2n-1} + a_{2n}c_{2n+1}, \quad \hat{c}_n = c_{2n}c_{2n-1} \qquad (0 \leq n \leq N)$$

*and the odd component*

$$\check{a}_n = a_{2n+1}a_{2n+2}, \quad \check{b}_n = a_{2n+1}c_{2n+2} + c_{2n+1}a_{2n}, \quad \check{c}_n = c_{2n+1}c_{2n} \qquad (0 \leq n < N).$$

*After the affine normalization $(\hat{a}_n, \hat{b}_n, \hat{c}_n) \mapsto (\hat{a}_n, \hat{b}_n - \hat{b}_0, \hat{c}_n)/\hat{a}_0$, and likewise for the odd component, these arrays satisfy (2.1) on the indices $0, \dots, N$ and $0, \dots, N - 1$. If both have nonnegative linearization, so does the symmetric array. More precisely, with $\hat{g}$ and $\check{g}$ the product coefficients of the components and terms with indices out of range omitted,*

$$g(2a, 2b; 2c) = \hat{g}(a, b; c), \qquad g(2a, 2b + 1; 2c + 1) = a_{2a}\,\check{g}(a, b; c) + c_{2a}\,\check{g}(a - 1, b; c),$$

*the coefficients with $m + n + k$ odd vanish, and $g(2a + 1, 2b + 1; 2c) \int R_{2c}^2\, d\mu = g(2c, 2a + 1; 2b + 1) \int R_{2b+1}^2\, d\mu$. Conversely, nonnegative linearization of the symmetric array implies that of its even component.*

*Proof.* By symmetry $R_{2n}(t) = P_n(t^2)$ and $R_{2n+1}(t) = tL_n(t^2)$ for polynomials $P_n$, $L_n$ of degree $n$. Applying (2.2) twice gives, with $x = t^2$,

$$xP_n = \hat{a}_n P_{n+1} + \hat{b}_n P_n + \hat{c}_n P_{n-1}, \qquad xL_n = \check{a}_n L_{n+1} + \check{b}_n L_n + \check{c}_n L_{n-1}.$$

The row sums of the components equal $a_{2n} + c_{2n} = 1$ and $a_{2n+1} + c_{2n+1} = 1$; the coefficients $\hat{a}_n, \check{a}_n$ are positive below the last index and vanish there, and $\hat{c}_n, \check{c}_n > 0$ for $n \geq 1$. Hence the normalized arrays, which describe the same polynomials in the variable $(x - \hat{b}_0)/\hat{a}_0$ (respectively $(x - \check{b}_0)/\check{a}_0$), satisfy (2.1). The terminal relation $tR_{2N} = c_{2N} R_{2N-1}$ on the support $\Sigma$ of the symmetric system, applied once and twice, gives the terminal relations of $P$ on $\{t^2 : t \in \Sigma\}$ and of $L$ on $\{t^2 : t \in \Sigma,\ t \neq 0\}$. These sets have $N+1$ and $N$ points, because $\Sigma$ is symmetric with $2N+1$ points and contains 0. By Lemma 5.1 they are the supports of the two components.

The identity $tR_{2n} = a_{2n} R_{2n+1} + c_{2n} R_{2n-1}$ gives $P_n = a_{2n} L_n + c_{2n} L_{n-1}$ on the support of $L$; for $n = N$ it is the terminal relation, with $a_{2N} = 0$. Hence, on $\Sigma \setminus \{0\}$,

$$R_{2a} R_{2b+1} = t\big(a_{2a} L_a L_b + c_{2a} L_{a-1} L_b\big) = \sum_c \big(a_{2a} \check{g}(a, b; c) + c_{2a} \check{g}(a-1, b; c)\big) R_{2c+1},$$

and both sides vanish at $t = 0$. Similarly $R_{2a} R_{2b} = (P_a P_b)(t^2) = \sum_c \hat{g}(a, b; c) R_{2c}$ on $\Sigma$. Uniqueness of expansions on $\Sigma$ gives the displayed formulas. The vanishing for $m + n + k$ odd and the relation for two odd indices follow from the symmetry of the measure and from $g(m, n; k) \int R_k^2 \, d\mu = \int R_m R_n R_k \, d\mu$, exactly as in the proof of Theorem 7.1. The converse holds because $\hat{g}(a, b; c) = g(2a, 2b; 2c)$. ∎

Combined with Theorem 2.1, the decomposition gives a parity-split criterion of the same shape as Szwarc's Theorem 3.3: separate conditions on the even-indexed and the odd-indexed coefficients.

**Corollary 7.9** *(Parity-split criterion).* *Let $N \geq 1$ and let a symmetric array ($b \equiv 0$) on the indices $0, \ldots, 2N$ satisfy (2.1). Suppose that*

1. *the odd downward coefficients are constant, $c_{2n+1} = p_0$, with $0 < p_0 \leq 1/2$;*
2. *the even downward coefficients $d_n = c_{2n}$ have positive increments $d_n - d_{n-1}$, $1 \leq n \leq N$, forming a geometric progression with ratio $q \geq 1$.*

*Then the array has nonnegative linearization. With $r = p_0/(1 - p_0)$ and $N \geq 2$, its normalized even and odd components are (2.4) with parameters $(N, q, r)$ and $(N - 1, q, r/q)$, respectively. For $N = 1$ the odd component consists only of $L_0 = 1$, the array is (7.2), and positivity follows directly from Theorem 7.1. If $1/2 < p_0 < 1$ in (i) and (ii) holds, then $g(2, 2; 2) < 0$.*

*Proof.* For $N = 1$ this is Theorem 7.1. Let $N \geq 2$. The even component has $\hat{a}_n = (1 - d_n)(1 - p_0)$, $\hat{b}_n = d_n(1 - p_0) + (1 - d_n)p_0$ and $\hat{c}_n = d_n p_0$, so $\hat{a}_0 = 1 - p_0$, $\hat{b}_0 = p_0$, and its normalization is $(1 - d_n,\ (1 - r)d_n,\ r d_n)$. This satisfies Theorem 2.1 with $\beta = (1 - r)/r \geq$

0. For the odd component, $\check{a}_n = (1-p_0)(1-d_{n+1})$, $\check{b}_n = (1-p_0)d_{n+1} + p_0(1-d_n)$ and $\check{c}_n = p_0 d_n$. Geometric increments give $d_{n+1} - d_1 = q d_n$, so the normalization is

$$\Big(\frac{1-d_{n+1}}{1-d_1},\ \frac{(q-r)d_n}{1-d_1},\ \frac{r d_n}{1-d_1}\Big).$$

Its ratio $\beta = (q-r)/r$ is constant and nonnegative, and its profile $q d_n/(1-d_1)$ has geometric increments with ratio $q$. Theorem 2.1 applies to both components, and Theorem 7.8 concludes. Lemma 2.2 identifies the parameters. For $p_0 > 1/2$, $g(2,2;2) = \hat{g}(1,1;1)$ is the normalized diagonal coefficient $(1-r)d_1 < 0$ (for $N = 1$ see Theorem 7.1). ∎

With $q = 1$ the corollary is Szwarc's system (7.2), and it recovers Theorem 7.1, including the Krawtchouk polynomials as its even component. For $q > 1$ it gives a three-parameter family of symmetric finite systems with nonnegative linearization whose even components are the dual $q$-Krawtchouk arrays of Corollary 6.2. Theorem 7.8 also transfers the perturbation criteria: a symmetric array has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Theorem 5.3, or those of Lemma 7.5 around a reference with a margin, provided the set $\Sigma$ of that lemma contains every coefficient that can be nonzero for the perturbed components, or the remaining coefficients are separately known to stay nonnegative. Lemma 7.5 controls only $\Sigma$: for example, starting from $K_{2,1}$, keeping $b \equiv 0$ and $c_2 = 1$, and setting $c_1 = \frac{1}{2} + \varepsilon$, $a_1 = \frac{1}{2} - \varepsilon$ with $0 < \varepsilon < \frac{1}{2}$ gives $g(2,2;2) = -2\varepsilon/(\frac{1}{2} - \varepsilon) < 0$, although the coefficients positive at the reference remain positive. Conversely, the even component of every array admitted by Theorem 7.6 is a nonsymmetric finite system with nonnegative linearization.

**Products of low total degree.** Szwarc's inequality method [7, Theorems 3.2 and 3.4] survives the passage to finite systems for all products whose two factors have total degree at most $N$. Only the terminal coefficients are excluded from its hypotheses. Write a finite system in a positive renormalization $P_n = R_n/\sigma_n$ ($\sigma_0 = 1$) as $xP_n = \gamma_n P_{n+1} + \beta_n P_n + \alpha_n P_{n-1}$, so that $\gamma_n = a_n\sigma_{n+1}/\sigma_n$, $\beta_n = b_n$, $\alpha_n = c_n\sigma_{n-1}/\sigma_n$ and $\gamma_N = 0$.

**Theorem 7.10** *(Low products).* *Let $a, b, c$ satisfy (2.1). Suppose that for some positive $\sigma$,*

1. *$\beta_n$ is nondecreasing for $0 \le n \le N-1$, and $\alpha_n$ is nondecreasing for $0 \le n \le N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \le n \le N-2$;*
3. *$\gamma_i \ge \alpha_j$ whenever $0 \le j \le i$ and $i + j \le N-2$.*

*Then $g(n,m;k) \ge 0$ whenever $n + m \le N$. Consequently a product coefficient can be negative only for a triple $(n,m,k)$ in which each two indices have sum greater than $N$.*

*Proof.* Coefficients in two positive normalizations differ by positive factors, so we may work with the polynomials $\widetilde{P}_n$ of [7, (3.9)], which satisfy $x\widetilde{P}_n = \alpha_{n+1}\widetilde{P}_{n+1} + \beta_n\widetilde{P}_n + \gamma_{n-1}\widetilde{P}_{n-1}$ with $\alpha_{N+1} = 0$. Fix $k$ and let $u(n,m)$ be the coefficient of $\widetilde{P}_k$ in $\widetilde{P}_n\widetilde{P}_m$. Multiplying the product by $x$ in the two ways shows $Hu = 0$ at every point of $\{0,\dots,N\}^2$, where

$H = L_1 - L_2$ is the operator of [7, §3.4] built from these coefficients, and $u(n,0) = \mathbf{1}_{n=k}$, $u(n,m) = u(m,n)$.

We prove $u(n,m) \geq 0$ for $n \geq m$, $n+m \leq N$, by induction on $m$, following [7, Theorem 3.4]. Let $v$ be the indicator of the points $(i,j)$ with $j < m$, $|i-n| < m-j$ and $n+m-i-j$ odd. Every point in the support of $v$ satisfies $j < i$ and $i+j \leq n+m-1 \leq N-1$, so $\langle Hu, v\rangle = 0$ involves only rows where the finite recurrence holds. Hence $0 = \langle u, H^*v\rangle$, and every point in the support of $H^*v$ has $i+j \leq n+m \leq N$. With $\alpha_0 = 0$ and $v$ set to zero outside its support, transposition gives

$$(H^*v)(i,j) = \alpha_i v(i-1,j) + \gamma_i v(i+1,j) + (\beta_i - \beta_j)v(i,j) - \alpha_j v(i,j-1) - \gamma_j v(i,j+1),$$

as in [7, §3.4]. At $(n,m)$ it is $-\alpha_m < 0$. At the other points it is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j,$$

with $j \leq i$. The first occurs only for $i \leq N-1$. The second occurs only at interior points, where $i+j \leq n+m-2 \leq N-2$. The third occurs on the edge $i+j = n+m$. The last occurs on the edge $i-j = n-m$, where $i+j \leq N-2$. Hypotheses (i)–(iii) make all of these nonnegative, and all such points have second coordinate less than $m$. By the induction hypothesis and $u(i,0) \geq 0$, $\alpha_m u(n,m) = \sum_{(i,j)\neq(n,m)} (H^*v)(i,j)u(i,j) \geq 0$. For the last statement, $g(n,m;k)$, $g(n,k;m)$ and $g(m,k;n)$ are positive multiples of $\int R_n R_m R_k \, d\mu$. ∎

**Corollary 7.11** *(Convex profiles).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1-d_i$ with $0 < r \leq 1$ and a convex profile $0 = d_0 < d_1 < \cdots < d_N = 1$, that is, with nondecreasing increments. Then $g(n,m;k) \geq 0$ whenever two of $n, m, k$ have sum at most $N$. This includes every array (2.4).*

*Proof.* Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1-d_n)$, so $\alpha$ and $\beta = (1-r)d$ are nondecreasing and $\alpha_n + \gamma_n = \sqrt{r}$ is constant. Convexity and $d_0 = 0$ give $d_i + d_j \leq d_{i+j}$, so $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \geq \sqrt{r}\,(1 - d_{i+j}) > 0$ for $i+j \leq N-2$. Theorem 7.10 applies. ∎

For such profiles, positivity can therefore fail only on triples with all pairwise sums above $N$. The convex counterexamples recorded in Section 10 fail at $(N,N,N)$, which is such a triple.

The cone argument extends to all products if the monotonicity conditions in (i)–(ii) are imposed through the terminal index $N$ and the edge inequalities in (iii) are imposed for every $0 \leq j \leq i \leq N-1$.

**Theorem 7.12** *(All products).* *Let $a, b, c$ satisfy (2.1), and put $\gamma_N = 0$. Suppose that for some positive $\sigma$,*

1. *$\beta_n$ and $\alpha_n$ are nondecreasing for $0 \leq n \leq N$;*
2. *$\alpha_n + \gamma_n$ is nondecreasing for $0 \leq n \leq N$;*
3. *$\gamma_i \geq \alpha_j$ whenever $0 \leq j \leq i \leq N-1$.*

*Then the system has nonnegative linearization.*

*Proof.* Use the notation of the proof of Theorem 7.10. Since $\alpha_{N+1} = 0$, the relation $x\widetilde{P}_N = \beta_N \widetilde{P}_N + \gamma_{N-1}\widetilde{P}_{N-1}$ is the terminal relation, which holds on the support. So $Hu = 0$ at every point of $\{0, \dots, N\}^2$, as noted there. We prove $u(n,m) \geq 0$ for all $n \geq m$ by induction on $m$. Let $v$ be the indicator of the points $(i,j)$ with $0 \leq i \leq N$, $j < m$, $|i-n| < m-j$ and $n+m-i-j$ odd. Then $0 = \langle Hu, v\rangle = \langle u, H^*v\rangle$, and $H^*v$ is given by the formula in that proof. At $i = N$ the term $\gamma_N v(N+1, j)$ is absent; this is the convention $\gamma_N = 0$. Hence $H^*v(n,m) = -\alpha_m$, and at every other point $(i,j)$ with $H^*v(i,j) \neq 0$ we have $j < m$, $j \leq i \leq N$, and $H^*v(i,j)$ is one of

$$\beta_i - \beta_j, \quad (\alpha_i + \gamma_i) - (\alpha_j + \gamma_j), \quad \alpha_i - \alpha_j, \quad \gamma_i - \alpha_j.$$

The last one occurs only on the edge $i - j = n - m$, where $i \leq n - 1 \leq N - 1$. Hypotheses (i)–(iii) make all of these nonnegative. The induction hypothesis then gives $\alpha_m u(n,m) \geq 0$, and the case $n < m$ follows by symmetry. ∎

**Corollary 7.13** *(Heavy top step).* *Let $b_i = (1-r)d_i$, $c_i = rd_i$, $a_i = 1 - d_i$ with $0 < r \leq 1$ and an increasing profile $0 = d_0 < d_1 < \cdots < d_N = 1$ with $d_{N-1} \leq \frac{1}{2}$. Then the system has nonnegative linearization.*

*Proof.* Take $\sigma_n = r^{n/2}$. Then $\alpha_n = \sqrt{r}\, d_n$ and $\gamma_n = \sqrt{r}\,(1 - d_n)$ for $n < N$, $\gamma_N = 0$, and $\beta_n = (1-r)d_n$. So $\alpha$ and $\beta$ are nondecreasing, and $\alpha_n + \gamma_n = \sqrt{r}$ for every $0 \leq n \leq N$, because $d_N = 1$. For $j \leq i \leq N-1$, $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j) \geq \sqrt{r}\,(1 - 2d_{N-1}) \geq 0$. Theorem 7.12 applies. ∎

The profile need not be convex. The corollary contains the arrays (2.4) with $q \geq 2$, since then $d_{N-1} = (q^{N-1} - 1)/(q^N - 1) \leq \frac{1}{2}$, but not the Krawtchouk arrays with $N \geq 3$. The bound $d_{N-1} \leq \frac{1}{2}$ cannot be relaxed in general: for $N = 2$ and $r = 1$, $g(2,2;2) = (1 - 2d_1)/(1 - d_1)$. For constant-ratio arrays with a convex profile, $d_i + d_j \leq d_{i+j} \leq 1$ whenever $i + j \leq N$, so the only terms that can be negative in the proof are the edge terms $\gamma_i - \alpha_j = \sqrt{r}\,(1 - d_i - d_j)$ with $i + j > N$. For the Krawtchouk arrays these edge coefficients are $\sqrt{r}\,(N - i - j)/N$; any negative weighted contributions they make to the cone identity must be compensated by its other terms.

**Deep triples and reversal.** Call $(n, m, k)$ deep if each two of its indices have sum greater than $N$. The reversed array $a'_i = c_{N-i}$, $b'_i = b_{N-i}$, $c'_i = a_{N-i}$ need not satisfy $a'_0 = 1$, $b'_0 = 0$. Its polynomials $R'_i$ are defined by the reversed recurrence from $R'_0 = 1$, so $R'_1 = (x - b_N)/c_N$, and they still satisfy $R'_i(1) = 1$; its matrix is $\Pi J \Pi$, where $\Pi$ reverses the order of the indices. Write $g'$ for its product coefficients.

**Lemma 7.14** *(Reversal identity).* *On the support, $R_N R'_t = R_{N-t}$ for $0 \leq t \leq N$.*

*Proof.* From $e_0^{\mathsf{T}} R_i(J) = e_i^{\mathsf{T}}$ for the array and for its reversal, $e_N^{\mathsf{T}} R'_t(J) = e_{N-t}^{\mathsf{T}}$. Hence $e_0^{\mathsf{T}} R_N(J) R'_t(J) = e_N^{\mathsf{T}} R'_t(J) = e_0^{\mathsf{T}} R_{N-t}(J)$. Both matrices commute with $J$ and $e_0$ is cyclic, so $R_N(J) R'_t(J) = R_{N-t}(J)$, which is the claim on the spectrum. ∎

**Theorem 7.15** *(Deep reduction).* *For every array satisfying (2.1) and every deep triple $(x, y, z)$,*

$$g(x, y; z) = \sum_{t=0}^{N} g'(N - x, N - y; t)\, g(N, N - t; z).$$

*Moreover $g(N, N; N - l)$ is the coefficient of $R'_l$ in $R_N$. Hence all coefficients are nonnegative if (a) $g \geq 0$ on non-deep triples, (b) $g'(i, j; t) \geq 0$ whenever $i + j < N$, and (c) $R_N$ is a nonnegative combination of $R'_0, \ldots, R'_N$.*

*Proof.* By Lemma 7.14, $R_x = R_N R'_{N-x}$ and $R_y = R_N R'_{N-y}$ on the support. Expanding $R'_{N-x} R'_{N-y} = \sum_t g'(N - x, N - y; t) R'_t$ and using Lemma 7.14 again gives $R_x R_y = \sum_t g'(N - x, N - y; t) R_N R_{N-t}$. Expanding $R_N R_{N-t}$ gives the formula. Writing $R_N = \sum_l e_l R'_l$, Lemma 7.14 gives $R_N^2 = \sum_l e_l R_{N-l}$, so $e_l = g(N, N; N - l)$. For the last statement, $(N - x) + (N - y) < N$, so (b) makes the weights $g'$ nonnegative. It remains to see $g(N, n; z) \geq 0$. If $(N, n, z)$ is not deep this is (a). Otherwise $n + z > N$, and $g(N, n; z)$ is a positive multiple of $g(n, z; N) = \sum_t g'(N - n, N - z; t)\, g(N, N - t; N)$; here $g(N, N - t; N)$ is a positive multiple of $g(N, N; N - t)$, which is nonnegative by (c). ∎

**Theorem 7.16** *(Palindromic criterion).* *Let $a, b, c$ satisfy (2.1) with $a_n = c_{N-n}$ and $b_n = b_{N-n}$ for all $n$, and satisfy the hypotheses of Theorem 7.10 for some positive $\sigma$. Then the system has nonnegative linearization.*

*Proof.* The reversed array equals the array, so $R' = R$ and $g' = g$. Theorem 7.10 gives $g \geq 0$ on all non-deep triples, in particular $g'(i, j; t) \geq 0$ for $i + j < N$. Since $R_N = R'_N$ is a single reversed polynomial, $g(N, N; \cdot) \geq 0$. Theorem 7.15 gives the deep triples. ∎

**Corollary 7.17** *(Monotone palindromic systems).* *Let $b \equiv 0$, $a_n = 1 - c_n$, and let $0 = c_0 < c_1 \leq c_2 \leq \cdots \leq c_N = 1$ satisfy $c_n + c_{N-n} = 1$. Then the system has nonnegative linearization. This includes the symmetric Krawtchouk array $c_n = n/N$.*

*Proof.* The array is palindromic. With $\sigma \equiv 1$, $\alpha_n = c_n$ is nondecreasing, $\beta \equiv 0$ and $\alpha_n + \gamma_n = 1$. If $j \leq i$ and $i + j \leq N - 2$, then $j < N - i$, so $\alpha_j = c_j \leq c_{N-i} = 1 - c_i = \gamma_i$. Theorem 7.16 applies. ∎

Within the palindromic subclass $b \equiv 0$, $c_n + c_{N-n} = 1$, the hypotheses of Corollary 7.17 are monotonicity inequalities. They contain the symmetric Krawtchouk endpoint, where the perturbation criterion of Section 5 gives no neighbourhood and where the convex counterexamples of Section 10 show that convexity alone does not suffice.

Combined with Theorem 7.8, Corollary 7.17 gives an inequality criterion for symmetric systems: an array with $b \equiv 0$ on $0, \ldots, 2N$ has nonnegative linearization whenever its two normalized components satisfy the hypotheses of Corollary 7.17. For Szwarc's system (7.2) at $p_0 = 1/2$ the components are the symmetric Krawtchouk arrays on $N + 1$ and $N$ points (Corollary 7.9 with $q = r = 1$), so this case is covered by monotonicity inequalities within the palindromic subclass.

The reversal route cannot reach systems whose diagonal increases at the top once $N \geq 3$. If $b_{N-1} < b_N$, the reversed array has $g'(1, 1; 1) = (b'_1 - b'_0)/a'_0 = (b_{N-1} - b_N)/c_N < 0$,

and condition (b) of Theorem 7.15 requires $g'(1,1;1) \geq 0$ when $1+1 < N$. In particular it fails for the Krawtchouk arrays with $r < 1$ and $N \geq 3$. Szwarc's array (7.2) has $b \equiv 0$, so this argument does not apply to it. Its reversal replaces $p_0$ by $1-p_0$ and gives $g'(2,2;2) = (2p_0-1)/(Np_0) < 0$ for $p_0 < 1/2$, which violates condition (b) for the array of length $2N+1$ when $4 < 2N$, that is, $N \geq 3$. Miclo's palindromic criterion [5, Theorem 31] requires $2^n P(n,n+1)$ to be nonincreasing on $[0, N/2]$, which the symmetric Krawtchouk chain violates for $N > 2$.

## 8. Log-concave profiles

Theorem 2.1 requires the increments of the profile $d_i = b_i + c_i$ to be geometric. This section replaces that equality by inequalities on the Newton coefficients of the profile, for arrays with constant ratio $b_i/c_i$. Write

$$\nu_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} d_i \quad (1 \leq k \leq N), \qquad d_n = \sum_{k=1}^{n} \binom{n}{k} \nu_k \quad (0 \leq n \leq N);$$

below we put $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$. We say that $(\nu_k)_{1 \leq k \leq N}$ is *log-concave without internal zeros* if all $\nu_k \geq 0$, the indices with $\nu_k > 0$ form an interval $\{1, \dots, K\}$, and $\nu_k^2 \geq \nu_{k-1}\nu_{k+1}$ for $2 \leq k \leq N-1$. Note $\nu_1 = d_1 > 0$.

**Theorem 8.1** *(Log-concave Newton criterion).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \leq 1$, and put $a_i = 1 - d_i$, $b_i = (1-r)d_i$, $c_i = r d_i$. Then the system has nonnegative linearization.*

The hypotheses are inequalities in the profile. They hold for the Krawtchouk arrays ($\nu = (1/N, 0, \dots, 0)$), for all arrays of Theorem 2.1 ($\nu_k$ proportional to $(q-1)^{k-1}$, log-concave with equality). The hypotheses also hold for every admissible convex quadratic profile: $\nu_1 > 0$, $\nu_2 \geq 0$ when $N \geq 2$, and $\nu_k = 0$ for $3 \leq k \leq N$. The hypotheses exclude the known counterexamples with constant ratio. The convex counterexamples recorded in Section 10 have Newton coefficients of both signs. For the combinations $(1-\lambda)K_{N,r} + \lambda G_{N,q,r}$ discussed there, with $q > 1$, $N \geq 3$ and $0 < \lambda < 1$, one has $\nu_1 = (1-\lambda)/N + \lambda/[N]_q$ and $\nu_k = \lambda(q-1)^{k-1}/[N]_q$ for $2 \leq k \leq N$, so $\nu_2^2 - \nu_1\nu_3 = -\lambda(1-\lambda)(q-1)^2/(N[N]_q) < 0$.

**A rescaled system and a kernel.** Fix $r$ and put $t = (1-r)/\sqrt{r} \geq 0$. If $p_i$ is defined by $p_0 = 1$ and $y p_i = a_i p_{i+1} + t d_i p_i + d_i p_{i-1}$ $(0 \leq i < N)$, then $R_i(x) = r^{i/2} p_i(x/\sqrt{r})$, so the coefficients $H_{mn}^k$ in $p_m p_n = \sum_k H_{mn}^k p_k$ on the support satisfy $g(m,n;k) = r^{(m+n-k)/2} H_{mn}^k$. It suffices to show $H_{mn}^k \geq 0$. Put

$$e_i = \prod_{j=0}^{i-1} \frac{a_j}{d_{j+1}}, \qquad E_i = e_i p_i.$$

Then $y E_i = d_{i+1}E_{i+1} + t d_i E_i + a_{i-1}E_{i-1}$ on the support for $0 \leq i \leq N$, with the terms $d_{N+1}E_{N+1}$ and $a_{-1}E_{-1}$ omitted; at $i = N$ this is the terminal relation. Write $E_m E_n =$

$\sum_k L^k_{mn}E_k$, so that $H^k_{mn} = L^k_{mn}e_k/(e_me_n)$, and

$$K_k(z,w) = \sum_{m,n=0}^{N} L^k_{mn} z^m w^n.$$

Let $\theta_z = z\,\partial_z$ and let $d(\theta) = \sum_k \nu_k \binom{\theta}{k}$ act on polynomials by $d(\theta_z)z^i = d_i z^i$. For $F(z) = \sum_{i=0}^N E_i z^i$ the recurrence reads $yF(z) = \mathcal{L}_z F(z)$ with

$$\mathcal{L}_z = (z^{-1} + t - z)\, d(\theta_z) + z,$$

because $d_0 = 0$ removes the term $z^{-1}E_0$ and $1 - d_N = 0$ removes the term $z^{N+1}E_N$. Applying this to $F(z)F(w) = \sum_k K_k(z,w)E_k$ in the two variables separately and comparing coefficients of the basis $E_k$ gives $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$. Since $K_k$ is symmetric, $K_k = \sum_{\ell,a\ge 0} \omega_{\ell,a} S^a Z^\ell$ with $S = z + w$, $Z = zw$, and $K_k(z,0) = z^k$ gives $\omega_{0,a} = \mathbf{1}_{a=k}$. As $K_k$ has degree at most $N$ in $z$, comparing the coefficients of $z^{\ell+a}w^\ell$ by induction on $\ell$ gives $\omega_{\ell,a} = 0$ whenever $\ell + a > N$.

**The kernel recurrence.** Put $f_n(\ell) = \sum_{h\ge 0} \binom{\ell}{h} \nu_{n+h}$ with $\nu_0 = 0$ and $\nu_k = 0$ for $k > N$, so $f_0(\ell) = d_\ell$, and

$$W_{p,q}(\ell, a) = \sum_{n\ge p} \binom{n-p+q}{q} \binom{a}{n} f_n(\ell) \ge 0,$$

and $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$.

**Lemma 8.2.** *For all $\ell, a \ge 0$,*

$$\begin{aligned} d_{\ell+1}\omega_{\ell+1,a} = a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\ge 0} (-1)^j \Big( & t\,A_j(\ell-j, a+2j+1)\,\omega_{\ell-j,a+2j+1} \\ & + B_j(\ell-j, a+2j+2)\,\omega_{\ell-j,a+2j+2} \\ & + C_j(\ell-j-1, a+2j+2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned} \tag{8.1}$$

*where $a_i = 1 - d_i$ is extended by the polynomial $d_i$ for $i > N$ and terms with a negative index are omitted.*

*Proof.* Since $S^aZ^\ell = \sum_i \binom{a}{i} z^{\ell+i} w^{\ell+a-i}$, Vandermonde's identity $\binom{\ell+i}{k} = \sum_{p+q=k} \binom{\ell}{q}\binom{i}{p}$ and $\binom{a}{i}\binom{i}{p} = \binom{a}{p}\binom{a-p}{i-p}$ give

$$d(\theta_z)\, S^aZ^\ell = \sum_{p,q\ge 0} \nu_{p+q} \binom{\ell}{q}\binom{a}{p} z^p S^{a-p} Z^\ell.$$

Hence $(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell$ equals $(z-w)S^aZ^\ell$ plus $\sum_{p,q} \nu_{p+q}\binom{\ell}{q}\binom{a}{p} S^{a-p}Z^\ell$ times $(z^{p-1} - w^{p-1}) + t(z^p - w^p) - (z^{p+1} - w^{p+1})$. Divide by $z - w$, using $\chi_n = (z^{n+1} - w^{n+1})/(z - w) = \sum_{0\le j\le \lfloor n/2\rfloor} (-1)^j \binom{n-j}{j} S^{n-2j} Z^j$ for $n \ge 0$, $\chi_{-1} = 0$ and $(z^{-1} - w^{-1})/(z - w) =$

$-Z^{-1}$. Since $\sum_q \binom{\ell}{q}\nu_{p+q} = f_p(\ell)$, the quotient is

$$-d_\ell S^a Z^{\ell-1} + a_{\ell+a} S^a Z^\ell + \sum_{j\ge 0}(-1)^j\Big(tA_j(\ell,a)S^{a-2j-1}Z^{\ell+j} + B_j(\ell,a)S^{a-2j-2}Z^{\ell+j} + C_j(\ell,a)S^{a-2j-2}Z^{\ell+j+1}\Big).$$

Here $-d_\ell S^a Z^{\ell-1}$ comes from $p=0$; the term $a_{\ell+a}S^aZ^\ell$ combines $S^aZ^\ell$ with the parts $j=0$ of $-\chi_p$, which sum to $-d_{\ell+a}S^aZ^\ell$ by Vandermonde; and the three families from $\chi_{p-2}$, $t\chi_{p-1}$ and $-(\chi_p - S^p)$, whose coefficients $\binom{p-2-j}{j}$, $\binom{p-1-j}{j}$ and $\binom{p-1-j}{j+1}$ are those of $B_j$, $A_j$ and $C_j$. Summing over $\ell, a$ with weights $\omega_{\ell,a}$ and taking the coefficient of $S^aZ^\ell$ gives (8.1), because the monomials $S^aZ^\ell$ are linearly independent. ∎

**The log-concavity inequality.**

**Lemma 8.3.** *Under the hypotheses of Theorem 8.1, for $i, l \ge 1$ and $\ell \ge 0$,*

$$f_i(\ell)\, f_l(\ell+1) \;\ge\; d_{\ell+1}\, f_{i+l}(\ell).$$

*Proof.* For a nonnegative finitely supported sequence $u$ that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_iu_{i+2}) + (u_iu_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(0, \nu_1, \nu_2, \dots)$ has this property, and $f_i(\ell)$ is obtained from it by applying $u \mapsto v$ $\ell$ times. For such a sequence $g$, balanced products dominate: $g_ig_l \ge g_0g_{i+l}$ and $g_ig_{l+1} \ge g_1g_{i+l}$ for $i, l \ge 1$. Adding these with $g = f(\ell)$ and using $g_l + g_{l+1} = f_l(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1) = d_{\ell+1}$ gives the claim. ∎

**Lemma 8.4.** *Under the hypotheses of Theorem 8.1, for all $j \ge 1$ and $\ell, a \ge 0$,*

$$\begin{aligned} A_0(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}A_j(\ell,a),\\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}B_j(\ell,a),\\ B_0(\ell,a)C_{j-1}(\ell+1,a-2) &\ge 4d_{\ell+1}C_j(\ell,a), \end{aligned}$$

*with $W_{p,q}(\ell,a) = 0$ for $a < 0$.*

*Proof.* Use $A_0(\ell,a) \ge af_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}f_2(\ell)$, expand the second factor, apply Lemma 8.3 to $f_1(\ell)f_n(\ell+1)$ and $f_2(\ell)f_n(\ell+1)$, and use $a\binom{a-1}{n} = (n+1)\binom{a}{n+1}$ and $\binom{a}{2}\binom{a-2}{n} = \binom{n+2}{2}\binom{a}{n+2}$. After division by $d_{\ell+1}$ the three left sides are bounded below by $\sum_m \gamma_m\binom{a}{m}f_m(\ell)$ with

$$\gamma_m = \binom{m+1}{2}\binom{m-j-2}{j-1}, \qquad \binom{m}{2}\binom{m-j-3}{j-1}, \qquad \binom{m}{2}\binom{m-j-2}{j},$$

while the right sides are $4\sum_m \gamma'_m\binom{a}{m}f_m(\ell)$ with $\gamma'_m = \binom{m-j-1}{j}$, $\binom{m-j-2}{j}$ and $\binom{m-j-1}{j+1}$. Where $\gamma'_m > 0$ the quotients are

$$\frac{jm(m+1)}{2(m-j-1)}, \qquad \frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)},$$

on $m \geq 2j+1$, $m \geq 2j+2$ and $m \geq 2j+2$ respectively. They increase with $j$ for fixed $m$; at $j = 1$ their minima over integers are 5, 5 and 6. All quotients exceed 4, and $f_m(\ell) \geq 0$. ∎

*Proof of Theorem 8.1.* Let $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, on which $\omega$ is supported. Read (8.1) as moves into the node $(\ell+1, a)$, each divided by $d_{\ell+1} > 0$: the straight move from $(\ell, a)$ with weight $a_{\ell+a} \geq 0$ on $\Omega$, and for each $j$ the three moves from $(\ell - j, a + 2j + 1)$, $(\ell - j, a + 2j + 2)$, $(\ell - j - 1, a + 2j + 2)$ with weights $tA_j$, $B_j$, $C_j$. Every move raises $\ell$. Let $\mathcal{U}$ be the straight moves, $\mathcal{P}$ the moves with $j$ even and $\mathcal{Q}$ the moves with $j$ odd, all with nonnegative weights, so that $\omega = \beta + (\mathcal{U} + \mathcal{P} - \mathcal{Q})\omega$ with $\beta = \omega|_{\ell=0}$.

We claim $\mathcal{P}^2 \geq 4\mathcal{Q}$ entrywise. A move of $\mathcal{Q}$ with index $j$ from $x = (\ell, a)$ is matched by the two-step paths in $\mathcal{P}^2$ that start with $A_0$ or $B_0$ from $x$ (index 0, landing at $(\ell+1, a-1)$ or $(\ell+1, a-2)$, which lie in $\Omega$) and continue with the move of the same type and index $j-1$ (even), as in Lemma 8.4: $A_j$ by $A_0$ then $B_{j-1}$ and $B_0$ then $A_{j-1}$; $B_j$ by $B_0$ then $B_{j-1}$; $C_j$ by $B_0$ then $C_{j-1}$. These paths end at the target of the matched move, carry the same power of $t$, and have the extra denominator $d_{\ell+1}$ of the intermediate node. Distinct moves of $\mathcal{Q}$ have distinct displacements, so their paths are distinct, and Lemma 8.4 gives the claim.

Now show by induction on the level that $\omega \geq 0$ and $\omega \geq \frac{1}{2}\mathcal{P}\omega$. Both hold at $\ell = 0$. If they hold below a node $y$ at level $\ell + 1 \geq 1$, then, all weights being nonnegative, $(\mathcal{P}\omega)(y) \geq \frac{1}{2}(\mathcal{P}^2\omega)(y) \geq 2(\mathcal{Q}\omega)(y)$, and

$$\omega(y) = (\mathcal{U}\omega)(y) + (\mathcal{P}\omega)(y) - (\mathcal{Q}\omega)(y) \geq \tfrac{1}{2}(\mathcal{P}\omega)(y) \geq 0.$$

Hence all $\omega_{\ell,a} \geq 0$, so $K_k$ has nonnegative coefficients in $z, w$, so $L^k_{mn} \geq 0$, $H^k_{mn} \geq 0$ and $g(m, n; k) \geq 0$. ∎

*Remark* 8.5. The argument is valid coefficientwise in $t$. With independent parameters in $yp_i = a_i p_{i+1} + u d_i p_i + v d_i p_{i-1}$, weighted homogeneity ($\deg y = \deg u = 1$, $\deg v = 2$) then shows that each linearization coefficient is a polynomial in $(u, v)$ with nonnegative coefficients: for $D = m + n - k \geq 0$ it has the form $\sum_{0 \leq j \leq \lfloor D/2 \rfloor} h_j u^{D-2j} v^j$ with $h_j \geq 0$, and it vanishes for $D < 0$. Consequently every $g(m, n; k)$ of the array of Theorem 8.1, as a function of $r$, equals $\sum_j h_j r^j (1-r)^{D-2j}$; multiplying the $j$-th term by $(r + (1-r))^j = 1$ shows that its coefficients in the Bernstein basis of degree $D$ are $\binom{D}{s}^{-1} \sum_{j \leq s \leq 2j} h_j \binom{j}{s-j} \geq 0$. The diagnostics `lc_kernel_check.py` verify Lemmas 8.2–8.4 and the matching $\mathcal{P}^2 \geq 4\mathcal{Q}$ in exact arithmetic on random log-concave profiles.

**Varying ratio.** The kernel argument also works in the stochastic normalization itself, with independent diagonal and downward coefficients. Let $a_i + b_i + c_i = 1$ for $0 \leq i \leq N$, with $b_0 = c_0 = 0$, $a_N = 0$, $a_i > 0$ for $i < N$ and $c_i > 0$ for $i \geq 1$. Write

$$\kappa_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} c_i, \qquad \beta_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} b_i \qquad (1 \leq k \leq N),$$

with $\kappa_k = \beta_k = 0$ for $k = 0$ and $k > N$, and put $\varphi_n(\ell) = \sum_{h\ge 0}\binom{\ell}{h}\kappa_{n+h}$ and $\psi_n(\ell) = \sum_{h\ge 0}\binom{\ell}{h}\beta_{n+h}$, so that $\varphi_0(\ell) = c_\ell$ and $\psi_0(\ell) = b_\ell$ for $\ell \le N$.

**Theorem 8.6** *(Varying ratio).* *Let $(\kappa_k)$ be log-concave without internal zeros and $\beta_k \ge 0$ for all $k$. Suppose that for $0 \le \ell \le N-1$*

$$\psi_1(\ell)\,\varphi_n(\ell+1) \ge c_{\ell+1}\,\psi_{n+1}(\ell) \quad (n \ge 2), \qquad \varphi_2(\ell)\,\psi_n(\ell+1) \ge c_{\ell+1}\,\psi_{n+2}(\ell) \quad (n \ge 1).$$

*Then the system has nonnegative linearization.*

**Corollary 8.7.** *If $(\kappa_k)$ is log-concave without internal zeros and $\beta_k = w_k\kappa_k$ for $k \ge 1$ with $w_1 \ge w_2 \ge \cdots \ge 0$, then the system has nonnegative linearization. The ratio $c_i/(b_i + c_i)$ is nondecreasing in $i \ge 1$, and it is constant when $w$ is constant on the support of $\kappa$; these are the arrays of Theorem 8.1.*

For example, if $c$ is a convex quadratic in $i$ ($\kappa_1, \kappa_2 > 0$, $\kappa_k = 0$ for $k \ge 3$) and $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$ with $w_1 > w_2 \ge 0$, scaled so that $b_N + c_N = 1$, the ratio $c_i/(b_i + c_i)$ increases strictly. The hypotheses exclude the counterexample recorded in Section 10, whose diagonal coefficients $b = (0, 1, 1, 1)/91$ have Newton coefficients of both signs. They also exclude strictly increasing linear ratios $c_i/(b_i + c_i)$ with a linear profile, whose diagonal coefficients are concave; Proposition 8.8 below covers these.

*Proof of Theorem 8.6.* Put $e_i = \prod_{j<i} a_j/c_{j+1}$ and $E_i = e_i R_i$. Then $xE_i = c_{i+1}E_{i+1} + b_iE_i + a_{i-1}E_{i-1}$ on the support. The out-of-range terms are omitted: at the endpoints, $xE_0 = c_1E_1$ and $xE_N = b_NE_N + a_{N-1}E_{N-1}$. The series $F = \sum_i E_i z^i$ satisfies $xF = \mathcal{L}_z F$ with

$$\mathcal{L}_z = z^{-1}c(\theta_z) + b(\theta_z) + z\big(1 - b(\theta_z) - c(\theta_z)\big),$$

where $c(\theta) = \sum_k \kappa_k\binom{\theta}{k}$ and $b(\theta) = \sum_k \beta_k\binom{\theta}{k}$. As in Section 8, $K_k = \sum_{m,n} L^k_{mn} z^m w^n$, with $E_mE_n = \sum_k L^k_{mn}E_k$, satisfies $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$, and $K_k = \sum \omega_{\ell,a} S^a Z^\ell$ with $\omega_{0,a} = \mathbf{1}_{a=k}$ and $\omega_{\ell,a} = 0$ for $\ell + a > N$. The computation in the proof of Lemma 8.2 gives

$$\frac{(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell}{z - w} = S^aZ^\ell + \sum_{p\ge 0}\binom{a}{p}S^{a-p}Z^\ell\Big(\varphi_p(\ell)(\chi_{p-2} - \chi_p) + \psi_p(\ell)(\chi_{p-1} - \chi_p)\Big),$$

with $\chi_{-2} = -Z^{-1}$ and $\chi_{-1} = 0$. Collecting terms as there, with $W_{p,q}$ formed from $\varphi$ and $W^\psi_{p,q}$ from $\psi$, gives

$$c_{\ell+1}\omega_{\ell+1,a} = a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\ge 0}(-1)^j\Big(A^\psi_j(\ell - j, a + 2j + 1)\,\omega_{\ell-j,a+2j+1} + B_j(\ell - j, a + 2j + 2)\,\omega_{\ell-j,a+2j+2} + \big(C_j + C^\psi_j\big)(\ell - j - 1, a + 2j + 2)\,\omega_{\ell-j-1,a+2j+2}\Big),$$

where $A^\psi_j = W^\psi_{2j+1,j}$, $C^\psi_j = W^\psi_{2j+2,j+1}$, and $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$. The straight weight is $1 - c_{\ell+a} - b_{\ell+a} = a_{\ell+a}$, extended polynomially beyond $N$, and $\beta_0 = 0$. Up to this point no hypothesis of the theorem has been used: the recurrence holds for every array in the stochastic normalization. Under the hypotheses, all weights are nonnegative on $\Omega$.

For $j \ge 1$, $0 \le \ell \le N-1$ and $a \ge 0$, the following detour inequalities hold, with weights having a negative second argument interpreted as zero:

$$\begin{aligned} A_0^{\psi}(\ell,a)B_{j-1}(\ell+1,a-1) + B_0(\ell,a)A_{j-1}^{\psi}(\ell+1,a-2) &\ge 4c_{\ell+1}A_j^{\psi}(\ell,a), \\ B_0(\ell,a)B_{j-1}(\ell+1,a-2) &\ge 4c_{\ell+1}B_j(\ell,a), \\ B_0(\ell,a)\big(C_{j-1}+C_{j-1}^{\psi}\big)(\ell+1,a-2) &\ge 4c_{\ell+1}\big(C_j+C_j^{\psi}\big)(\ell,a). \end{aligned}$$

The proof of Lemma 8.4 applies, with the lower bounds $A_0^{\psi}(\ell,a) \ge a\psi_1(\ell)$ and $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$. Lemma 8.3 for $\varphi$ handles the products $\varphi_2(\ell)\varphi_n(\ell+1)$. The two hypotheses of the theorem handle $\psi_1(\ell)\varphi_n(\ell+1)$ for $n \ge 2$ and $\varphi_2(\ell)\psi_n(\ell+1)$ for $n \ge 1$, which are the indices that occur. The moves $C_j$ and $C_j^{\psi}$ have the same displacement and are treated as one move. With this, the proof of Theorem 8.1 gives $\omega \ge 0$, hence $L_{mn}^k \ge 0$ and $g(m,n;k) = L_{mn}^k e_k/(e_m e_n) \ge 0$. ∎

*Proof of Corollary 8.7.* Set $w_0 = w_1$; since $\kappa_0 = 0$, this leaves $\beta_0 = 0$. Fix $\ell$ and write $G_i = \varphi_i(\ell)$, $H_i = \psi_i(\ell)$ and $u_x = \binom{\ell}{x-i}$. For $i \ge 0$,

$$H_iG_{i+1} - H_{i+1}G_i = \sum_{x,y \ge i} u_x u_y \kappa_x \kappa_{y+1}(w_x - w_{y+1}).$$

Terms with $x \le y$ are nonnegative, and terms with $x = y+1$ vanish. The map $(x,y) \mapsto (y+1, x-1)$ is a bijection from the terms with $x \ge y+2$ onto those with $i+1 \le x \le y$; the remaining terms have $x = i$. A term and its image sum to $\kappa_x\kappa_{y+1}(w_{y+1} - w_x)(u_{y+1}u_{x-1} - u_x u_y) \ge 0$, because $w$ is nonincreasing and the zero-extended binomial coefficients are log-concave. Hence $H_{i+1}G_i \le H_iG_{i+1}$.

Since the support of $(G_i)$ is an interval and $\beta_k = 0$ where $\kappa_k = 0$, chaining gives $H_{n+1}G_1 \le H_1G_{n+1}$, $H_{n+2}G_n \le H_nG_{n+2}$ and $H_{n+2}G_{n+1} \le H_{n+1}G_{n+2}$. Lemma 8.3 with $(i,l) = (1,n)$ and $(2,n)$ reads $G_1(G_n + G_{n+1}) \ge CG_{n+1}$ and $G_2(G_n + G_{n+1}) \ge CG_{n+2}$, where $C = G_0 + G_1 = c_{\ell+1}$ and $G_n + G_{n+1} = \varphi_n(\ell+1)$. Since $G_1 \ge \kappa_1 > 0$, $H_1(G_n + G_{n+1}) \ge CH_1G_{n+1}/G_1 \ge CH_{n+1}$. If $G_{n+2} > 0$, then $G_2(H_n + H_{n+1}) \ge G_2H_{n+2}(G_n + G_{n+1})/G_{n+2} \ge CH_{n+2}$; if $G_{n+2} = 0$, then $H_{n+2} = 0$. These are the two hypotheses of Theorem 8.6. The case $i = 0$ of the first display, with $\beta_0 = \kappa_0 = 0$, reads $b_{\ell+1}c_\ell \le b_\ell c_{\ell+1}$; for $1 \le \ell < N$, division by $c_\ell c_{\ell+1} > 0$ shows that $b_i/c_i$ is nonincreasing and $c_i/(b_i + c_i)$ is nondecreasing. If $w$ is constant on the support of $\kappa$, then $b$ is proportional to $c$, and with $d = b + c$ the array is that of Theorem 8.1. ∎

For arrays that are quadratic in the index, concave diagonal coefficients are allowed.

**Proposition 8.8** *(Quadratic arrays).* *Let $b_n = \beta_1 n + \beta_2\binom{n}{2}$ and $c_n = \kappa_1 n + \kappa_2\binom{n}{2}$ $(0 \le n \le N)$ define an array as above. If $\kappa_2 \ge 0$, $\kappa_2 + \beta_2 \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, then the system has nonnegative linearization.*

*Proof.* In the recurrence of the proof of Theorem 8.6, which holds for every array, $\varphi_n = \psi_n = 0$ for $n \ge 3$. So all moves with $j \ge 1$ vanish, and the moves with $j = 0$ have the weights $A_0^{\psi}(\ell,a) = \sum_{n \ge 1}\binom{a}{n}\psi_n(\ell) = b_{\ell+a} - b_\ell$, $B_0(\ell,a) = \binom{a}{2}\kappa_2$ and $(C_0 + C_0^{\psi})(\ell,a) =$

$\binom{a}{2}(\kappa_2+\beta_2)$. These and the straight weights $a_{\ell+a}$ are nonnegative on $\Omega$, so $\omega \geq 0$ by induction on $\ell$, and $g(m,n;k) \geq 0$ as in the proof of Theorem 8.6. ∎

With the linear profile $d_n = n/N$ and the ratio $c_n/d_n = r + \kappa n/N$ ($1 \leq n \leq N$, $r > 0$, $\kappa \geq 0$), one has $\kappa_2 = 2\kappa/N^2$ and $\kappa_2 + \beta_2 = 0$ for $N \geq 2$. For $N = 1$, property (P) follows directly from $R_1^2 = c_1 R_0 + b_1 R_1$ when $b_1 \geq 0$. So every linearly increasing ratio with a linear profile and nondecreasing $b$ has property (P); $b$ is nondecreasing exactly when $\kappa \leq (1-r)N/(2N-1)$. Exact computations (`runs/varratio/quadratic.py`) show that each of the three hypotheses is needed: dropping any one of them gives quadratic arrays with a negative coefficient at $N \leq 3$.

**Linearly increasing ratio.** For curved profiles, linearly increasing ratios need not satisfy the hypotheses of Theorem 8.6: the Newton coefficients of $b$ of order at least 3 can be negative, and then some moves of even index in the recurrence of its proof have negative weight and no two-step detour. A resummation of the positive moves handles this.

**Theorem 8.9** *(Linearly increasing ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \leq 1$ and $\kappa \geq 0$, and put $\rho_i = r + \kappa i/N$, $a_i = 1 - d_i$, $b_i = (1-\rho_i)d_i$ and $c_i = \rho_i d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization. The condition that $b$ be nondecreasing is equivalent to $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \leq N(1-r)$.*

For the linear profile $d_i = i/N$ the condition reads $\kappa \leq \frac{N}{2N-1}(1-r)$, and these are arrays of Proposition 8.8. For a convex profile $d_{N-1} \leq (N-1)/N$, so the admitted range of $\kappa$ contains $[0, \frac{N}{2N-1}(1-r)]$. When $r < 1$, this containment is strict for a nonlinear profile. For example, $N = 4$, $d_i = (2^i - 1)/15$, $r = 1/2$ and $\kappa = 2/5$ are admitted, with $b = (0, 2/75, 3/50, 7/75, 1/10)$, although $(2N-1)\kappa/N = 7/10 > 1 - r$.

*Proof.* Write $s = 1 - r$ and $t = \kappa/N$, and let $f_n(\ell) = \Delta^n d(\ell)$ and $\nu_i = \Delta^i d(0)$ be as in Section 8, with $\nu_{-1} = \nu_0 = 0$. For $0 \leq i < N$ put $R_i = d_i/f_1(i)$; then $f_1(i) \geq \nu_1 > 0$, and log-concavity of $(f_n(i))_{n\geq 0}$ gives $R_{i+1} - R_i = \big(f_1(i)^2 - d_i f_2(i)\big)/\big(f_1(i)f_1(i+1)\big) \geq 0$. Since $b_{i+1} - b_i = f_1(i)\big(s - t(i+1) - tR_i\big)$ and the bracket is nonincreasing in $i$, $b$ is nondecreasing exactly when its last increment is nonnegative, that is, when $t(N + R_{N-1}) \leq s$ with $R_{N-1} = d_{N-1}/(1-d_{N-1})$; this is the stated condition. In particular $tN \leq s$. Since $c_i = (r+ti)d_i$ and $b_i = (s-ti)d_i$, the Leibniz rule for differences gives, for $\ell \geq 0$ and $1 \leq n \leq N - \ell$, which includes every index occurring in the moves on $\Omega$,

$$\varphi_n(\ell) = (r + t(\ell+n))f_n(\ell) + tnf_{n-1}(\ell), \qquad \psi_n(\ell) = h_n(\ell) - v_n(\ell),$$

with $h_n(\ell) = (s - t(\ell+n))f_n(\ell)$ and $v_n(\ell) = tnf_{n-1}(\ell)$; both are nonnegative when $\ell + n \leq N$. (Outside this range the formulas need not hold, because the Newton coefficients of $b$ and $c$ are set to zero beyond $N$.) Extend $\nu$ by zero outside $1 \leq i \leq N$ and put $\widehat{\kappa}_i = p_i\nu_i + q_i\nu_{i-1}$ ($i \geq 1$), $\widehat{\kappa}_0 = 0$, with $p_i = r + ti$ and $q_i = ti$; the Newton sequence $\kappa$ of $c$ is the truncation of $\widehat{\kappa}$ at $N$. Then

$$\widehat{\kappa}_i^2 - \widehat{\kappa}_{i-1}\widehat{\kappa}_{i+1} = p_i^2(\nu_i^2 - \nu_{i-1}\nu_{i+1}) + q_i^2(\nu_{i-1}^2 - \nu_{i-2}\nu_i) + q_{i-1}p_{i+1}(\nu_i\nu_{i-1} - \nu_{i-2}\nu_{i+1}) + t^2(\nu_{i-1}\nu_{i+1} + \nu_{i-2}\nu_i + 2\nu_i\nu_i$$

and each term is nonnegative by log-concavity of $\nu$. The positive support of $\widehat{\kappa}$ is that of $\nu$ when $t = 0$ and extends it by one index when $t > 0$, an interval in both cases. Truncation at $N$ preserves log-concavity and interval support, so $\kappa$ is log-concave without internal zeros, and Lemma 8.3 applies to $f$ and to $\varphi$.

*Moves.* We use the recurrence of the proof of Theorem 8.6, which holds for every array. For $j \geq 1$ split $A_j^{\psi} = H_j - V_j$, where $H_j = W^h_{2j+1,j}$ and $V_j = W^v_{2j+1,j}$ are formed from $h_n$ and $v_n$ in place of $\psi_n$; both are nonnegative on $\Omega$. Keep $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ as one move; it is nonnegative because $b$ is nondecreasing. The weights $B_j$, formed from $\varphi$, and $C_j + C_j^{\psi} = W^f_{2j+2,j+1}$, formed from $f$, are nonnegative. Write $C'_j = C_j + C_j^{\psi}$.

*Detours.* All detour inequalities below are asserted for source nodes $(\ell, a) \in \Omega$; in the mixed products the occurring indices satisfy $n \leq a - 2$, hence $\ell + n + 2 \leq N$. Since $\varphi_2(\ell) \geq (r + t(\ell+1))f_2(\ell)$ and $c_{\ell+1} = (r + t(\ell+1))d_{\ell+1}$, Lemma 8.3 for $f$ gives

$$\varphi_2(\ell)f_n(\ell+1) \geq c_{\ell+1}f_{n+2}(\ell), \quad \varphi_2(\ell)h_n(\ell+1) \geq c_{\ell+1}h_{n+2}(\ell), \quad \varphi_2(\ell)v_n(\ell+1) \geq \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell)\ (n \geq 2),$$

using $s - t(\ell + n + 1) \geq s - t(\ell + n + 2) \geq 0$ for the second and Lemma 8.3 with indices $(2, n-1)$ for the third. As in the proof of Lemma 8.4, with $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell)$, for $0 \leq \ell \leq N - 1$ and second factors at $(\ell + 1, a - 2)$,

$$B_0B_{j-1} \geq 4c_{\ell+1}B_j\ (j \geq 1), \quad B_0H_{j-1} \geq 6c_{\ell+1}H_j\ (j \geq 2), \quad B_0V_{j-1} \geq 6c_{\ell+1}V_j\ (j \geq 2), \quad B_0C'_{j-1} \geq 6c_{\ell+1}C'_j\ (j$$

The first uses Lemma 8.3 for $\varphi$. The coefficient ratios are, on $m \geq 2j+2$, $2j+1$, $2j+1$ and $2j+2$ respectively,

$$\frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{jm(m-1)}{2(m-j-1)}, \qquad \frac{j(m-1)(m-2)}{2(m-j-1)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)};$$

they increase with $j$ and are at least 5, 10, 6 and 6. Finally,

$$A_0^{\psi}(\ell, a)B_0(\ell+1, a-1) + B_0(\ell, a)A_0^{\psi}(\ell+1, a-2) \geq c_{\ell+1}\big(4H_1(\ell, a) - 2V_1(\ell, a)\big). \tag{8.2}$$

Only $a \geq 3$ needs proof. Put $u_n = f_n(\ell)$, $y_n = \binom{a}{n}u_n$, $D = u_0 + u_1 = d_{\ell+1}$, $\alpha = s - t(\ell + a)$, $X = \sum_{n\geq 1} y_n$, $Y = \sum_{n\geq 2} y_n$, $X' = \sum_{j\geq 1}\binom{a-2}{j}(u_j + u_{j+1})$, $Y' = \sum_{j\geq 2}\binom{a-1}{j}(u_j + u_{j+1})$ and $J = \sum_{n=2}^{a-1}\binom{a-1}{n}u_n$, so that $X = au_1 + Y$, $X' = (a-2)u_1 + J$ and $Y' = Y - (a-1)u_2$. Then $A_0^{\psi}(\ell, a) = \alpha X - tau_0$ and $A_0^{\psi}(\ell+1, a-2) = (\alpha + t)X' - t(a-2)D$; both are nonnegative, and the two weights $B_0$ are at least $(r + t(\ell+1))Y'$ and $(r + t(\ell+1))Y$. With $H_* = \sum_{n\geq 3}(n-2)y_n$ and $T = \sum_{n\geq 2}(n-3)(a-n)y_n$ one has $H_1 = \sum_{n\geq 3}(n-2)(\alpha + t(a-n))y_n$ and, after a shift of index, $V_1 = t\sum_{n\geq 2}(n-1)(a-n)y_n$, so $4H_1 - 2V_1 = 4\alpha H_* + 2tT$. Since $c_{\ell+1} = (r + t(\ell+1))D$, it suffices to show

$$(\alpha X - tau_0)Y' + Y\big((\alpha + t)X' - t(a-2)D\big) \geq D(4\alpha H_* + 2tT). \tag{8.3}$$

The coefficient of $\alpha$ in the difference of the two sides is $K = XY' + YX' - 4DH_*$. By Lemma 8.3, $u_i(u_j + u_{j+1}) \geq Du_{i+j}$ for $i, j \geq 1$, so $XY' \geq au_1Y' \geq D\sum_{n\geq 3} ny_n$ and $YX' \geq \binom{a}{2}u_2X' \geq D\sum_{n\geq 3}\binom{n}{2}y_n$; hence $K \geq \frac{D}{2}\sum_{n\geq 3}\big((n-3)(n-4) + 4\big)y_n \geq 0$, and (8.3) holds when $t = 0$. Let $t > 0$. Monotonicity of $b$ at $\ell + a - 1$ gives $\alpha \geq tR_{\ell+a-1} \geq$

$tR_\ell = tu_0/u_1$, so it suffices to prove (8.3) at $\alpha = tu_0/u_1$. Both sides are homogeneous of degree two in $(u_n)$; take $u_1 = 1$ and write $z = u_0$ and $q = u_2$. If $q = 0$, then $u_n = 0$ for $n \geq 2$ and both sides vanish. Otherwise $0 \leq z \leq 1/q$ by log-concavity. At $\alpha = tz$ the two weights $A_0^\psi$ are $tzY$ and $t(z+1)J$, and (8.3) divided by $t$ reads $F(z) \geq 0$, where

$$F(z) = zYY' + (z+1)YJ - (z+1)(4zH_* + 2T) = (1-qz)F(0) + qzF(1/q) + 4H_* z(1/q - z),$$

the second form holding because $F$ is quadratic with leading coefficient $-4H_*$. It remains to show $F(0) \geq 0$ and $F(1/q) \geq 0$. Put $L_3 = \frac{3}{2}$, $L_n = \frac{n(n+2)}{6}$ for $n \geq 4$, and $E_n = \frac{2(n^2-5)}{n+1}$. In $YJ$ keep, for each $n$, the product with indices $(2,2)$ if $n = 3$, and the products with indices $(2, n-1)$ and $(3, n-2)$ if $n \geq 4$; by log-concavity with $u_1 = 1$, $u_i u_{n+1-i} \geq u_n$, and $\binom{a-1}{j} \geq \binom{a-i}{j}$, so $YJ \geq \sum_{n=3}^{a-1} L_n (a-n) y_n$. In $Y$ keep the term $\binom{a}{2} q$; as $q(u_{n-1} + u_n) \geq (q+1)u_n$ for $n \geq 3$, $YY' \geq (q+1)\sum_{n=3}^{a} \frac{n(a-1)}{2} y_n$. Since $L_n \geq 2(n-3)$,

$$F(0) = YJ - 2T \geq 2(a-2)y_2 + \sum_{n=3}^{a-1} \big(L_n - 2(n-3)\big)(a-n)y_n \geq 0.$$

Using $u_n \leq q u_{n-1}$ and a shift of index, $\frac{4}{q}H_* + 2T \leq \sum_{n=2}^{a-1} E_n (a-n) y_n$, where $E_2 < 0$. As $L_n + \frac{n}{2} \geq E_n$ for $n \geq 3$,

$$\frac{F(1/q)}{1+1/q} = YJ + \frac{YY'}{q+1} - \frac{4}{q}H_* - 2T \geq \sum_{n=3}^{a-1} \Big(L_n + \frac{n}{2} - E_n\Big)(a-n)y_n \geq 0.$$

The numerical facts used are $L_3 = \frac{3}{2} \geq 0$, $L_3 + \frac{3}{2} = 3 \geq E_3 = 2$, and, for $n \geq 4$, $6(L_n - 2(n-3)) = (n-5)^2 + 11$ and $6(n+1)(L_n + \frac{n}{2} - E_n) = n^3 - 6n^2 + 5n + 60 > 0$.

*Resummation.* Regard the moves as operators on functions on $\Omega$, each divided by $c$ at the destination level; all strictly raise $\ell$, and inequalities between operators are entrywise. Let $P$ and $Q$ be the sums of the $B_j$ with $j$ even and odd, $C = C_+ - C_-$ with $C_\pm$ the sums of the $C'_j$ with $j$ even and odd, $H_+$ and $H_-$ the sums of the $H_j$ with $j \geq 2$ even and $j \geq 3$ odd, $V_+$ and $V_-$ the sums of the $V_j$ with $j$ odd and $j \geq 2$ even, $A_0$ the operator of $A_0^\psi$, and $U$ the straight moves. The recurrence reads $\omega = \beta + (U + P - Q + C + A)\omega$ with $A = A_0 + H_+ - H_- - H_1 + V_+ - V_-$ and $\beta = \omega|_{\ell=0} \geq 0$. The inequalities above give

$$P^2 \geq 4Q, \qquad H_+ P \geq 6H_-, \qquad V_+ P \geq 6V_-, \qquad C_+ P \geq 6C_-, \qquad A_0 P + P A_0 + 2V_1 \geq 4H_1.$$

Let $G = (I - P/2)^{-1} = \sum_{m=0}^{N} (P/2)^m \geq 0$. Then $PG = GP = 2(G - I)$ and $(I - P/6)G = \frac{2}{3}G + \frac{1}{3}I$, so $G(H_+ - H_-)G \geq GH_+(I - P/6)G \geq 0$, $GCG \geq 0$ and $G(V_+ - V_-)G \geq \frac{2}{3}GV_1G$. Hence

$$GAG \geq G\big(A_0 - H_1 + \tfrac{2}{3}V_1\big)G \geq G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \geq 0.$$

With $R = U + P^2/4 - Q \geq 0$, the operator $M = G(R + C + A)G$ is nonnegative and strictly raises $\ell$. Since $(I - P/2)^2\omega = \beta + (R + C + A)\omega$, the function $v = (I - P/2)\omega$ satisfies $v = G\beta + Mv$. Therefore $\omega = G\sum_{q=0}^{N} M^q G\beta \geq 0$, and $g(m, n; k) \geq 0$ as in the proof of Theorem 8.6. ∎

The diagnostics `runs/varratio/affine_verify.py` and `runs/varratio/verify_consultH.py` check in exact arithmetic the steps of this proof, the nonnegativity of $M$, the identity $\omega = G\sum_q M^q G\beta$ and property (P) on random profiles with $3 \le N \le 11$, including $\kappa$ at the largest value for which $b$ is nondecreasing.

For profiles whose Newton coefficients vanish from order 3 on, every increasing convex quadratic ratio with nondecreasing $b$ is admissible.

**Proposition 8.10** *(Quadratic profiles).* *Let $d_i = \nu_1 i + \nu_2\binom{i}{2}$ with $\nu_1 > 0$, $\nu_2 \ge 0$ and $d_N = 1$, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \ge 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

As in the proof of Theorem 8.9, $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - (t + \tau i)R_i\big)$ with a nonincreasing bracket, so $b$ is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N-1)R\big) \le 1 - r$, where $R = d_{N-1}/(1 - d_{N-1})$.

*Proof.* We use the recurrence of the proof of Theorem 8.6. Here $f_2 = \nu_2$ and $f_n = 0$ for $n \ge 3$. Fix $\ell$ and write $u = d_\ell$, $v = f_1(\ell)$, $w = \nu_2$ and $p = \rho(\ell + 1)$, so that $c_{\ell+1} = p(u + v)$ and, by convexity, $u \le \ell v$. For $\ell + n \le N$ the Leibniz rule gives $\varphi_n(\ell) = \rho(\ell + n)f_n + n(t + \tau(\ell + n - 1))f_{n-1} + \binom{n}{2}\tau f_{n-2}$, all at $\ell$. Hence $\varphi_n = \psi_n = 0$ for $n \ge 5$, $\varphi_3 \ge 0$, $\varphi_4 = 6\tau w \ge 0$, and $\psi_n = f_n - \varphi_n = -\varphi_n \le 0$ for $n = 3, 4$. On $\Omega$ the moves are therefore: $A_0^\psi = b_{\ell+a} - b_\ell \ge 0$; $A_1^\psi = \sum_{n=3,4}(n-2)\binom{a}{n}\psi_n \le 0$, which enters with the sign $-1$; $A_j^\psi = 0$ for $j \ge 2$; $B_0 \ge 0$, $B_1 = \binom{a}{4}\varphi_4 \ge 0$, and $B_j = 0$ for $j \ge 2$; $C_0 + C_0^\psi = W_{2,1}^f \ge 0$, and $C_j + C_j^\psi = W_{2j+2,j+1}^f = 0$ for $j \ge 1$; and the straight moves are nonnegative. The only negative contribution comes from $B_1$. Now $\varphi_2(\ell) \ge pw + \tau\big(u + 2(\ell + 1)v\big)$ and $\varphi_2(\ell + 1) \ge pw + \tau\big(u + v + 2(\ell + 2)(v + w)\big)$, and the two brackets together exceed $6(u + v)$ by $-4u + (4\ell + 1)v + (2\ell + 4)w \ge 0$. Hence $\varphi_2(\ell)\varphi_2(\ell + 1) \ge 6p\tau w(u + v) = c_{\ell+1}\varphi_4(\ell)$, and with $B_0 \ge \binom{a}{2}\varphi_2$ and $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$, $B_0(\ell, a)B_0(\ell + 1, a - 2) \ge 6c_{\ell+1}B_1(\ell, a)$. In the notation of the resummation above, $P = B_0$, $Q = B_1$, $P^2 \ge 4Q$, $H_j = 0$, $V_1 = -A_1^\psi \ge 0$, and all other operators are nonnegative, so $M \ge 0$, $\omega \ge 0$, and (P) follows as before. ∎

Proposition 8.10 extends to all profiles with log-concave Newton coefficients.

**Theorem 8.11** *(Convex quadratic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \ge 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 8.9 ($\tau = 0$) and Proposition 8.10. As there, $b$ is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N-1)R\big) \le 1 - r$ with $R = d_{N-1}/(1 - d_{N-1})$. The Newton coefficients of $c$ need not be log-concave: for $d_i = i/4$ and $\rho(x) = \frac{1}{2} + \binom{x}{2}/40$ they are $(1/8, 1/80, 3/160, 0)$. The proof below replaces their log-concavity by a weighted detour inequality.

*Proof.* Write $f_n(\ell) = \Delta^n d(\ell)$, $\Delta\rho(x) = t + \tau x$ and $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 8.9. Since $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - \Delta\rho(i)R_i\big)$ and $b$ is

nondecreasing, $1-\rho(m+1) \ge \Delta\rho(m)R_m$ for $0 \le m < N$, and $\rho(i) \le 1$ for $i \le N$. For $\ell+n \le N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n \;=\; \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \binom{n}{2}\tau f_{n-2}, \qquad \psi_n \;=\; h_n - v_n - w_n,$$

with $h_n = (1-\rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $w_n = \binom{n}{2}\tau f_{n-2}$; here $\varphi_n$, $h_n$, $v_n$ and $w_n$ are nonnegative, while $\psi_n$ need not be.

*Moves.* In the recurrence of the proof of Theorem 8.6, keep $A_0^\psi = b_{\ell+a} - b_\ell \ge 0$ and split $A_j^\psi = H_j - V_j - W_j$ for $j \ge 1$, where $H_j$, $V_j$ and $W_j$ are $W_{2j+1,j}$ formed from $h_n$, $v_n$ and $w_n$; the weights $B_j$, formed from $\varphi$, and $C'_j = C_j + C_j^\psi = W^f_{2j+2,j+1}$ are nonnegative. With the signs $(-1)^j$, the moves $W_j$ behave like the $V_j$.

*Detours of $H$, $V$, $W$ and $C'$.* All detours are asserted for sources $(\ell,a) \in \Omega$, with second factors at $(\ell+1, a-2)$, marked by a prime. Since $\varphi_2(\ell) \ge \rho(\ell+1)f_2(\ell)$ and $c_{\ell+1} = \rho(\ell+1)d_{\ell+1}$, Lemma 8.3 for $f$ gives, for $n \le a-2$,

$$\varphi_2 f'_n \ge c_{\ell+1}f_{n+2}, \quad \varphi_2 h'_n \ge c_{\ell+1}h_{n+2}, \quad (n+1)(n+2)\varphi_2 v'_n \ge n^2 c_{\ell+1}v_{n+2}, \quad (n+1)(n+2)\varphi_2 w'_n \ge n(n-1)c_{\ell+1}$$

the last two for $n \ge 3$. The second uses $1-\rho(\ell+n+1) \ge 1-\rho(\ell+n+2) \ge 0$; the third uses Lemma 8.3 with indices $(2, n-1)$ and $(n+1)\Delta\rho(\ell+n) - n\Delta\rho(\ell+n+1) = t+\tau\ell \ge 0$; the fourth uses indices $(2, n-2)$. As in the proof of Lemma 8.4, with $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$ and $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$,

$$B_0 H'_{j-1} \ge 10c_{\ell+1}H_j, \qquad B_0 V'_{j-1} \ge \tfrac{9}{2}c_{\ell+1}V_j, \qquad B_0 W'_{j-1} \ge 3c_{\ell+1}W_j \quad (j \ge 2), \qquad B_0\, C'_{j-1}(\ell+1, a-2) \ge 6c_{\ell}$$

The coefficient ratios are $\frac{jm(m-1)}{2(m-j-1)}$, $\frac{j(m-2)^2}{2(m-j-1)}$ and $\frac{j(m-2)(m-3)}{2(m-j-1)}$ on $m \ge 2j+1$, and $\frac{(j+1)m(m-1)}{2(m-j-1)}$ on $m \ge 2j+2$; they increase with $j$, and at the smallest $j$ the bounds are $(m-5)(m-6) \ge 0$, $(m-5)(2m-7) \ge 0$, $m \ge 5$ and $(m-3)(m-4) \ge 0$.

*Detour of $B$.* We show, for $n \ge 2$ and $\ell+n+2 \le N$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \;\ge\; 8(n-1)\,c_{\ell+1}\varphi_{n+2}(\ell). \tag{8.4}$$

Granting this, $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$, (8.4) with $n = m-2$, $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$ and $\binom{m}{2}\frac{8(m-3)}{m(m-1)} = 4(m-3)$ give

$$B_0(\ell,a)B_{j-1}(\ell+1,a-2) \;\ge\; 4c_{\ell+1}\sum_{m\ge 2j+2}(m-3)\binom{m-j-3}{j-1}\binom{a}{m}\varphi_m(\ell) \;\ge\; 4c_{\ell+1}B_j(\ell,a)$$

for $j \ge 1$, because $(m-3)\binom{m-j-3}{j-1} - \binom{m-j-2}{j} = \binom{m-j-3}{j-1}\frac{(j-1)(m-2)}{j} \ge 0$. (With constant 1 in place of $\frac{(n+1)(n+2)}{8(n-1)}$, (8.4) can fail: for $N=20$, $d_i = (2^i-1)/(2^{20}-1)$, $r=\frac12$, $t=0$, $\tau = 10^{-3}$ one gets $\varphi_2(16)\varphi_2(17) < c_{17}\varphi_4(16)$.)

To prove (8.4), put $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$, and $p = \rho(\ell+1)$, $K_n = (n+1)(n+2)$, $L_n = 8(n-1)$. Both sides are homogeneous of degree two in $u$; take $u_1 = 1$ and write $z = u_0$, $q = u_2$. By convexity and log-concavity, $0 \le z \le \ell$ and $zq \le 1$. If $q = 0$ or $u_n = 0$, then $\varphi_{n+2}(\ell) = 0$. Otherwise put $y = u_{n+1}/u_n \in [0,q]$. Log-concavity gives

$u_{n+2} \le y^2 u_n$, and for $n \ge 3$ also $u_{n-1} \ge u_n/q$ and $u_{n-2} \ge u_n/q^2$. Hence $\varphi_n(\ell+1) \ge u_n \mathcal{L}(y)$ and $\varphi_{n+2}(\ell) \le u_n \mathcal{T}(y)$, where

$$\mathcal{T}(y) \;=\; \rho(\ell+n+2)y^2 + (n+2)\Delta\rho(\ell+n+1)y + \binom{n+2}{2}\tau,$$

$\mathcal{L}(y) = \rho(\ell+n+1)(1+y) + n\Delta\rho(\ell+n)(1+q^{-1}) + \binom{n}{2}\tau(q^{-1}+q^{-2})$ for $n \ge 3$, and $\mathcal{L}(y) = \rho(\ell+3)(1+y) + 2\Delta\rho(\ell+2)(1+q^{-1}) + \tau(z+1)q^{-1}$ for $n = 2$ (exact). In both cases $\mathcal{L}(y) = A + By$ with $A \ge B \ge 0$, so $\mathcal{L}(0) \ge \mathcal{L}(q)/(1+q)$; and $\mathcal{T}(q) \ge (1+q)\mathcal{T}(0)$ because $(n+2)\Delta\rho(\ell+n+1) \ge (n+2)(n+1)\tau$. The function $\mathcal{F}(y) = K_n\varphi_2(\ell)\mathcal{L}(y) - L_n p(z+1)\mathcal{T}(y)$ is concave with $\mathcal{F}(0) \ge \mathcal{F}(q)/(1+q)$, so it suffices that $\mathcal{F}(q) \ge 0$, which is (8.4) for the configuration $u_0 = z$, $u_k = q^{k-1}$ $(k \ge 1)$.

Next we reduce to $z = \min(\ell, 1/q)$. For $n \ge 3$, only $\varphi_2(\ell) = A_2 + \tau z$ and $c_{\ell+1} = p(z+1)$ depend on $z$, where $A_2 = \rho(\ell+2)q + 2\Delta\rho(\ell+1) \ge \tau$, so $\varphi_2(\ell)/(z+1)$ is nonincreasing in $z$. For $n = 2$, write also $\varphi_2(\ell+1) = G_2 + \tau(z+1)$ with $G_2 = \rho(\ell+3)q(1+q) + 2\Delta\rho(\ell+2)(1+q)$. For $z \le Z \le \ell$ the quotient $\varphi_2(\ell)\varphi_2(\ell+1)/(z+1)$ at $z$ minus its value at $Z$ equals $(Z-z)\big(G_2(A_2-\tau)/((z+1)(Z+1)) - \tau^2\big) \ge 0$, since $G_2(A_2-\tau) \ge 2(\ell+2)(2\ell+1)\tau^2 \ge (\ell+1)^2\tau^2$.

For the remaining configurations we use the following polynomial inequality. For an integer $n \ge 2$ and reals $h \ge 1$, $p, \tau \ge 0$ and $s \ge (h-1)\tau$, put $A = p + (2h-1)s + h^2\tau$, $B = p + nh(s+\tau) + \binom{n}{2}h^2\tau$ and $C = p + ((n+2)h-1)s + \binom{n+2}{2}h^2\tau$. Then

$$K_n AB - L_n pC \;\ge\; 0. \tag{8.5}$$

Indeed, $K_nAB - L_npC$ is nondecreasing in $s$ (its $ps$-coefficient is $a_n((n+2)h-1)$ with $a_n = n^2 - 5n + 10 > 0$, and its other $s$-dependent coefficients are nonnegative), and at $s = (h-1)\tau$ it equals $a_np^2 + C_n(h)p\tau + D_n(h)\tau^2$, where $C_n(h) = \frac{1}{2}\big(h^2U_n + h(10n^2+14n-60) + 2a_n\big)$, $U_n = n^4 - 4n^3 - 21n^2 + 12n + 60$ and $D_n(h) = \frac{1}{2}h^2n(n+1)^2(n+2)(3h^2-3h+1)$. For $n \ge 7$ all three coefficients are nonnegative, since $U_n = k^4 + 24k^3 + 189k^2 + 502k + 144$ with $k = n-7$. For $2 \le n \le 6$, $4a_nD_n - C_n^2$ is a polynomial in $v = h-1$ with positive coefficients:

| $n$ | $v^0$ | $v^1$ | $v^2$ | $v^3$ | $v^4$ |
|---|---|---|---|---|---|
| 2 | 576 | 2880 | 5616 | 4992 | 1664 |
| 3 | 1520 | 6240 | 9744 | 7200 | 2160 |
| 4 | 6300 | 27000 | 42660 | 30600 | 8604 |
| 5 | 25200 | 126000 | 229500 | 184800 | 56000 |
| 6 | 62720 | 376320 | 774144 | 677376 | 216576 |

so $a_np^2 + C_np\tau + D_n\tau^2 = a_n\big(p + \frac{C_n\tau}{2a_n}\big)^2 + \frac{4a_nD_n - C_n^2}{4a_n}\tau^2 \ge 0$.

If $1/q \le \ell$, take $z = 1/q$, $h = 1 + 1/q$ and $s = t + \tau\ell \ge (h-1)\tau$. For the configuration above, $\mathcal{F}(q)$ divided by $q(1+q)$ equals $K_nAB - L_npC$, which is nonnegative by (8.5).

If $\ell < 1/q$, take $z = \ell$, and write $w = 1/q - \ell > 0$, $h_0 = \ell + 1$, $h = h_0 + w$ and $s = t + \tau\ell$. With $A(w) = p + (2h-1)s + \tau\big((\ell+2)h - \ell - 1\big)$ and $B(w)$, $C(w)$ as above (with this $h$), $\mathcal{F}(q)$ divided by $q^2$ equals $\mathcal{E}_n(w) = K_n(h_0+w)A(w)B(w) - L_nh_0pC(w)$ for $n \ge 3$, and $\mathcal{E}_2(w) - 12A(w)\tau w(\ell+w)^2$ for $n = 2$. At $w = 0$ both equal $h_0(K_nAB - L_npC)$ with

$h = h_0$, which is nonnegative by (8.5). Expanding in $w$, with $\bar A = A(0) - p$ and $\bar B = B(0) - p$, every coefficient of $w$, $w^2$, $w^3$ and $w^4$ is a sum of nonnegative terms: for $n \ge 3$,

$$[w] = K_n p^2 + p(a_1 t + b_1 \tau) + K_n\big(\bar A\bar B + h_0(\eta\bar B + \bar A\theta)\big),$$
$$[w^2] = p(a_2 t + b_2\tau) + K_n\big(\eta\bar B + \bar A\theta + h_0(\eta\theta + \bar A\epsilon)\big), \qquad [w^3] = K_n\big(\eta\theta + A(0)\epsilon + h_0\eta\epsilon\big), \qquad [w^4] = K_n\eta\epsilon,$$

where $\eta = 2s + (\ell+2)\tau$, $\theta = n(s+\tau) + n(n-1)\tau h_0$, $\epsilon = \binom{n}{2}\tau$, $a_1 = (n+2)\big(2\ell(n^2 - n + 6) + 2n^2 - 3n + 11\big)$, $a_2 = K_n(n+2)$, $b_1 = \frac{n+2}{2}(\ell^2 P_n + \ell Q_n + R_n)$ and $b_2 = \frac{K_n}{2}\big(3n(n-3)(\ell+1) + 14\ell + 12\big)$, with $P_n = 3k^3 + 15k^2 + 6k + 8$, $Q_n = 6k^3 + 30k^2 + 20k + 16$ and $R_n = 3k^3 + 15k^2 + 16k + 16$ for $k = n - 3$; for $n = 2$ the coefficients of $w$, $w^2$ and $w^3$ are

$$12p^2 + 4(16\ell+13)pt + 8(\ell+1)(2\ell+3)p\tau + 48(\ell+1)(3\ell+2)t^2 + 12(36\ell^3 + 92\ell^2 + 76\ell + 21)t\tau$$
$$\quad + 12(27\ell^4 + 93\ell^3 + 114\ell^2 + 62\ell + 13)\tau^2,$$
$$48pt + 12(2\ell+3)p\tau + 24(6\ell+5)t^2 + 12(36\ell^2 + 67\ell + 29)t\tau + 12(27\ell^3 + 78\ell^2 + 67\ell + 19)\tau^2,$$
$$12\big(2s + (\ell+2)\tau\big)\big(2t + (3\ell+5)\tau\big).$$

This proves (8.4).

*Key inequality.* We show, for $(\ell, a) \in \Omega$ with $a \ge 3$,

$$A_0^\psi(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^\psi(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1 - 2V_1 - W_1\big)(\ell,a). \quad (8.6)$$

Use the notation of the proof of Theorem 8.9 ($u_n = f_n(\ell)$, $y_n$, $D$, $X$, $Y$, $X'$, $Y'$, $J$, $H_*$, $T$), and put $\alpha = 1 - \rho(\ell+a)$, $s = \Delta\rho(\ell+a-1)$, $p = \rho(\ell+1)$ and $Z = \sum_{n\ge2}\binom{a-n}{2}y_n$. Then $\alpha \ge sR_{\ell+a-1} \ge su_0/u_1$ and $s \ge \tau(a-1)$. Backward differences give

$$A_0^\psi(\ell,a) = \alpha X - sau_0 + \tau\tbinom{a}{2}u_0, \qquad A_0^\psi(\ell+1,a-2) = (\alpha+s)X' - s(a-2)D + \tau\tbinom{a-1}{2}D,$$

both nonnegative, and $B_0(\ell,a) \ge pY$, $B_0(\ell+1,a-1) \ge pY'$, $c_{\ell+1} = pD$. Since $1 - \rho(\ell+n) = \alpha + (a-n)s - \tau\binom{a-n}{2}$, reindexing gives $4H_1 - 2V_1 = 4\alpha H_* + 2sT + 4\tau Z$ and $W_1 = \tau\sum_{n\ge1} n\binom{a-n}{2}y_n$. The argument of the proof of Theorem 8.9 with $s$ in place of $t$, keeping the surplus of the two endpoints ($F(0) \ge 2(a-2)y_2$ and $F(1/q) \ge \frac23(1+1/q)(a-2)y_2$, from the term $n = 2$ and $E_2 = -\frac23$), shows

$$(\alpha X - sau_0)Y' + Y\big((\alpha+s)X' - s(a-2)D\big) - D(4\alpha H_* + 2sT) \ge \tfrac23 sD(a-2)y_2$$

for $\alpha u_1 \ge su_0$; the right side of this bound is linear in $z = u_0/u_1$ and is subtracted before the interpolation. With $s \ge \tau(a-1)$, the difference of the two sides of (8.6) divided by $p$ is therefore at least

$$\tau\Big(\tbinom{a}{2}u_0Y' + D\Big[\tbinom{a-1}{2}y_1 + \tfrac{(a-2)(a+11)}{6}y_2 + (2a-5)y_3 + \sum_{n\ge4}\Big(\tbinom{a-1}{2} + (n-4)\tbinom{a-n}{2}\Big)y_n\Big]\Big) \ge 0.$$

*Resummation.* With $W_\pm$ the sums of the $W_j$ with $j$ odd and $j \ge 2$ even, the recurrence reads as in the proof of Theorem 8.9 with $A = A_0 - H_1 + (H_+ - H_-) + (V_+ - V_-) + (W_+ - W_-)$, and the inequalities above give $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge \frac92 V_-$, $W_+P \ge 3W_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 + W_1 \ge 4H_1$. Since $(I - P/\mu)G =$

$(1-\frac{2}{\mu})G+\frac{2}{\mu}I$, one has $G(H_+-H_-)G\ge 0$, $GCG\ge 0$, $G(V_+-V_-)G\ge \frac{5}{9}GV_1G$ and $G(W_+-W_-)G\ge \frac{1}{3}GW_1G$, hence

$$GAG\ge G\big(A_0-\tfrac{1}{4}(A_0P+PA_0)+\tfrac{1}{18}V_1+\tfrac{1}{12}W_1\big)G=\tfrac{1}{2}(GA_0+A_0G)+\tfrac{1}{18}GV_1G+\tfrac{1}{12}GW_1G\ge 0.$$

With $U+P^2/4-Q\ge 0$ this gives $M\ge 0$, and (P) follows as before. ∎

The diagnostics `runs/varratio/verify_consultI.py`, `runs/varratio/verify_consultI_poly.py` and `runs/varratio/verify_consultI_local.py` check in exact arithmetic (8.4), the summed $B$-detour, (8.6), the detours of $H$, $V$, $W$ and $C'$, the identities of the key inequality, $M\ge 0$ and (P) on random members with $4\le N\le 13$, the certificate (8.5) symbolically, and (8.4) on random log-concave local configurations with $\ell\le 200$.

**Ratios of higher degree.** For profiles of Newton degree at most two the ratio can have any degree, provided the Newton coefficients of $c$ are nonnegative and log-concave without internal zeros; no key inequality is needed.

**Theorem 8.12** *(Low-degree profiles).* *Let $d_i=\nu_1 i+\nu_2\binom{i}{2}$ with $\nu_1>0$, $\nu_2\ge 0$ and $d_N=1$, let $c_0=0$, and suppose that the Newton coefficients $\kappa_k=\Delta^k c(0)$, $1\le k\le N$, are log-concave without internal zeros. Put $a_i=1-d_i$ and $b_i=d_i-c_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

*Proof.* We use the recurrence of the proof of Theorem 8.6, which holds for every array, with $\varphi_n(\ell)=\Delta^n c(\ell)$, $\psi_n(\ell)=\Delta^n b(\ell)$ and $f_n(\ell)=\Delta^n d(\ell)$ for the polynomial extensions. Since $f_n=0$ for $n\ge 3$, every move $A^{\psi}_j$ with $j\ge 1$, which involves only $n\ge 2j+1\ge 3$, equals $-\widetilde{B}_j$ with $\widetilde{B}_j=W_{2j+1,j}$ formed from $\varphi$; the moves $C_j+C^{\psi}_j=W^{f}_{2j+2,j+1}$ vanish for $j\ge 1$, and $C_0+C^{\psi}_0=\binom{a}{2}\nu_2\ge 0$. The weight $A^{\psi}_0(\ell,a)=b_{\ell+a}-b_\ell$ is nonnegative on $\Omega$, as are the straight weights and all $B_j$ and $\widetilde{B}_j$, since $(\varphi_n(\ell))_n$ is obtained from $(\kappa_k)$ by the shifts of Lemma 8.3 and is therefore nonnegative and log-concave without internal zeros. Lemma 8.3 for $\varphi$ gives, as in Lemma 8.4, $B_0(\ell,a)B_{j-1}(\ell+1,a-2)\ge 4c_{\ell+1}B_j(\ell,a)$ for $j\ge 1$, and with $B_0(\ell,a)\ge\binom{a}{2}\varphi_2(\ell)$ also $B_0(\ell,a)\widetilde{B}_{j-1}(\ell+1,a-2)\ge 4c_{\ell+1}\widetilde{B}_j(\ell,a)$ for $j\ge 2$, because the coefficient quotients $jm(m-1)/(2(m-j-1))$ on $m\ge 2j+1$ are at least 10. In the resummation of the proof of Theorem 8.9 take $H=0$ and $V_j=\widetilde{B}_j$; the key inequality is then trivial, and $M\ge 0$ gives (P). ∎

**Corollary 8.13.** *In Theorem 8.12 let $c_i=\rho(i)d_i$ with $\rho(x)=\sum_k\rho_k\binom{x}{k}$, where $(\rho_k)_{k\ge 0}$ is nonnegative and log-concave without internal zeros and $\rho_0>0$. Then the Newton coefficients of $c$ are log-concave without internal zeros. Hence every such ratio, of any degree, with nondecreasing $b$ gives nonnegative linearization on these profiles.*

*Proof.* Put $s_k=\rho_k+\rho_{k-1}$ ($\rho_{-1}=0$), which is log-concave without internal zeros, and $h=\nu_2/2$. The Leibniz rule gives $\kappa_k=ke_k$ for $k\ge 1$, with $e_k=(\nu_1+h(k-1))s_k+h(k-$

$1)s_{k-1}$. Write $\alpha = \nu_1 + h(k-1)$ and $\beta = h(k-1)$; then

$$e_k^2 - e_{k-1}e_{k+1} = \alpha^2(s_k^2 - s_{k-1}s_{k+1}) + \beta^2(s_{k-1}^2 - s_{k-2}s_k) + (\beta - h)(\alpha + h)(s_{k-1}s_k - s_{k-2}s_{k+1}) + h^2(s_{k-1}s_{k+1} + s_{k-2}s_k + 2s_{k-1}s_k),$$

and every term is nonnegative for $k \geq 2$ by log-concavity of $s$. So $\kappa_k^2 \geq k^2 e_{k-1}e_{k+1} \geq \kappa_{k-1}\kappa_{k+1}$. If $\kappa_k = 0$ with $k \geq 1$, then $s_k = 0$, so $\rho_k = 0$; as $\rho_0 > 0$ and $\rho$ has no internal zeros, $\rho_m = 0$ for $m \geq k$, and $\kappa_{k+1} = 0$. ■

For example, on the linear profile $d_i = i/N$ the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ with $r > 0$ and $\varepsilon \geq 0$, whose Newton coefficients $r\varepsilon^k$ are geometric, give nonnegative linearization whenever $b$ is nondecreasing. Nonnegativity of the ratio's Newton coefficients alone does not suffice: the cubic counterexample in Section 10 has the linear profile $d_i = i/4$ and the ratio with Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$, which are nonnegative and satisfy the log-concavity inequalities but have internal zeros. Theorem 8.11 and Corollary 8.13 are not comparable: the former admits convex quadratic ratios with $t^2 < r\tau$ on every log-concave Newton profile, the latter ratios of every degree on profiles of Newton degree at most two. The diagnostic `runs/varratio/thmQ_check.py` checks in exact arithmetic the formula $\kappa_k = ke_k$, the log-concavity of $\kappa$, Lemma 8.3 for $\varphi$ with $i = 2$, both detours, $M \geq 0$ and (P) on random members with $3 \leq N \leq 10$, including geometric $(\rho_k)$.

**Products of Newton sequences.** The proof of Corollary 8.13 used only that the Newton coefficients of $c = \rho d$ are log-concave. This holds for every profile.

**Proposition 8.14** *(Products).* *Let $(\rho_i)_{i\geq 0}$ and $(\nu_j)_{j\geq 0}$ be nonnegative and log-concave without internal zeros, extended by zero at negative indices, and let $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$ for $x \in \mathbb{Z}_{\geq 0}$, where the sums are finite. Then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of $fg$ are nonnegative and log-concave without internal zeros.*

*Proof.* The product rule $\binom{x}{i}\binom{x}{j} = \sum_k \binom{k}{i}\binom{i}{k-j}\binom{x}{k}$ gives $\kappa_k = \sum_p H_k(p)\rho_{p_1}\nu_{p_2}$, where $H_k(i,j) = \binom{k}{i}\binom{i}{k-j} = [y^i w^j](y + w + yw)^k$ vanishes unless $i, j \leq k \leq i + j$. If either sequence vanishes identically, so does $\kappa$. Otherwise, if $\rho$ and $\nu$ are supported on $[p_0, p_1]$ and $[q_0, q_1]$, with $p_1, q_1 \leq \infty$, then $\kappa_k > 0$ exactly for $\max(p_0, q_0) \leq k \leq p_1 + q_1$, so $\kappa$ has no internal zeros. For log-concavity fix $k \geq 1$; then

$$\kappa_k^2 - \kappa_{k-1}\kappa_{k+1} = \sum_{c\in\mathbb{Z}^2}\sum_{p\in\mathbb{Z}^2} T_c(p)\, u_c(p_1)\, w_c(p_2), \qquad T_c(p) = H_k(p)H_k(c-p) - H_{k-1}(p)H_{k+1}(c-p),$$

with $u_c(i) = \rho_i\rho_{c_1-i}$ and $w_c(j) = \nu_j\nu_{c_2-j}$. By log-concavity without internal zeros, $u_c$ is symmetric about $c_1/2$ and nondecreasing on $i \leq c_1/2$, so $u_c = \sum_{0\leq a\leq c_1/2}(u_c(a) - u_c(a-1))\mathbf{1}_{[a,c_1-a]}$ with nonnegative coefficients, and likewise $w_c$. It therefore suffices to show $D_c(B) = \sum_{p\in B} T_c(p) \geq 0$ for every box $B = [a, c_1 - a] \times [b, c_2 - b]$ with $a \leq c_1/2$, $b \leq c_2/2$.

Let $E = \{(1,0), (0,1), (1,1)\}$, so that $H_k = \sum_{e\in E} H_{k-1}(\cdot - e)$. Expanding $H_k(p)$,

$H_k(c-p)$ and $H_{k+1}(c-p)$ in this way gives

$$D_c(B) \;=\; \sum_{e,e'\in E} \big(V_{e+e'}(B-e) - V_{e+e'}(B)\big), \qquad V_s(p) \;=\; H_{k-1}(p)H_{k-1}(c-s-p),$$

where $V_s(X) = \sum_{p\in X} V_s(p)$. Fix $s = e+e'$, write $V = V_s$, $m = c-s$, $t = k-1$ and $\mu = m/2$. With $n_1 = 2t-m_2$, $n_2 = 2t-m_1$, $n_3 = m_1+m_2-2t$ one has $V(x,y) = \frac{t!^2}{n_1!n_2!n_3!}\binom{n_1}{t-y}\binom{n_2}{t-x}\binom{n_3}{x+y-t}$ (and $V=0$ if some $n_i<0$). Three properties follow.

(T1) For every interval $W$, the marginal $G_W(x) = \sum_{y\in W} V(x,y)$ is a constant times $\binom{n_2}{t-x}\sum_{z\in t-W}\binom{n_1}{z}\binom{n_3}{x-z}$, a product of a binomial row and a convolution of two log-concave sequences without internal zeros; so $G_W$ is log-concave without internal zeros. The same holds with the roles of the coordinates exchanged.

(T2) $V(c-s-p) = V(p)$, and $V$ is reverse regular of order two: $V(x,y)V(x',y') \le V(x,y')V(x',y)$ for $x\le x'$, $y\le y'$, by log-concavity of $\binom{n_3}{\cdot}$. Taking $x' = 2\mu_1 - x$ and $y, y'$ symmetric about $\mu_2$, central symmetry turns this into $V(x,y)^2 \le V(x,y')^2$. Hence $V(x,y') \ge V(x,y)$ whenever $x\le\mu_1$, $y\le y'$ and $y+y' = 2\mu_2$; and symmetrically in the coordinates.

(T3) If $G$ is log-concave without internal zeros and $G(x) \le G(2\mu_1-x)$ for all $x\le\mu_1$, then $G$ is nondecreasing on $x\le\mu_1$: otherwise $G(x-1) > G(x)$ for some $x\le\mu_1$, and log-concavity makes $G$ nonincreasing from $x-1$ on, so $G(2\mu_1-x+1) \le G(x) < G(x-1)$.

Write $I = [a, c_1-a]$, $J = [b, c_2-b]$, $X(I',J') = V(I'\times J')$, and telescope along rows or columns. By the symmetry $H_k(i,j) = H_k(j,i)$ it suffices to treat five ordered pairs.

$e = e' = (1,0)$: $\mu = c/2-(1,0)$ and $V(B-e)-V(B) = G_J(a-1) - G_J(a-2)$ by central symmetry. Since $J$ is symmetric about $\mu_2$, $G_J$ is symmetric about $\mu_1$, hence nondecreasing on $x\le\mu_1$ by (T1) and (T3); and $a-1\le\mu_1$.

$e = (1,0)$, $e' = (0,1)$: $\mu = (c-(1,1))/2$ and $V(B-e)-V(B) = V(a-1, c_2-b) - V(a-1,b-1) \ge 0$ by (T2), since $a-1<\mu_1$ and $(c_2-b)+(b-1) = 2\mu_2$.

$e = (1,0)$, $e' = (1,1)$: $\mu = c/2-(1,\frac12)$ and

$$V(B-e)-V(B) \;=\; \big[V(a-1,c_2-b) - V(a-1,b-1)\big] + \big[G_{J-1}(a-1) - G_{J-1}(a-2)\big].$$

The first bracket is nonnegative by (T2), as $a-1\le\mu_1$. For the second, $G_{J-1}(2\mu_1-x) - G_{J-1}(x) = V(x,c_2-b) - V(x,b-1) \ge 0$ for $x\le\mu_1$ by (T2), so (T1) and (T3) apply.

$e = (1,1)$, $e' = (1,0)$: the same $V$, and $V(B-e)-V(B) = \big[X(I-1,J-1) - X(I-1,J)\big] + \big[V(B-(1,0)) - V(B)\big]$. The first bracket vanishes by central symmetry, since $I-1$ is symmetric about $\mu_1$; the second is the previous case.

$e = e' = (1,1)$: $\mu = c/2-(1,1)$ and $V(B-e)-V(B) = \big[K(b-1)-K(b-2)\big] + \big[X(a-1,J) - X(a-1,J-2)\big] + \big[G_{J-2}(a-1) - G_{J-2}(a-2)\big]$ with $K(y) = \sum_{x\in I-1} V(x,y)$. Since $I-1$ is symmetric about $\mu_1$, $K$ is symmetric about $\mu_2$ and nondecreasing on $y\le\mu_2$ by (T1) and (T3); and $b-1\le\mu_2$. The second bracket is $[V(a-1,c_2-b) - V(a-1,b-2)] + [V(a-1,c_2-b-1) - V(a-1,b-1)]$, two differences of the form in

(T2), since both pairs of second coordinates sum to $2\mu_2 = c_2 - 2$. For the third, $G_{J-2}(2\mu_1 - x) - G_{J-2}(x) = X(x, J) - X(x, J-2) \geq 0$ for $x \leq \mu_1$ by (T2), so (T1) and (T3) give $G_{J-2}(a-1) \geq G_{J-2}(a-2)$.

Each of the nine ordered-step contributions $V_{e+e'}(B-e) - V_{e+e'}(B)$ is nonnegative, so $D_c(B) \geq 0$, which proves the proposition. (The individual $T_c(p)$ need not be nonnegative.) ∎

In the language of random sets: if $S$ and $T$ are independent random subsets of $[n]$ whose laws are invariant under permutations and whose cardinalities are ultra-log-concave of order $n$, then so is $|S \cup T|$, since $\Pr(|S \cup T| = k) = \binom{n}{k}\kappa_k$ for $\Pr(S = A) = \rho_{|A|}$ and $\Pr(T = A) = \nu_{|A|}$. This closure theorem is due to Walkup [6, Theorem 2], who proves that the double-binomial convolution $h_k = \sum_i \sum_r \binom{k}{i}\binom{i}{r}\rho_i \nu_{k-r}$, which is $\kappa_k$ after the substitution $j = k - r$, preserves log-concavity; Proposition 8.14 expresses it in the Newton basis, and the proof above is self-contained. Hoggar's theorem is the analogue for the power basis. The diagnostics `runs/np/two_step.py` and `runs/np/case_proofs.py` check in exact arithmetic the identity for $D_c(B)$, the sign of each of its nine terms for $k \leq 9$, and every bracket of the case analysis, (T1) and (T2) for $k \leq 6$.

**Log-concave ratios on log-concave profiles.** Corollary 8.13 extends to every log-concave Newton profile. Besides Proposition 8.14, the proof uses a sign criterion for quadratic forms in log-concave sequences.

**Lemma 8.15.** *Let $Q(u) = \sum_{0 \leq i \leq j} q_{ij} u_i u_j$ with $q_{0j} \leq 0$ for all $j$ and $q_{ij} \geq 0$ for $2 \leq i \leq j$, and put $N_W(Q) = \sum_{i \leq j,\, i+j=W} q_{ij}$. If $N_W(Q) \geq 0$ for every $W$, then $Q(u) \geq 0$ for every nonnegative sequence $(u_n)_{n \geq 0}$ that is log-concave without internal zeros.*

*Proof.* Fix $W$ and put $p_i = u_i u_{W-i}$ for $0 \leq i \leq W/2$ and $p_{-1} = 0$. Log-concavity without internal zeros makes $p$ nondecreasing. With $T_k = \sum_{k \leq i \leq W/2} q_{i,W-i}$ we get $\sum_{i \leq W/2} q_{i,W-i} p_i = \sum_{k \leq W/2} T_k (p_k - p_{k-1})$. Here $T_0 = N_W(Q) \geq 0$, $T_1 = N_W(Q) - q_{0W} \geq 0$, and $T_k \geq 0$ for $k \geq 2$. ∎

The weight sums are the coefficients of $Q$ on geometric sequences: $Q(1, r, r^2, \ldots) = \sum_W N_W(Q) r^W$. This Abel argument, with the reduction to the full sum when the negative terms come first, is that of Wang and Yeh [9, Lemma 2.1 and Corollary 2.6].

**Theorem 8.16** *(Log-concave ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k \geq 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i)) d_i$ and $c_i = \rho(i) d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 8.9 and Corollary 8.13. It also contains the case $t^2 \geq r\tau$ of Theorem 8.11. The exponential ratios $\rho(x) = r(1 + \varepsilon)^x$ are admitted on every log-concave Newton profile. The proof is computer-assisted in two places, both by exact rational arithmetic:

- the signs of finitely many explicit polynomials in the weight-sum certificates of step (vi);
- the key inequality (8.8) for windows $a \le 5$, where step (vii) uses exact linear-programming certificates.

*Proof. Setup.* Only $\rho_0, \dots, \rho_N$ enter, and truncating $(\rho_k)$ after index $N$ keeps it log-concave without internal zeros, so we may assume it finitely supported. Write $f_n(\ell) = \Delta^n d(\ell)$, $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 8.9, and $P_k(x) = \Delta^k \rho(x) = \sum_h \binom{x}{h} \rho_{k+h}$. As in the proof of Theorem 8.11, $1 - \rho(m+1) \ge \Delta\rho(m) R_m$ for $0 \le m < N$, and $0 < \rho_0 \le \rho(i) \le 1$ for $i \le N$; $\rho$ is nondecreasing.

For every integer $x \ge 0$, the sequence $(P_k(x))_k$ arises from $(\rho_k)$ by the shifts of Lemma 8.3, so it is log-concave without internal zeros. Since $\rho(x+1)^2 - \rho(x)\rho(x+2) = P_1(x)^2 - P_0(x)P_2(x) \ge 0$, the positive sequence $(\rho(n))_{n\ge 0}$ is log-concave. (The function $x \mapsto \rho(x)$ of a real variable need not be: $\rho(x) = 1 + x/2 + x^2/2$ has $(\log \rho)''(0) > 0$.) Log-concavity in $k$ gives $\rho(y)P_{k+1}(y) \le P_1(y)P_k(y)$. Combining these facts, for integers $0 \le x \le y$ and $k \ge 1$,

$$\rho(x)P_{k-1}(y+1) \le \rho(x+1)P_{k-1}(y), \qquad \rho(x)P_k(y) \le \Delta\rho(x)P_{k-1}(y). \tag{8.7}$$

Indeed, log-concavity of $x \mapsto \rho(x)$ gives $\rho(x)\rho(y+1) \le \rho(x+1)\rho(y)$, equivalently $\rho(x)P_1(y) \le \Delta\rho(x)\rho(y)$. For the first inequality, $\rho(y)P_{k-1}(y+1) = \rho(y)P_{k-1}(y) + \rho(y)P_k(y) \le \rho(y+1)P_{k-1}(y)$; multiply by $\rho(x)$ and divide by $\rho(y) > 0$. For the second, $\rho(x)\rho(y)P_k(y) \le \rho(x)P_1(y)P_{k-1}(y) \le \Delta\rho(x)\rho(y)P_{k-1}(y)$. By Proposition 8.14, the Newton coefficients of $c = \rho d$ are log-concave without internal zeros, and so is $(\varphi_n(\ell))_n$ for every $\ell$. For $\ell + n \le N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n = \sum_{j=0}^{n} \binom{n}{j} P_j(\ell + n - j) f_{n-j}, \qquad \psi_n = h_n - v_n - \widehat{w}_n,$$

with $h_n = (1 - \rho(\ell+n)) f_n$, $v_n = n\Delta\rho(\ell+n-1) f_{n-1}$ and $\widehat{w}_n = \sum_{j=2}^{n} w_n^{(j)}$, where $w_n^{(j)} = \binom{n}{j} P_j(\ell+n-j) f_{n-j}$. Here $\varphi_n$, $h_n$, $v_n$, $w_n^{(j)}$ and $\widehat{w}_n$ are nonnegative, while $\psi_n$ need not be.

*Moves and detours.* Split the moves as in the proof of Theorem 8.11, with $W_j$ now formed from $\widehat{w}_n$. Lemma 8.3 for $\varphi$ gives $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \ge 4c_{\ell+1}B_j(\ell, a)$ for $j \ge 1$, as in Lemma 8.4. The detours of $H$ and $C'$, with constants 10 and 6, hold verbatim: they used only $\varphi_2(\ell) \ge \rho(\ell+1) f_2(\ell)$, Lemma 8.3 for $f$, and $0 \le \rho \le 1$ nondecreasing.

For $V$, the term $\rho(\ell+2)f_2$ of $\varphi_2$, the first inequality of (8.7) with $k = 2$, $x = \ell + 1$, $y = \ell + n$, and Lemma 8.3 for $f$ with indices $(2, n-1)$ give

$$\varphi_2(\ell) v_n(\ell+1) \ge \tfrac{n}{n+2}\, c_{\ell+1} v_{n+2}(\ell).$$

For $\widehat{w}$, expand $\varphi_2(\ell) = \rho(\ell+2)f_2 + 2\Delta\rho(\ell+1)f_1 + P_2(\ell)f_0$, and pair its three terms with the components of $\widehat{w}_n(\ell+1)$:

- $\rho(\ell+2)f_2$ with component $i \le n$, by the first inequality of (8.7) and indices $(2, n-i)$, gives at least $\binom{n}{i}\binom{n+2}{i}^{-1} c_{\ell+1} w_{n+2}^{(i)}(\ell)$;

- $2\Delta\rho(\ell+1)f_1$ with component $i \le n$, by the second inequality of (8.7) and indices $(1, n-i)$, gives at least $2\binom{n}{i}\binom{n+2}{i+1}^{-1} c_{\ell+1} w_{n+2}^{(i+1)}(\ell)$;

- $P_2(\ell) f_0$ with component $n$ gives at least $c_{\ell+1} w_{n+2}^{(n+2)}(\ell)$, since $P_2(\ell)P_n(\ell+1) \ge (P_0 + P_1)(\ell)P_{n+2}(\ell)$ by log-concavity in $k$, and $f_0(\ell+1) = d_{\ell+1}$.

When $n - i = 0$ the index condition is trivial. Collecting by target,

$$\varphi_2(\ell)\widehat{w}_n(\ell+1) \ge c_{\ell+1} \sum_{i=2}^{n+2} \theta_{n,i} w_{n+2}^{(i)}(\ell),$$

where $\theta_{n,2} = \binom{n}{2}\binom{n+2}{2}^{-1}$, $\theta_{n,i} = 1 - \frac{i(i-1)}{(n+1)(n+2)}$ for $3 \le i \le n$, $\theta_{n,n+1} = \frac{2}{n+2}$ and $\theta_{n,n+2} = 1$.

As in the proof of Theorem 8.11, the coefficient quotients are $\frac{j(m-1)(m-2)}{2(m-j-1)}$ for $V$, and $\theta_{m-2,i}\frac{jm(m-1)}{2(m-j-1)}$ for $\widehat{W}$, on $m \ge 2j+1$, $j \ge 2$. The first is at least 6, by $(m-4)(m-5) \ge 0$ at $j = 2$. The second exceeds 2: it is at least $\frac{j(m-2)(m-3)}{2(m-j-1)} \ge m-2$ for $i = 2$, at least $\frac{j(2m-3)}{m-j-1}$ for $3 \le i \le m-2$, equal to $\frac{j(m-1)}{m-j-1}$ for $i = m-1$, and at least 10 for $i = m$. Hence

$$B_0 V'_{j-1} \ge 6c_{\ell+1} V_j \qquad \text{and} \qquad B_0 W'_{j-1} \ge 2c_{\ell+1} W_j \qquad (j \ge 2).$$

*Resummation.* We prove below, for $(\ell, a) \in \Omega$ with $a \ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^{\psi}(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1 - 2V_1\big)(\ell,a). \tag{8.8}$$

With the notation of the proof of Theorem 8.11, the inequalities above give $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge 6V_-$, $W_+P \ge 2W_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 \ge 4H_1$. Since $(I - P/\mu)G = (1 - \frac{2}{\mu})G + \frac{2}{\mu}I$, one has $G(H_+ - H_-)G \ge 0$, $GCG \ge 0$, $G(W_+ - W_-)G \ge 0$ and $G(V_+ - V_-)G \ge \frac{2}{3}GV_1G$, so

$$GAG \ge G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \ge 0,$$

and (P) follows as before.

*Proof of (8.8): notation.* Use the notation of the proof of Theorem 8.9: $u_n = f_n(\ell)$, which is log-concave without internal zeros, $y_n$, $D$, $X$, $X'$, $Y$, $Y'$, $J$, $H_*$. Put also $A = d_{\ell+a}$, $B = d_{\ell+a-1}$, $F = A - B = f_1(\ell+a-1)$, $X'' = B - d_\ell$ and $P_k = P_k(\ell)$. Since $c_{\ell+n} = d_{\ell+n}\sum_k \binom{n}{k}P_k$ for $n \le a$, the following are linear in $P$:

- $B_0(\ell,a) = \sum_k \beta_k P_k$ and $B_0(\ell+1,a-1) = \sum_k \beta'_k P_k$, where

  $\beta_0 = Y$, $\beta_1 = a(A-D)$, $\beta_k = \binom{a}{k}A$ $(k \ge 2)$,
  $\beta'_0 = Y'$, $\beta'_1 = aA - 2(a-1)d_{\ell+2} + (a-2)D$, $\beta'_2 = \binom{a}{2}A - (a-1)d_{\ell+2}$, $\beta'_k = \binom{a}{k}A$ $(k \ge 3)$;

- $c_{\ell+a} - c_\ell = \sum_k \chi_k P_k$ and $c_{\ell+a-1} - c_{\ell+1} = \sum_k \chi'_k P_k$;

- $c_{\ell+a} - c_{\ell+a-1} = \sum_k \delta_k P_k$ with $\delta_k = \binom{a}{k}F + \binom{a-1}{k-1}B$;

- $H_1 = H_* - \sum_k \zeta_k P_k$ with $\zeta_k = \sum_{n\ge 3}(n-2)\binom{n}{k}y_n$, and $V_1 = \sum_k \nu_k P_k$ with $\nu_k = \sum_{n\ge 3}(n-2)n\binom{n-1}{k-1}\binom{a}{n}u_{n-1}$;
- $c_{\ell+1} = D(P_0 + P_1)$.

As $A_0^\psi(\ell, a) = X - (c_{\ell+a} - c_\ell)$ and $A_0^\psi(\ell+1, a-2) = X' - (c_{\ell+a-1} - c_{\ell+1})$, the difference of the two sides of (8.8) is $\mathrm{Key}(P) = \mathrm{Lin}(P) - \mathrm{Quad}(P)$. Here $\mathrm{Lin}(P) = \sum_k l_k P_k$ with $l_k = X\beta_k' + \beta_k X' - 4[k \le 1]DH_*$, and Quad is a quadratic form. We use repeatedly:

$$AX'' \ge (a-1)d_\ell F, \qquad AX' \ge (a-2)DF, \qquad u_0 F \le u_1 B, \qquad FD \le u_1 A. \tag{8.9}$$

The first two hold because $A/F \ge R_{\ell+a-1} \ge R_{\ell+i}$ gives $(A/F)f_1(\ell+i) \ge d_{\ell+i}$ for $i \le a-1$; summing over $0 \le i \le a-2$, respectively $1 \le i \le a-2$, gives them. The last two follow from $u_0 u_{m+1} \le u_1 u_m$, as $F = \sum_m \binom{a-1}{m}u_{m+1}$, $B = \sum_m \binom{a-1}{m}u_m$ and $D = u_0 + u_1$.

*(i) Reduction to a quadratic form in $P$.* By monotonicity of $b$ at $\ell + a - 1$, $\Delta c := c_{\ell+a} - c_{\ell+a-1} \le F$. With $\Phi(P) := \Delta c \cdot \mathrm{Lin}(P) - F\,\mathrm{Quad}(P)$ one has $\mathrm{Key} = (1 - \Delta c/F)\mathrm{Lin} + \Phi/F$, so it suffices that every $l_k \ge 0$ and $\Phi \ge 0$. With $\chi_k = \binom{a}{k}A - [k=0]d_\ell$ and $\chi_k' = \binom{a-1}{k}B - \binom{1}{k}D$, so that $E_m^1 = X\delta_m - F\chi_m$, $E_m^2 = X'\delta_m - F\chi_m'$ and $E_m^3 = 2F\nu_m - 4(H_*\delta_m - F\zeta_m)$, expansion gives $\Phi(P) = \sum_{k,m} M_{k,m}P_kP_m$ with

$$M_{k,m} = \beta_k' E_m^1 + \beta_k E_m^2 + [k \le 1]D\,E_m^3,$$

and explicitly $E_0^1 = E_0^2 = E_0^3 = 0$ and, for $m \ge 1$,

$$E_m^1 = \tbinom{a-1}{m-1}AX'' - \tbinom{a-1}{m}d_\ell F, \qquad E_m^2 = \tbinom{a-1}{m-1}AX' - \Big(\tbinom{a-1}{m} - [m=1]\Big)DF,$$
$$E_m^3 = 2F\nu_m - 4\tbinom{a-1}{m-1}BH_* - 4FZ_m, \qquad Z_m = \sum_{n\ge 3}(n-2)\Big(\tbinom{a}{m} - \tbinom{n}{m}\Big)y_n.$$

Since $(P_k)$ is log-concave without internal zeros, the Abel summation in the proof of Lemma 8.15, applied to every anti-diagonal of $\Phi$, shows $\Phi \ge 0$ if every central sum $S_{\sigma,i} = \sum_{k=i}^{\sigma-i} M_{k,\sigma-k}$, $0 \le i \le \sigma/2$, is nonnegative. As $M_{k,0} = 0$, for $\sigma \ge 3$

$$S_{\sigma,0} = S_{\sigma,1} + M_{0,\sigma} \qquad \text{and} \qquad S_{\sigma,1} = S_{\sigma,2} + M_{1,\sigma-1} + M_{\sigma-1,1}.$$

Since $\binom{a-1}{m} = \frac{a-m}{m}\binom{a-1}{m-1}$ and $\frac{a-m}{m(a-1)} \le \frac{a-m}{m(a-2)} \le \frac{1}{2}$ for $m \ge 2$, (8.9) gives, for $m \ge 2$,

$$E_m^1 \ge \tfrac{1}{2}\tbinom{a-1}{m-1}AX'' \ge 0 \qquad \text{and} \qquad E_m^2 \ge \tfrac{1}{2}\tbinom{a-1}{m-1}AX' \ge 0.$$

Also every $\beta_k \ge 0$ and $\beta_k' \ge 0$; for $\beta_2'$ use $\binom{a}{2}A \ge (a-1)d_{\ell+2}$. Finally $E_1^1 = AX'' - (a-1)d_\ell F \ge 0$ and $E_1^2 = AX' - (a-2)DF \ge 0$.

*(ii) The central sums with $i \ge 2$, and the coefficients $l_k$.* For $k, m \ge 2$, $M_{k,m} = \beta_k' E_m^1 + \beta_k E_m^2 \ge 0$, so $S_{\sigma,i} \ge 0$ for $i \ge 2$. Next, $l_0 = K \ge 0$ by the proof of Theorem 8.9, and $l_k \ge 0$ for $k \ge 2$ termwise. By convexity, $A - D \ge (a-1)f_1(\ell+1) = (a-1)(d_{\ell+2} - D)$, so $\beta_1' \ge (a-2)(A-D)$ and

$$l_1 \ge L := (A-D)\big((a-2)X + aX'\big) - 4DH_*.$$

*(iii) The central sums with $\sigma \le 2$.* Take $P = (P_0, P_1, 0, \dots)$ with $P_0 \ge 0$ and $P_1 > 0$, and put $\lambda = F/\Delta c(P)$. Then $\mathrm{Key}(\lambda P) = (\lambda/\Delta c)\Phi(P)$, and $\lambda P$ is the linear ratio $\tilde\rho(\ell + x) = \lambda(P_0 + P_1 x)$. Its $\tilde b = (1-\tilde\rho)d$ satisfies $\Delta\tilde b(\ell + a - 1) = 0$. So $\alpha := 1 - \tilde\rho(\ell + a) = tR_{\ell+a-1} \ge tu_0/u_1$ with $t = \lambda P_1$. For such $\alpha$ and $t$, the proof of Theorem 8.9 shows (8.3). Its two weights $A_0^\psi$ are nonnegative, since $X \ge au_1$ and $X' \ge (a-2)u_1$, and $B_0(\ell, a) \ge \tilde\rho(\ell + 1)Y$, $B_0(\ell+1, a-1) \ge \tilde\rho(\ell+1)Y'$, since $\beta_1 \ge Y$ and $\beta_1' \ge Y'$. Hence $\mathrm{Key}(\lambda P) \ge 0$, that is, $\Phi(P) = P_1(M_{0,1}P_0 + M_{1,1}P_1) \ge 0$. This gives $S_{1,0} = M_{0,1} \ge 0$ and $S_{2,1} = M_{1,1} \ge 0$. The sum $S_{2,0} = M_{0,2} + M_{1,1}$ is treated in (v).

*(iii′) $M_{1,m} + M_{m,1} \ge 0$ for $m \ge 2$ and $a \ge 6$.* Put $c_m' = \binom{a-1}{m-1}$. Then

$$M_{1,m} + M_{m,1} = \big[\beta_1 E_m^2 - 4c_m' DBH_*\big] + \big[\beta_1' E_m^1 - DF(4Z_m - 2\nu_m)\big] + \beta_m' E_1^1 + \beta_m E_1^2,$$

and the last two terms are nonnegative.

- First bracket: $E_m^2 \ge \frac12 c_m' AX'$ and $A \ge B$, so it is nonnegative if $R := a(A-D)X' - 8DH_* \ge 0$.
- Second bracket: $Z_m \le \binom{a}{m}H_* \le \frac{a}{2}c_m' H_*$, $\nu_m \ge 0$, $F \le A$, $E_m^1 \ge \frac12 c_m' AX''$ and $\beta_1' \ge (a-2)(A-D)$. So it is nonnegative if $R_2 := (a-2)(A-D)X'' - 4aDH_* \ge 0$.

*(iv) $M_{0,s} \ge 0$ for $s \ge 3$.* Put $c' = \binom{a-1}{s-1}$ and $c'' = \binom{a-1}{s}$. Then

$$M_{0,s} = Y'E_s^1 + YE_s^2 + DE_s^3 = c'BK'' + FG_s,$$

where

$$K'' = Y'X'' + YX' - 4DH_*, \qquad G_s = c'(Y'X'' + YX') - c''(d_\ell Y' + DY) + D(2\nu_s - 4Z_s).$$

Note $u_1 = d_{\ell+1} - d_\ell > 0$. If $K'' \ge 0$, then $u_1 B \ge u_0 F$ gives $u_1 M_{0,s} \ge F(c'u_0 K'' + u_1 G_s)$, and

$$c'u_0K'' + u_1G_s = D\,Q_s + c''u_1^2 Y', \qquad Q_s = c'(Y'X'' + YX') - 4c'u_0H_* + u_1(2\nu_s - 4Z_s) - c''u_1(Y + Y').$$

So $M_{0,s} \ge 0$ once $K'' \ge 0$ and $Q_s \ge 0$.

*(v) $S_{2,0} \ge 0$ for $a \ge 5$.* With $c'' = \binom{a-1}{2}$, the definitions give the identity

$$S_{2,0} = \beta_1' E_1^1 + \big[\beta_1 E_1^2 - c''F(d_\ell Y' + DY)\big] + B\,K_a + F\,W,$$

where $K_a = (a-1)K'' - 4DH_*$ and $W = (a-1)(Y'X'' + YX') + D(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

- Since $\beta_1 = a(X' + F)$, $E_1^2 = AX' - (a-2)DF$, $d_\ell \le D$ and $FD \le u_1 A$, the bracket is at least $A\Psi$, where $\Psi = a(A-D)J - c''u_1(Y + Y')$. Recall $J = X' - (a-2)u_1$ and $X' + F = A - D$.
- If $K_a \ge 0$, then $u_1(BK_a + FW) \ge F(u_0K_a + u_1W) = FD\widehat{Q}$, where $\widehat{Q} = (a-1)(Y'X'' + YX') - 4au_0H_* + u_1(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

Hence $u_1 S_{2,0} \geq u_1\beta_1' E_1^1 + u_1 A\Psi + FD\widehat{Q} \geq 0$ once $K_a$, $\Psi$ and $\widehat{Q}$ are nonnegative.

*(vi) Weight sums.* Steps (ii)–(v) reduce (8.8) to the nonnegativity of the quadratic forms $L$, $R$, $R_2$, $K''$, $Q_s$ $(3 \leq s \leq a)$, $K_a$, $\Psi$ and $\widehat{Q}$ in the sequence $u$. In each of them:

- the only negative coefficients come from multiples of $u_0 H_*$, $u_1 H_*$, $u_1 Z_s$ and $u_1(Y + Y')$, so they sit at index pairs $(0, j)$ and $(1, j)$;
- the positive parts $X$, $X'$, $X''$, $J$, $Y$, $Y'$ and $A - D$ contain no $u_0$.

So Lemma 8.15 applies, and it suffices that every weight sum be nonnegative. At $u_n = r^n$, with $p = 1 + r$, one has $d_{\ell+n} \mapsto p^n$. Hence

$$X \mapsto p^a - 1, \quad X' \mapsto p^{a-1} - p, \quad X'' \mapsto p^{a-1} - 1, \quad Y \mapsto p^a - 1 - ar, \quad H_* \mapsto arp^{a-1} - 2p^a + 2 + ar,$$

and $r\nu_s \mapsto sG^{(s)}$, $Z_s \mapsto \binom{a}{s} H_* - G^{(s)}$, where

$$G^{(s)} = \tbinom{a}{s}\big((s-2)r^s p^{a-s} + (a-s)r^{s+1}p^{a-s-1}\big) + [s = 1]ar.$$

Extracting $[r^W]$ from these products of powers of $p$ and $r$ gives each weight sum as a finite combination of binomial coefficients; for the forms other than $L$ and for $W \geq 4$ it has the form $N_W = \pi\binom{2a-1}{W} + \binom{a}{W-1}\varrho(a, W)$, with $\pi > 0$ and $\varrho$ rational, and $W = 3$ is treated directly (for $\Psi$ it contains the extra term $\binom{a-1}{2}(a-1)$). For $Q_s$ we bound $N_W(Q_s)/c'$ below, for $W \geq 4$, by its value with the nonnegative $G^{(s)}$-terms omitted and $s = 3$ in the two remaining $s$-dependent negative terms, whose magnitudes decrease as $s$ grows. For $W = 3$,

$$N_3(Q_s) = c'\tfrac{(a-1)(a-2)(2a-3)}{6} + 6\tbinom{a}{3}[s = 3] - c''(a-1)^2,$$

which is $(a-1)(a-2)\big(\frac{a(a-1)}{12} + a\big)$ for $s = 3$ and at least $c'(a-1)\frac{a^2+a}{12}$ for $s \geq 4$. The certificates are as follows.

- For $W \geq a + 2$, all negative terms vanish.
- For $6 \leq W \leq a + 1$, bound $\binom{2a-1}{W}$ below by the Vandermonde terms $\binom{a}{k}\binom{a-1}{W-k}$ with $k \in \{W, W-1, W-2, W-3, 0, 1, 2\}$. These are distinct. Substituting $a = W - 1 + x$ and $W = 6 + y$, the cleared numerator of the resulting lower bound for $N_W/\binom{a}{W-1}$ is a polynomial in $(x, y)$ with nonnegative coefficients.
- For $W \in \{3, 4, 5\}$, $N_W$ is a polynomial in $a$ whose coefficients are nonnegative after $a \mapsto a + a_0$. Here $a_0 = 5$ for $K_a$, $a_0 = 6$ for $R_2$, and $a_0 = 3$ otherwise.
- For $W \leq 2$, the weight sums are 0 or positive.

The coefficient signs are verified in exact arithmetic by `runs/varratio/key_weight_cert.py`, which fails unless every certificate passes; `runs/varratio/key_weight_closed.py` checks the closed forms against the coefficient expansions for $5 \leq a \leq 30$. For $L$, the bound $\binom{2a}{W} \geq \binom{2a-1}{W}$ gives $N_W(L) \geq \frac{2(a-1)}{a} N_W(R) \geq 0$ for $W \geq 3$; also $N_2(L) = 2a(a-1)(a-2) > 0$, and $N_W(L) = 0$ for $W \leq 1$.

*(vii) Small windows.* Steps (iii′) and (v) need $a \geq 6$ and $a \geq 5$. For $3 \leq a \leq 5$, every central sum $S_{\sigma,i}$, a cubic polynomial in $u$, and every $l_k$ is certified nonnegative on log-concave $u$. The certificates are exact rational combinations of $u_k(u_{i+1}u_{j-1} - u_iu_j)$, respectively $u_{i+1}u_{j-1} - u_iu_j$ $(i+1 \leq j-1)$, and monomials (`runs/varratio/key_phi_ustep.py`, `runs/varratio/key_lin_cert.py`; these cover all $a \leq 10$). This proves (8.8) and the theorem. ∎

The diagnostics check in exact arithmetic the following.

- The closed forms of the central sums: `runs/varratio/key_S_high.py` and `runs/varratio/key_S_low.py`, symbolically for $a \leq 8$.
- The identities and reductions of steps (iii′)–(v) on random log-concave $u$: `runs/varratio/key_L1_decomp.py`, `runs/varratio/key_L1_quad.py`, `runs/varratio/key_S20_proof.py` and `runs/varratio/key_L2_chain.py`.
- The Abel tails of every form: `runs/varratio/key_L1_quad.py` ($a \leq 70$), `runs/varratio/key_S20_quad.py` and `runs/varratio/key_S20_proof.py` ($a \leq 40$).
- The $\widehat{w}$ detour: `runs/varratio/what_pointwise.py`.
- (P) on random members, including steep exponential ratios: `runs/varratio/steep_exp_test.py` and `runs/varratio/g_farfield.py`.

For cubic ratios, log-concavity of the Newton coefficients can be replaced by a single inequality.

**Theorem 8.17** *(Cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t, \tau, \sigma \geq 0$ and $r\sigma \leq t\tau$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If $b$ is nondecreasing, then the system has nonnegative linearization.*

If $(r, t, \tau, \sigma)$ is log-concave without internal zeros, then $r\sigma \leq t\tau$: for $t\tau > 0$ this follows from $t^2\tau^2 \geq r\tau \cdot t\sigma$, and otherwise $\sigma = 0$. Theorem 8.17 also admits cubic ratios with $t^2 < r\tau$ or $\tau^2 < t\sigma$, which Theorem 8.16 does not cover. The failing cubic examples of Section 10 violate $r\sigma \leq t\tau$. The proof is computer-assisted: the inequalities (8.11), (8.12) and (8.10) and the inequalities $\Phi_2^{\mathrm{loc}}, \Phi_3^{\mathrm{loc}} \geq 0$ below are reduced to finitely many families of explicit polynomial inequalities, each certified in exact rational arithmetic by the nonnegativity of all coefficients.

*Proof. Setup.* As in the proof of Theorem 8.11, $R_i = d_i/f_1(i)$ is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and $\rho$ is nondecreasing with $0 < \rho(i) \leq 1$ for $i \leq N$; here $\Delta\rho(x) = t + \tau x + \sigma\binom{x}{2}$ and $\Delta^2\rho(x) = \tau + \sigma x$. For $\ell + n \leq N$ the Leibniz rule gives, all at $\ell$,

$$\varphi_n = \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \tbinom{n}{2}\Delta^2\rho(\ell+n-2)f_{n-2} + \tbinom{n}{3}\sigma f_{n-3}.$$

Since $\binom{\ell+n-1}{2} = \binom{\ell}{2} + \ell(n-1) + \binom{n-1}{2}$ and $n\binom{n-1}{2} = \binom{n}{2}(n-2) = 3\binom{n}{3}$, this gives $\psi_n = h_n - v_n - e_n$ with $h_n = (1-\rho(\ell+n))f_n$,

$$v_n = n\Gamma_n(\ell)f_{n-1}, \quad \Gamma_n(\ell) = t + \tau(\ell+n-1) + \sigma\Big(\binom{\ell}{2} + \ell(n-1)\Big), \qquad e_n = \binom{n}{2}(\tau+\sigma\ell)f_{n-2} + \sigma\binom{n}{3}\big(3f_{n-1} + 3f_{n-$$

all nonnegative. Write $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$; then $(u_k)$ is log-concave, and $u_0 \le \ell u_1$ by convexity.

*Moves.* Split the moves as in the proof of Theorem 8.11, with $V_j$ formed from $v_n$ and with a third family $E_j = W_{2j+1,j}$ formed from $e_n$ in place of $W_j$.

*Detours of $H$, $V$ and $C'$.* The detours of $H$ and $C'$ hold verbatim, with constants 10 and 6. For $V$,

$$(n+1)\Gamma_n(\ell+1) - n\Gamma_{n+2}(\ell) = t + \tau\ell + \sigma\Big(\binom{\ell}{2} + n^2 - 1\Big) \ge 0,$$

so Lemma 8.3 with indices $(2, n-1)$ gives $(n+1)(n+2)\varphi_2 v_n' \ge n^2 c_{\ell+1} v_{n+2}$ for $n \ge 3$, and $B_0 V_{j-1}' \ge \frac{9}{2} c_{\ell+1} V_j$ for $j \ge 2$ as before. (With the full $\sigma$-part $n\sigma\binom{\ell+n-1}{2}f_{n-1}$ in $v_n$, the corresponding difference is negative for $\ell < n+1$; without any $\sigma$-part in $v_n$, the reduction of the key inequality below fails for large $\ell$.)

*Detour of $E$.* We show, for $n \ge 3$ and $\ell + n + 2 \le N$,

$$(n+1)(n+2)\Big(\varphi_2(\ell) + \frac{n}{3}\varphi_3(\ell)\Big)e_n(\ell+1) \ge 3(n-1)\, c_{\ell+1} e_{n+2}(\ell). \tag{8.10}$$

Granting this, $B_0(\ell, a) \ge \binom{a}{2}\varphi_2(\ell) + \binom{a}{3}\varphi_3(\ell)$, $\binom{a}{3}\binom{a-2}{n} = \frac{a-2}{3}\binom{a}{2}\binom{a-2}{n}$ with $a - 2 \ge n$ whenever $\binom{a}{n+2} \ne 0$, and $\binom{a}{2}\binom{a-2}{n} = \binom{a}{n+2}\binom{n+2}{2}$ give, for $j \ge 2$,

$$B_0(\ell, a)E_{j-1}(\ell+1, a-2) \ge \tfrac{3}{2} c_{\ell+1} \sum_{m \ge 2j+1} (m-3)\binom{m-j-2}{j-1}\binom{a}{m} e_m(\ell) \ge 3 c_{\ell+1} E_j(\ell, a),$$

because $(m-3)\binom{m-j-2}{j-1} - 2\binom{m-j-1}{j} = \binom{m-j-2}{j-1}\frac{(j-2)(m-1)}{j} \ge 0$.

*Detour of $B$.* We show, for $n \ge 2$, $\ell + n + 2 \le N$ and $(\ell, n) \ne (0, 2)$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \ge 8(n-1)\, c_{\ell+1}\varphi_{n+2}(\ell), \tag{8.11}$$

and, if $N \ge 4$,

$$\big(12\varphi_2(0) + 8\varphi_3(0)\big)\varphi_2(1) \ge 8 c_1 \varphi_4(0). \tag{8.12}$$

Granting these, the argument after (8.4) gives $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \ge 4 c_{\ell+1} B_j(\ell, a)$ for $j \ge 1$. The only exception is the term $m = 4$ for $\ell = 0$, $j = 1$. There $B_0(0, a) \ge \binom{a}{2}\varphi_2(0) + \binom{a}{3}\varphi_3(0)$, $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$ and $\binom{a}{3}\binom{a-2}{2} = 2(a-2)\binom{a}{4} \ge 4\binom{a}{4}$ reduce the term to (8.12). Inequality (8.11) fails at $(\ell, n) = (0, 2)$ in general.

*Proof of (8.11), (8.12) and (8.10).*

- *Quadratic forms.* In each, the difference of the two sides is a quadratic form $\sum \kappa_{\beta\gamma}\rho_\beta\rho_\gamma$ in $(\rho_0, \rho_1, \rho_2, \rho_3) = (r, t, \tau, \sigma)$. Its coefficients are bilinear forms in the $u_k$, with coefficients polynomial in $\ell$ and $n$. By $r\sigma \le t\tau$, the form is nonnegative if $\kappa_{t\tau} + \min(\kappa_{r\sigma}, 0) \ge 0$ and all coefficients other than $\kappa_{r\sigma}$ are nonnegative.

- *The quadratic part.* For (8.11) the coefficients $\kappa_{rr}, \kappa_{r\tau}, \kappa_{\tau\tau}$ need not be checked: the form with $t = \sigma = 0$ is (8.4) for the ratio $r + \tau\binom{x}{2}$. The left side of (8.12) exceeds that of (8.11) at $(0, 2)$ by $8\varphi_3(0)\varphi_2(1)$, a form with nonnegative coefficients. So only $\kappa_{t\sigma}$ changes there, and it is checked again with $u_0 = 0$.

- *Reduction.* Each remaining coefficient inequality is reduced as in the proof of (8.4). Take $u_1 = 1$, $q = u_2$ and $z = u_0 \in [0, \min(\ell, 1/q)]$, and for (8.10) also $x = u_3/u_2 \le q$, since $\varphi_3$ involves $u_3$; put $x = q$ for (8.11). The second factor involves $u_{n-3}, \dots, u_{n+2}$, respectively $u_{n-3}, \dots, u_{n+1}$ for (8.10). The two smallest indices carry nonnegative coefficients and the largest a nonpositive one.

  - Normalize $u_{n-1} = 1$. By log-concavity, $u_{n-2} \ge 1/x$ and $u_{n-3} \ge 1/x^2$, and $u_n = c \le x$, $u_{n+1} = cy$ and $u_{n+2} \le cy^2$ with $y \le c$ (for (8.10), $u_{n+1} \le c^2$). Concavity in $y$ and in $c$ reduces each coefficient inequality to its endpoints.
  - Where the two blocks of indices overlap ($n \le 3$ for (8.11), $n \le 4$ for (8.10)), the reduction is done exactly in the free indices. Where $u_0$ occurs in both factors, a coefficient need not be affine in $z$; then $z$ is written as $\zeta \min(\ell, 1/q)$ with $\zeta \in [0, 1]$.

- *Certificates.* The resulting polynomial inequalities are certified in three regimes: $z = 0$; $z = \ell$ with $q = 1/(\ell + w)$; and $z = 1/q$ with $\ell = 1/q + w$. In each regime all coefficients are nonnegative, in Bernstein form on $[0, 1]$ in $c/q$, $x/q$ and $\zeta$, with $n$ symbolic above the overlap.

  - The scripts are `runs/cubic/bdetour_cert.py`, `runs/cubic/corner_cert.py` and `runs/cubic/edetour_cert.py`.
  - At $(\ell, n) = (0, 2)$ the endpoint value of $\kappa_{t\sigma}$ in (8.11) is $-8(q+1)(q^2-q-5)$. With the extra term of (8.12) it becomes $8(6q^4s^2 - 4q^3s^2 + 29q^3s + 11q^2s + 33q^2 + 36q + 11)$ with $s = x/q \in [0, 1]$, whose Bernstein coefficients in $s$ have only nonnegative coefficients.

*Key inequality.* We show, for $(\ell, a) \in \Omega$ with $a \ge 3$,

$$A_0^{\psi}(\ell, a)B_0(\ell+1, a-1) + B_0(\ell, a)A_0^{\psi}(\ell+1, a-2) \ge c_{\ell+1}\big(4H_1 - 2V_1 - E_1\big)(\ell, a). \quad (8.13)$$

Use the notation of the proof of Theorem 8.9 ($u_n$, $y_n$, $D$, $X$, $Y$, $X'$, $Y'$, $J$, $H_*$, $K$), and put $\alpha = 1 - \rho(\ell + a)$, $s = \Delta\rho(\ell + a - 1)$ and $z = u_0/u_1$.

- *The parameter $\alpha$.* We have $\alpha \ge sR_{\ell+a-1} \ge sz$. Every occurrence of 1 in the two sides lies in $\alpha$, through $1 - \rho(\ell + n) = \alpha + \rho(\ell + a) - \rho(\ell + n)$. The coefficient of $\alpha$ in the difference of the two sides is $XB_0(\ell+1, a-1) + B_0(\ell, a)X' - 4c_{\ell+1}H_* \ge \rho(\ell+1)K \ge 0$. So it suffices to take $\alpha = sz$.

- *A quadratic form in $\rho$.* Then both sides are quadratic forms in $(\rho_\beta)$, where $\rho = \sum_\beta \rho_\beta e_\beta$ and $e_\beta(x) = \binom{x}{\beta}$. Write $[e_\beta]$ for the contribution of a single basis ratio, and put $s_\beta = \Delta e_\beta(\ell + a - 1)$.

– By convexity of $e_\beta$, $A_0^\psi[e_\beta](\ell,a) \ge z s_\beta(X - a u_1) \ge 0$ and $A_0^\psi[e_\beta](\ell+1,a-2) \ge s_\beta D(X' - (a-2)u_1) \ge 0$.

– Moreover $B_0[e_\gamma](\ell,a) \ge e_\gamma(\ell+1)Y$, $B_0[e_\gamma](\ell+1,a-1) \ge e_\gamma(\ell+1)Y'$ and $c_{\ell+1}[e_\gamma] = e_\gamma(\ell+1)D$.

Hence the coefficient of $\rho_\beta\rho_\gamma$ in the difference is at least $e_\gamma(\ell+1)\Phi_\beta + e_\beta(\ell+1)\Phi_\gamma$ for $\beta \ne \gamma$, and at least $e_\beta(\ell+1)\Phi_\beta$ for $\beta = \gamma$. Here

$$\Phi_\beta = A_0^\psi[e_\beta](\ell,a)\, Y' + Y\, A_0^\psi[e_\beta](\ell+1,a-2) - D\,(4H_1 - 2V_1 - E_1)[e_\beta],$$

and $\Phi_0 = 0$. So (8.13) needs no condition linking $r$ and $\sigma$, and it remains to show $\Phi_1, \Phi_2, \Phi_3 \ge 0$.

- *Local decomposition.* Expand in the local basis: $\binom{\ell+k}{2} = \binom{\ell}{2} + \ell k + \binom{k}{2}$ and $\binom{\ell+k}{3} = \binom{\ell}{3} + \binom{\ell}{2}k + \ell\binom{k}{2} + \binom{k}{3}$. With the splittings of $\Gamma_n$ and $e_n$ above, this gives

$$\Phi_2 = \ell\Phi_1 + \Phi_2^{\mathrm{loc}}, \qquad \Phi_3 = \tbinom{\ell}{2}\Phi_1 + \ell\Phi_2^{\mathrm{loc}} + \Phi_3^{\mathrm{loc}}.$$

Here $\Phi_\beta^{\mathrm{loc}}$ is $\Phi_\beta$ computed with $\rho(\ell+k)$ replaced by $\binom{k}{\beta}$. Its moves $V_1$ are formed with $\Gamma_n$ replaced by $n-1$ for $\beta = 2$ and by 0 for $\beta = 3$, and its moves $E_1$ from $\binom{n}{2}f_{n-2}$, respectively $\binom{n}{3}(3f_{n-1} + 3f_{n-2} + f_{n-3})$.

- $\Phi_1$ *and* $\Phi_2^{\mathrm{loc}}$. With $u_1 = 1$, $\Phi_1$ is the function $F(z)$ of the proof of Theorem 8.9, which is nonnegative for $0 \le z \le 1/q$. $\Phi_2^{\mathrm{loc}}$ is the $\tau$-part of (8.6) with $s = a-1$, nonnegative for $0 \le z \le 1/q$ by the proof of Theorem 8.11.

- $\Phi_3^{\mathrm{loc}}$*: two endpoints.* Take $u_1 = 1$, $s = \binom{a-1}{2}$, $\kappa = \frac{s(2a-3)}{3}$ and $T_3 = \sum_{n\ge3}(n-2)\big(\binom{a}{3} - \binom{n}{3}\big)y_n$. Since $E_1 \ge 0$,

$$\Phi_3^{\mathrm{loc}} \ge z\Big(sY + 2\tbinom{a}{3}\Big)Y' + (z+1)Y(sJ+\kappa) - 4(z+1)\big(szH_* + T_3\big).$$

The right side is concave in $z$, so the endpoints $z = 0$ and $z = 1/q$ suffice.

- $\Phi_3^{\mathrm{loc}}$*: coefficient bounds.* In $YJ$ keep the products with indices $(2, n-1)$ for $n \ge 3$ and $(3, n-2)$ for $n \ge 4$, and use $u_iu_j \ge u_{i+j-1}$.

– At $z = 0$, the coefficient of $y_n$ ($n \ge 3$) is then at least

$$\kappa + s\Big(\frac{n(a-1)}{2} + [n \ge 4]\frac{(a-1)(a-2)n(n-1)}{6(a-n+1)}\Big) - 4(n-2)\Big(\tbinom{a}{3} - \tbinom{n}{3}\Big).$$

– At $z = 1/q$, multiply by $q/(1+q)$. In $sYY'/(1+q)$ keep $\binom{a}{2}q$ from $Y$, and use $q(u_j + u_{j+1}) \ge (1+q)u_{j+1}$ and $u_n \le qu_{n-1}$. The coefficient of $y_2$ is then $\kappa - \frac{2s(a-2)}{3} = \frac{s}{3}$, and that of $y_n$ ($n \ge 3$) is at least the expression above plus $\frac{sn(a-1)}{2} - s\big(\frac{10(n+1)}{3} - 8\big)\frac{a-n}{n+1}$.

With $a = n+1+m$ and $n = 3$ or $n = 4+p$, the numerators of these bounds are polynomials in $(m,p)$ with nonnegative coefficients (`runs/cubic/phi3loc_cert.py`); for $n = a$ the bounds are nonnegative directly. The same argument proves $\Phi_2^{\mathrm{loc}} \geq 0$ independently (`runs/cubic/phi2loc_cert.py`). At $z = 1/q$, as $q \to 0$, both the right side of the display and this bound tend to $\frac{1}{3}s\binom{a}{2}$.

*Resummation.* Proceed as in the proof of Theorem 8.11, with $E$ in place of $W$. The inequalities $P^2 \geq 4Q$, $H_+P \geq 10H_-$, $V_+P \geq \frac{9}{2}V_-$, $E_+P \geq 3E_-$, $C_+P \geq 6C_-$ and $A_0P + PA_0 + 2V_1 + E_1 \geq 4H_1$ give $GAG \geq \frac{1}{2}(GA_0 + A_0G) + \frac{1}{18}GV_1G + \frac{1}{12}GE_1G \geq 0$, and (P) follows. ∎

The condition $r\sigma \leq t\tau$ enters only through (8.11) and (8.10), each time through $\kappa_{t\tau} + \kappa_{r\sigma} \geq 0$. This coefficientwise argument cannot be pushed past $r\sigma \leq t\tau$. On geometric profiles at $(\ell, n) = (0,3)$ the quotient $(\kappa_{r\sigma} + \kappa_{t\tau})/|\kappa_{r\sigma}|$ of (8.11) equals $\frac{4q^4+30q^3+54q^2+35q+6}{q(4q^3+11q^2+9q+1)}$, which tends to 1 as $q \to \infty$. The diagnostics check in exact arithmetic the following.

- The splitting of $\psi_n$, every detour quotient and the key inequality (8.13) on random members with $N \leq 16$: `runs/cubic/final_ingredients.py`. The minimal quotients found were 9.1 for $B$, 11.1 for $V$, 4.8 for $E$, 23 for $H$ and 9.9 for $C'$, and 2.5 for the key ratio.
- (P) on random members at the boundary $r\sigma = t\tau$, in full tables for $4 \leq N \leq 14$ and in the coefficients $g(m, N; k)$ for $15 \leq N \leq 60$: `runs/cubic/c3_property_P.py`.
- The polarized coefficients behind (8.11), (8.10) and (8.13): `runs/cubic/kappa_sym.py`, `runs/cubic/edetour_polar.py` and `runs/cubic/key_polar.py`. The identities of the local decomposition: `runs/cubic/phi_loc.py`.

## 9. Limits of inequality criteria

Theorems 7.10 and 7.16 control every triple except the deep ones by inequalities, Theorem 8.1 controls all triples of constant-ratio arrays with log-concave Newton profiles, and Section 10 records shape conditions that fail on deep triples. This section shows that part of the failure is structural near the symmetric endpoint. For uniformly sampled polynomial deformation directions, and for the analytic and boundary-layer extensions specified below, admission at every sufficiently large symmetric Krawtchouk system imposes two first-order boundary equalities. This does not exclude arbitrary uniform inequality criteria, nor families of admissible arrays that are not of these forms. Write $K_{N,r}$ for the Krawtchouk array (2.4) with $q = 1$, and use the coordinates $b_1, \dots, b_{N-1}, c_1, \dots, c_N$ of a stochastic array with $b_0 = 0$, $a_0 = 1$, $a_N = 0$, $b_N = 1 - c_N$ and $a_n = 1 - b_n - c_n$ otherwise. An affine change of $x$, together with the positive normalization $R_n(x_0) = 1$ at the largest support point $x_0$, reduces every finite system to this form. A direction $v = (\delta b_1, \dots, \delta b_{N-1}, \delta c_1, \dots, \delta c_N)$ changes these coordinates; we put $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$.

**Lemma 9.1** *(Corner value).* *For $0 < r \leq 1$, the array $K_{N,r}$ has $g(N, N; N) = (1-r)^N$.*

*Proof.* In the model of Lemma 5.2, $U_N = y_1 \cdots y_N$ is the only normalized basis element containing $y_{\{1,\dots,N\}}$, and $U_N^2 = \prod_i \big((1-r)y_i + r\big)$ has coefficient $(1-r)^N$ at that monomial. ∎

Thus the corner coefficient vanishes at $r = 1$, where it is a structural zero, together with the other perimeter and parity zeros of the palindromic array. The next lemma computes its first-order change exactly.

**Lemma 9.2** *(Corner identity).* *Let $N \geq 2$ and $s_N = (-1)^{\lfloor N/2 \rfloor + 1} \binom{N-1}{\lfloor (N-1)/2 \rfloor}^{-1}$. Put $F = \delta b + 2\delta c$ if $N$ is even and $F = \delta b$ if $N$ is odd. Then*

$$\frac{d}{d\varepsilon}\, g(N,N;N)\Big|_{K_{N,1}+\varepsilon v,\ \varepsilon=0} = s_N \sum_{n=0}^{N} (-1)^n \binom{N}{n} F(n).$$

*Proof.* At $K_{N,1}$ the nodes are $\lambda_j = 1 - 2j/N$, with masses $w_j = 2^{-N}\binom{N}{j}$ and $\pi_n = \binom{N}{n}$ in Lemma 5.1. Moreover $\binom{N}{n} R_n(\lambda_j)$ is the binary Krawtchouk polynomial, so that

$$\sum_{j=0}^{N} w_j R_n(\lambda_j) s^j = 2^{-N}(1+s)^{N-n}(1-s)^n. \tag{9.1}$$

The vectors $u_j = (R_n(\lambda_j))_n$ and $v_j = (w_j \pi_n R_n(\lambda_j))_n$ are right and left eigenvectors of $J$ with $v_i^{\mathsf{T}} u_j = \mathbf{1}_{i=j}$. Since $R_N(\lambda_j) = (-1)^j$, the matrix $\Pi = R_N(J)$ reverses the indices. Let $J_\varepsilon = J + \varepsilon E$ be the perturbed recurrence matrix and $R_N^\varepsilon$ the perturbed polynomial. The matrix $R_N^\varepsilon(J_\varepsilon)$ commutes with $J_\varepsilon$ and has first row $e_N^{\mathsf{T}}$, by (5.1). Its derivative $X$ at $\varepsilon = 0$ therefore satisfies $[J, X] = [\Pi, E]$ and $e_0^{\mathsf{T}} X = 0$. In the eigenbasis, $\widehat{X}_{ij} = v_i^{\mathsf{T}} X u_j$ equals $((-1)^i - (-1)^j)\widehat{E}_{ij}/(\lambda_i - \lambda_j)$ for $i \neq j$, where $\widehat{E}_{ij} = v_i^{\mathsf{T}} E u_j$, and $\sum_i \widehat{X}_{ij} = 0$. By (5.1), and since $\pi_N = 1$, the derivative in the lemma is $h_N := e_N^{\mathsf{T}} X e_N = \sum_{i,j} (-1)^{i+j} \widehat{X}_{ij} w_j$. Eliminating $\widehat{X}_{jj}$ gives

$$h_N = -2 \sum_{i+j \text{ odd}} \widehat{X}_{ij} w_j = N \sum_{i \neq j} \widehat{E}_{ij} w_j \frac{(-1)^i - (-1)^j}{i-j}.$$

Perturbing row $n$ alone means $E = e_n \rho^{\mathsf{T}}$ and $\widehat{E}_{ij} = w_i \pi_n R_n(\lambda_i)\, \rho^{\mathsf{T}} u_j$. Write i for the imaginary unit and insert

$$\frac{(-1)^i - (-1)^j}{i-j} = -\pi \int_0^1 \sin \pi\big(\theta i + (1-\theta) j\big)\, d\theta,$$

which also vanishes for $i = j$. Then

$$h_N = -\pi N \pi_n \int_0^1 \operatorname{Im}\big(A(\theta) B(\theta)\big)\, d\theta,$$
$$A = \sum_i w_i R_n(\lambda_i) e^{\mathrm{i}\pi\theta i}, \qquad B = \sum_j w_j (\rho^{\mathsf{T}} u_j) e^{\mathrm{i}\pi(1-\theta) j},$$

and both sums are given by (9.1). Put $\phi = \pi\theta/2$.

- Coordinate $b_n$, $1 \le n < N$, so $\rho = e_n - e_{n+1}$. Then

$$AB \;=\; -(-1)^n\,\mathrm{i}^{N-1}\cos^N\!\phi\,\sin^{N-1}\!\phi\,e^{-\mathrm{i}\phi},$$

  and $h_N = s_N(-1)^n\binom{N}{n}$. The integral reduces to $\int_0^{\pi/2}\cos^{N+1}\!\phi\,\sin^{N-1}\!\phi\,d\phi$ ($N$ even) or $\int_0^{\pi/2}\cos^N\!\phi\,\sin^N\!\phi\,d\phi$ ($N$ odd), and each equals $|s_N|/(2N)$.

- Coordinate $c_n$, $1 \le n < N$, so $\rho = e_{n-1} - e_{n+1}$. Now $B$ acquires the factor $e^{\mathrm{i}\phi}/\cos\phi$, and the integrand carries $\sin((N-1)\pi/2)$. The contribution vanishes for odd $N$. For even $N$ it is twice that of $b_n$, since the two beta integrals have ratio 2.

- Coordinate $c_N$, so $\rho = e_{N-1} - e_N$ in row $N$. The same computation with $n = N$ gives $s_N$.

Summing these contributions with the conventions $\delta b_0 = \delta c_0 = 0$ and $\delta b_N = -\delta c_N$ gives the stated alternating sum. ∎

**Definition 9.3.** A *uniform rule* is a pair of real functions $\beta, \gamma$ on $[0,1]$. It defines, for each $N$, the direction $v_N$ with $\delta b_n = \beta(n/N)$ for $1 \le n < N$ and $\delta c_n = \gamma(n/N)$ for $1 \le n \le N$. A set of arrays *admits* the rule at the symmetric endpoint if there is $N_0$ such that for every $N \ge N_0$ there is $\varepsilon_N > 0$ with $K_{N,1} + \varepsilon v_N$ in the set for all $0 < \varepsilon < \varepsilon_N$. Put

$$L_{\mathrm{even}} \;=\; \beta(0) + 2\gamma(0) + \beta(1) + \gamma(1), \qquad L_{\mathrm{odd}} \;=\; \beta(0) - \beta(1) - \gamma(1).$$

**Theorem 9.4** *(Forced equalities for uniform criteria).* *Let $\beta, \gamma$ be polynomials. If a set of arrays with property (P) admits the rule $(\beta, \gamma)$ at the symmetric endpoint, then $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$. If $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$, the first-order change of $g(N, N; N)$ along $v_N$ vanishes for every $N$ above the degrees of $\beta$ and $\gamma$.*

*Proof.* For a polynomial $p$ of degree less than $N$, $\sum_{n=0}^{N}(-1)^n\binom{N}{n}p(n/N) = 0$. Only boundary terms survive in Lemma 9.2: the derivative equals $-s_N L_{\mathrm{even}}$ for even $N$ and $-s_N L_{\mathrm{odd}}$ for odd $N$. The sign of $s_N$ changes from $N$ to $N+2$. If, for instance, $L_{\mathrm{even}} \neq 0$, then the derivative is negative for every large even $N$ in one residue class modulo 4. For such $N$, $g(N, N; N) < 0$ at $K_{N,1} + \varepsilon v_N$ for all small $\varepsilon > 0$, since the coefficient vanishes at $\varepsilon = 0$. ∎

The known positive deformations satisfy both equalities: the Krawtchouk direction $\beta(x) = x$, $\gamma(x) = -x$, palindromic deformations with $\beta = 0$, $\gamma$ odd about $\frac{1}{2}$ and $\gamma(0) = \gamma(1) = 0$ (Corollary 7.17), and the tangent direction $\beta = 0$, $\gamma(x) = x(x-1)$ of the geometric family at $q = 1$ (Theorem 2.1). The two functionals are independent already on constant rules, where $L_{\mathrm{even}} = 2\beta + 3\gamma$ and $L_{\mathrm{odd}} = -\gamma$. Consequently, for every $d \ge 0$, no set of arrays with property (P) admits all rules of a nonempty open subset of the polynomial rules of degree at most $d$. In this sense, a criterion that admits the symmetric Krawtchouk systems through polynomial uniform rules must impose the two equalities on them. The geometric criterion and Corollary 7.17 are of this mixed type.

The polynomial hypothesis can be replaced by analyticity on a fixed region. Let $U(x) = \int_0^1 \log|x - t|\,dt$ and $\Omega = \{x \in \mathbb{C} : U(x) \le -1\}$, the region of convergence of equispaced

interpolation. It contains $[0,1]$, has nonanalytic boundary points (with vertical tangents) at 0 and 1, and meets the line $\operatorname{Re} x = \frac{1}{2}$ at $\operatorname{Im} x = \pm 0.26276\ldots$.

**Lemma 9.5** *(Nörlund–Rice bound).* *If $f$ is analytic on an open set containing $\Omega$, then $\sum_{n=0}^{N}(-1)^n\binom{N}{n}f(n/N) = O(\sqrt{N}\,e^{-\eta N})$ for some $\eta > 0$.*

*Proof.* The residues of $1/\prod_{k=0}^{N}(x-k/N)$ give

$$\sum_{n=0}^{N}(-1)^n\binom{N}{n}f(n/N) = (-1)^N\frac{N!}{N^N}\,\frac{1}{2\pi i}\oint_\Gamma \frac{f(x)\,dx}{\prod_{k=0}^{N}(x-k/N)}$$

for any cycle $\Gamma$ in the domain of $f$ that winds once around $[0,1]$. Since $U$ is continuous and tends to $\infty$, the compact sets $\{U \le -1+t\}$ shrink to $\Omega$ as $t \downarrow 0$. Choose $\eta > 0$ with $\{U \le -1+3\eta\}$ inside the domain of $f$, and a bounded polygonal open set $V$ with $\{U \le -1+\eta\} \subset V$ and $\overline{V} \subset \{U < -1+3\eta\}$. The oriented boundary $\Gamma$ of $V$, possibly with several components, is a cycle outside $\{U \le -1+\eta\}$ that winds once around every node $k/N$ and bounds a region inside the domain of $f$. With $d = \operatorname{dist}(\Gamma,[0,1]) > 0$, the derivative of $t \mapsto \log|x-t|$ on $[0,1]$ is bounded by $1/d$ for $x \in \Gamma$, so a Riemann-sum estimate gives $\sum_{k=0}^{N}\log|x-k/N| \ge NU(x) - C_\Gamma \ge N(-1+\eta) - C_\Gamma$ with $C_\Gamma$ independent of $N$. Stirling's bound $N!/N^N \le e\sqrt{N}e^{-N}$ completes the proof. ∎

**Corollary 9.6.** *Let $\beta,\gamma$ be analytic on an open set containing $\Omega$. If a set of arrays with property (P) admits the rule $(\beta,\gamma)$ at the symmetric endpoint, then $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$. When these equalities hold, the first-order change of $g(N,N;N)$ along $v_N$ is $O(|s_N|\sqrt{N}\,e^{-\eta N})$; it need not vanish.*

*Proof.* By Lemma 9.5, the alternating sum of $\beta + 2\gamma$ or $\beta$ is $O(\sqrt{N}e^{-\eta N})$. The derivative in Lemma 9.2 is therefore $s_N(-L_{\mathrm{even}} + o(1))$ for even $N$ and $s_N(-L_{\mathrm{odd}} + o(1))$ for odd $N$, and the argument of Theorem 9.4 gives the equalities. The rule $\beta(x) = e^x - 1 - (e-1)x$, $\gamma = 0$ has $L_{\mathrm{even}} = L_{\mathrm{odd}} = 0$ and derivative $s_N(1-e^{1/N})^N \neq 0$. ∎

Deformations may also depend on $N$ in other ways. For each fixed $N$ the symmetric endpoint is a limit of systems whose product coefficients are strictly positive on the whole triangle; the corner nevertheless constrains families that are regular in $N$. The first statement rests on the following lemma, valid for every finite system. Write $\mu$ for the orthogonality measure and $x_0$ for its largest support point, so that $R_n(x_0) = 1$.

**Lemma 9.7** *(Lowering the top mass).* *Let $0 \le \varepsilon < \mu(\{x_0\})$, let $\mu_\varepsilon = \mu - \varepsilon\delta_{x_0}$, and let $R_n^\varepsilon$ be its polynomials, normalized at $x_0$. Put $\pi_n^\varepsilon = \bigl(\int (R_n^\varepsilon)^2\,d\mu_\varepsilon\bigr)^{-1}$ and $\pi_n = \pi_n^0$; these inverse norms are not rescaled to impose $\pi_0^\varepsilon = 1$. The recurrence coefficients in (1) are taken in the fixed coordinate $x$, without an $\varepsilon$-dependent affine normalization.*

1. *The recurrence coefficients change at rate $(\dot a_n, \dot b_n, \dot c_n) = \pi_n(a_n,\, c_n - a_n,\, -c_n)$.*

2. *If* $\int R_x R_y R_z \, d\mu = 0$, *then*

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \int R_x^\varepsilon R_y^\varepsilon R_z^\varepsilon \, d\mu_\varepsilon = 2 - T_x(y,z) - T_y(x,z) - T_z(x,y),$$
$$T_x(y,z) = \sum_{k \geq x} g(y,z;k).$$

*Proof.* Write $R_n^\varepsilon = R_n + \varepsilon Q_n + O(\varepsilon^2)$, with $\deg Q_n \leq n$ and $Q_n(x_0) = 0$. First-order orthogonality gives $\int Q_n p \, d\mu = p(x_0)$ for $\deg p < n$, hence $Q_n = \sum_{k<n} \pi_k (R_k - R_n)$, and $\pi_n^\varepsilon = \pi_n\big(1 + \varepsilon(2S_n + \pi_n)\big) + O(\varepsilon^2)$ with $S_n = \sum_{k<n} \pi_k$. For (1), differentiate $c_n = \pi_{n-1} \int x R_n R_{n-1} \, d\mu$ and $b_n = \pi_n \int x R_n^2 \, d\mu$, using the recurrence, the row sums $a_m + b_m + c_m = x_0$ and $\pi_{n-1} a_{n-1} = \pi_n c_n$; the terms with $S_n$ cancel, and $\dot{a}_n$ follows from the row sums. For (2), the derivative equals $-R_x R_y R_z(x_0) + \int (Q_x R_y R_z + R_x Q_y R_z + R_x R_y Q_z) \, d\mu$, and $\int Q_x R_y R_z \, d\mu = \sum_{k<x} g(y,z;k) = 1 - T_x(y,z)$. ∎

**Proposition 9.8** *(Interior points near the symmetric endpoint).* *For every* $N \geq 2$ *there is* $\Lambda_N < \infty$ *such that, for* $\Lambda > \Lambda_N$ *and all sufficiently small* $\varepsilon > 0$, *the measure of* $K_{N,1-\Lambda\varepsilon}$ *with its top mass lowered by* $\varepsilon$ *has* $g(m,n;k) > 0$ *for all* $|m-n| \leq k \leq \min(m+n, N)$. *Consequently, after the affine normalization of Section 2, a relatively open set of arrays near it has property (P).*

*Proof.* By the model of Lemma 5.2, $K_{N,r}$ has

$$g(m,n;k) = \sum_j H(m,n;j) \binom{j}{s} r^{j-s} (1-r)^s,$$
$$H(m,n;j) = \frac{\binom{m}{j}\binom{N-m}{n-j}}{\binom{N}{n}}, \qquad s = k - m - n + 2j,$$

where the sum is over $\max(0, m+n-N) \leq j \leq \min(m,n)$ with $0 \leq s \leq j$. At $r = 1$ only $s = 0$ survives, so a triangle coefficient of $K_{N,1}$ vanishes exactly when $m+n+k$ is odd or $m+n+k > 2N$. Both deformations are smooth in $(r, \varepsilon)$, so only these zeros need first-order positivity.

- If $m+n+k > 2N$, all three tails in Lemma 9.7(2) vanish, since $g(y,z;k) = 0$ for $k > 2N - y - z$. The top-mass rate is therefore $2\pi_k > 0$, and the rate of decreasing $r$ is nonnegative because $g \geq 0$ for $r \leq 1$.
- If $m+n+k$ is odd and at most $2N+1$, the rate of decreasing $r$ is the $s = 1$ term $H(m,n;j)\, j$ with $j = (1+m+n-k)/2$. It is positive, because $j \geq 1$ and $\max(0, m+n-N) \leq j \leq \min(m,n)$.

These cases cover all zeros. Let $\mathcal{Z}_N$ be the set of triangle triples whose coefficient vanishes at $K_{N,1}$. For $z \in \mathcal{Z}_N$, write $A_z$ for the top-mass rate and $B_z$ for the rate of decreasing $r$, and define

$$\Lambda_N = \max\Big(\{0\} \cup \Big\{-\frac{A_z}{B_z} : z \in \mathcal{Z}_N,\ A_z < 0\Big\}\Big).$$

Every denominator appearing here is positive by the preceding cases. Hence $\Lambda > \Lambda_N$ makes every reference-zero derivative strictly positive, and finiteness of $\mathcal{Z}_N$ gives a common threshold for $\varepsilon$.

Here the measure of $K_{N,r}$ is its probability orthogonality measure. Choose $\varepsilon$ so small that $r = 1 - \Lambda\varepsilon > 0$ and $\varepsilon < (r/(1+r))^N$, so that the lowered top mass remains positive. The affine coordinate $y = (1-\varepsilon)x + \varepsilon = R_1^{\varepsilon}(x)$ restores the basic recurrence normalization and leaves all product coefficients unchanged. Strict positivity on the triangle gives $b_n = g(1, n; n) > 0$ for $1 \le n \le N$. Lemma 7.5, with $\Sigma$ the whole triangle, therefore gives a relatively open neighborhood with property (P) in the affine space defined by the basic boundary and row-sum equalities. ∎

Exact rational computation of the maximum above gives $\Lambda_N = 4, 10, 18, 28, 52, 90, 165$ for $N = 4, \dots, 10$. These finite-size values do not establish an asymptotic growth law or a geometric thickness estimate. At $K_{N,1}$, in the fixed spectral coordinate, Lemma 9.7(1) gives

$$\dot{c}_n = -\frac{n}{N}\binom{N}{n}, \qquad \dot{b}_n = \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}.$$

After the affine normalization restoring $R_1 = x$, the top-mass direction is

$$\delta a_n = \frac{N-n}{N}\Big(\binom{N}{n} - 1\Big), \qquad \delta b_n = 1 + \Big(\frac{2n}{N} - 1\Big)\binom{N}{n}, \qquad \delta c_n = -\frac{n}{N}\Big(1 + \binom{N}{n}\Big).$$

Its binomial dependence on $N$ is not a uniform rule of Definition 9.3, consistently with Theorem 9.4.

**Proposition 9.9** *(Families depending on $N$).* *Let $v_N$ be directions such that a set with property (P) contains $K_{N,1} + \varepsilon v_N$ for all large $N$ and small $\varepsilon > 0$, and let $A_N$ be the alternating sum in Lemma 9.2 for $v_N$. Then $s_N A_N \ge 0$ for all large $N$.*

1. *Along either parity class, $A_N$ is not eventually strictly positive and not eventually strictly negative. If $A_N$ has an eventual weak sign along a parity class, it vanishes for all large $N$ of the residue class modulo 4 on which that sign conflicts with the sign of $s_N$.*
2. *Suppose $\delta b_n = \beta(n/N) + p_n + q_{N-n}$ and $\delta c_n = \gamma(n/N) + p'_n + q'_{N-n}$, with $\beta, \gamma$ as in Corollary 9.6 and finitely supported boundary layers $p, q, p', q'$ independent of $N$. Then $q'_0 = L_{\mathrm{even}} = L_{\mathrm{odd}}$, $p_k = q_k$ and $p'_k + q'_k = -p_k$ for $k \ge 1$.*

*Proof.* A negative first-order change of $g(N, N; N)$ excludes $K_{N,1} + \varepsilon v_N$ for small $\varepsilon$, so $s_N A_N \ge 0$. Part (1) follows because the sign of $s_N$ alternates along each parity class. In (2), write $D_N f = \sum_{n=0}^{N} (-1)^n \binom{N}{n} f(n/N)$. For large $N$, $A_N = D_N(\beta + 2\gamma) + P_e(N)$ for even $N$ and $A_N = D_N \beta + P_o(N)$ for odd $N$, where

$$P_e(N) = q'_0 - L_{\mathrm{even}} + \sum_{k \ge 1} (-1)^k \binom{N}{k} \big[p_k + q_k + 2(p'_k + q'_k)\big],$$

$$P_o(N) = q'_0 - L_{\mathrm{odd}} + \sum_{k \ge 1} (-1)^k \binom{N}{k} (p_k - q_k).$$

These are polynomials in $N$. A nonzero polynomial has an eventual strict sign, which the remainder $O(\sqrt{N}e^{-\eta N})$ of Lemma 9.5 cannot change, so part (1) forces $P_e \equiv P_o \equiv 0$. The polynomials $1, \binom{N}{1}, \binom{N}{2}, \dots$ are linearly independent, which gives the stated conditions. ∎

Thus a separate adjustment of the top coefficient $c_N$ reduces the two bulk equalities to the single equality $\gamma(0) + \beta(1) + \gamma(1) = 0$. For $k \geq 1$ and $N \geq k + K + 1$, where $K$ bounds the supports of $p, q, p', q'$, the boundary-layer contributions satisfy the palindromic first-order relations $\delta b_k = \delta b_{N-k}$ and $\delta a_k = \delta c_{N-k}$; the separate terminal adjustment need not itself be palindromic. Families not covered by part (1) must have $A_N$ change sign, or vanish, infinitely often within a parity class.

Criteria that cover $r < 1$ but not $r = 1$ are thin near $r = 1$.

**Proposition 9.10** *(Thinness near the symmetric endpoint).* *Fix $N \geq 2$ and a direction $v$ along which the first-order change $\dot{g}$ of $g(N, N; N)$ at $K_{N,1}$ is negative. There are $r_* < 1$ and $C, c > 0$ such that $K_{N,r} + \varepsilon v$ lacks property (P) whenever $r_* < r < 1$ and $C(1-r)^N \leq \varepsilon \leq c$.*

*Proof.* The function $G(r, \varepsilon) = g(N, N; N)$ at $K_{N,r} + \varepsilon v$ is a rational function of the recurrence entries whose denominators involve only upward coefficients, which are positive at $K_{N,1}$; so $G$ is smooth near $(1, 0)$. By Lemma 9.1, $G(r, 0) = (1-r)^N$. Put $d = |\dot{g}|$. On a small rectangle around $(1, 0)$, on which the basic sign conditions persist, $\partial_\varepsilon G \leq -d/2$, hence $G(r, \varepsilon) \leq (1-r)^N - d\varepsilon/2 < 0$ for $\varepsilon \geq 4(1-r)^N/d$. ∎

**Corollary 9.11** *(Optimal order of the stability radius).* *For $N \geq 2$ let $\rho_N(r)$ be the largest $\rho$ such that every array satisfying (2.1) with $\widetilde{b}_i \geq 0$ and with left side of (5.4) at most $\rho$ has property (P). Then $\rho_N(r)$ is of exact order $(1-r)^N$ as $r \to 1$.*

*Proof.* Theorem 5.3 gives $\rho_N(r) \geq \eta_N(r) = \frac{1}{2}((1-r)/(32N))^N$ for $r \geq \frac{1}{2}$. For the upper bound, use the terminal direction $\delta c_N = -\operatorname{sgn}(s_N)$, $\delta b_N = \operatorname{sgn}(s_N)$, with all other entries unchanged. By Lemma 9.2 its first-order change is $-|s_N| < 0$, and it moves the full array by exactly $\varepsilon$ in the norm of (5.4). Put $t = 1 - r$ and $\varepsilon = Ct^N$ as in Proposition 9.10. For $N \geq 2$ and $r$ near 1, $\varepsilon \leq t/2$ and $\varepsilon \leq r/2$, so $\widetilde{b}_N \geq t/2$ and $\widetilde{c}_N \geq r/2$; every other coefficient is unchanged. The array satisfies the basic conditions and $\widetilde{b}_i \geq 0$, lies at full-array distance $\varepsilon = C(1-r)^N$ from $K_{N,r}$, and lacks property (P). Hence $\rho_N(r) \leq C(1-r)^N$. The restriction $N \geq 2$ is needed: every admissible two-point array with $b_1 \geq 0$ has property (P). ∎

The order $(1-r)^N$ is attained already as $r \to 1$; for fixed $r$ the radius also decays exponentially in $N$. The deformation that shows this raises the top mass, the reverse of Lemma 9.7, and it is exact rather than first order.

**Proposition 9.12** *(Raising the top mass).* *Let $N \geq 1$ and $0 < r < 1$. The measure of $K_{N,r}$ has nodes $x_n = 1 - (1+r)n/N$ and masses $w_n = \binom{N}{n} r^{N-n}/(1+r)^N$. For $\tau > 0$ let $\mu^\tau$ be this measure with the mass at $x_0 = 1$ raised to $(1+\tau)w_0$. Then $\mu^\tau$ has*

$$g(N, N; N) = \frac{(1+\tau)^2(I - w_0) + w_0}{(1+\tau)(h - w_0) + w_0}, \qquad I = \big(r(1-r)\big)^N, \quad h = r^N,$$

*which is negative exactly when* $\tau > \tau_N^* = \big(1-(1-r^2)^N\big)^{-1/2} - 1$. *After the affine normalization of Section 2, its recurrence coefficients differ from those of* $K_{N,r}$ *by at most* $3\tau \max_n w_n$.

*Proof.* Write $\pi_n$ and $S_n = \sum_{j<n} \pi_j$ for $K_{N,r}$, and put $\varepsilon = \tau w_0$. The polynomial $(1+\varepsilon S_n)R_n - \varepsilon \sum_{j<n} \pi_j R_j$ has degree $n$ and equals 1 at $x_0$. For $\deg p < n$, its integral against $p$ with respect to $\mu^\tau = \mu + \varepsilon\delta_{x_0}$ is $-\varepsilon p(x_0) + \varepsilon p(x_0) = 0$, because $\sum_{j<n} \pi_j R_j$ reproduces $p(x_0)$. Hence it is the polynomial $R_n^\tau$ of $\mu^\tau$.

On the support, dual orthogonality gives $\sum_{j\le N} \pi_j R_j(x) = \mathbf{1}_{x=x_0}/w_0$, and in particular $S_N + \pi_N = 1/w_0$. Therefore $R_N^\tau = (1+\tau)R_N - \tau \mathbf{1}_{x_0}$ on the support. By Lemma 9.1 and $\pi_N = r^{-N}$, $\int R_N^3\, d\mu = I$; also $\int R_N^2\, d\mu = h$. Hence

$$\begin{aligned}\int (R_N^\tau)^3 d\mu^\tau &= (1+\tau)^3(I - w_0) + (1+\tau)w_0,\\ \int (R_N^\tau)^2 d\mu^\tau &= (1+\tau)^2(h - w_0) + (1+\tau)w_0,\end{aligned}$$

and their ratio is the stated formula. Since $r > r/(1+r) > r(1-r)$, we have $h > w_0 > I$. So the denominator is positive, and the numerator is negative exactly when $(1+\tau)^2 > w_0/(w_0 - I) = \big(1-(1-r^2)^N\big)^{-1}$.

For the coefficients, comparing leading coefficients and using detailed balance as in Lemma 9.7 gives

$$a_n^\tau = a_n \frac{1+\varepsilon S_n}{1+\varepsilon S_{n+1}}, \qquad c_n^\tau = c_n \frac{1+\varepsilon S_{n+1}}{1+\varepsilon S_n},$$

and the row sums stay 1. Since $\varepsilon\pi_n = \tau w_n$, each of $a_n$, $b_n$, $c_n$ changes by at most $\tau w_n$. Put $A_n = 1 + \varepsilon S_n$ and $t_n = \varepsilon\pi_n = \tau w_n$. Since $b_0^\tau = \varepsilon/(1+\varepsilon)$, the affine coordinate is $y = (1+\varepsilon)x - \varepsilon$, and the normalized coefficients are $\widetilde{a}_n = (1+\varepsilon)a_n^\tau$, $\widetilde{b}_n = (1+\varepsilon)b_n^\tau - \varepsilon$, $\widetilde{c}_n = (1+\varepsilon)c_n^\tau$. For $n \ge 1$ we have $A_n \ge 1+\varepsilon$, and

$$\begin{aligned}\widetilde{a}_n - a_n &= \varepsilon a_n - (1+\varepsilon)a_n t_n/A_{n+1},\\ \widetilde{c}_n - c_n &= \varepsilon c_n + (1+\varepsilon)c_n t_n/A_n,\\ \widetilde{b}_n - b_n &= -\varepsilon(a_n + c_n) + (1+\varepsilon)t_n\big(a_n/A_{n+1} - c_n/A_n\big).\end{aligned}$$

Because $a_n, c_n \ge 0$ and $a_n + c_n \le 1$, each absolute change is at most $\varepsilon + t_n \le 2\tau \max_j w_j$. Row zero is restored exactly. This proves, in particular, the stated bound $3\tau \max_j w_j$ for every $\tau > 0$. ∎

**Corollary 9.13** *(Exponentially small stability radius).* *Let* $0 < r < 1$, *and let* $\rho_N(r)$ *be as in Corollary 9.11. If* $(1-r^2)^N \le \frac{1}{2}$ *and* $3(1-r^2)^N \le (1-r)/N$, *then*

$$\rho_N(r) \le 3\tau_N^* \max_n w_n \le 3(1-r^2)^N.$$

*In particular* $\rho_N(r) = O\big((1-r^2)^N\big)$ *as* $N \to \infty$, *for every fixed* $r \in (0,1)$.

*Proof.* Put $x = (1-r^2)^N$ and $W = \max_n w_n$. For $0 < x \le \frac{1}{2}$,

$$(1+x)^2(1-x) - 1 = x(1 - x - x^2) > 0,$$

so $\tau_N^* < x$. Choose $\tau \in (\tau_N^*, x)$. Proposition 9.12 gives a negative corner coefficient and full-array distance at most $3\tau W < 3x \le (1-r)/N$. Thus $\widetilde{b}_n > 0$ for $n \ge 1$, while $\widetilde{b}_0 = 0$. The exact coefficient formulas and the positive affine scale give $\widetilde{a}_n > 0$ for $n < N$, $\widetilde{c}_n > 0$ for $n > 0$, and all the basic boundary and row-sum equalities. Letting $\tau \downarrow \tau_N^*$ proves $\rho_N(r) \le 3\tau_N^* W \le 3x$. For each fixed $r \in (0,1)$, both smallness conditions hold for all sufficiently large $N$, since $N(1-r^2)^N \to 0$. ■

Thus, for every fixed $r \in (0,1)$, no set of arrays with property (P) contains coefficient balls of a positive radius independent of $N$ around all $K_{N,r}$; Theorem 5.3 gives the lower bound $\rho_N(r) \ge \frac{1}{2}(\min(r, 1-r)/(32N))^N$. The deformation of Proposition 9.12 has the binomial profile $w_n$ and is not a uniform rule. Along the terminal row alone, the first-order change of $g(N, N; N)$ at $K_{N,r}$ has an exact contour-integral representation, obtained as in Lemma 9.2 with $R_N(J)$ having eigenvalues $(-r)^j$. The formal saddle contributions have exponential factor $(1 - r + r^2)/(1 + r)$, which exceeds $1 - r$ exactly when $r > \frac{1}{2}$. For $r > \frac{1}{2}$ this suggests an oscillatory first-order envelope, not a proved threshold for finite perturbations. No asymptotic estimate or uniform control of the nonlinear remainder is proved here. Exact computations (`fixedr_probe.py`) confirm failures of property (P) along the terminal row at sizes down to $10^{-15}$ for $r = 4/5$, $N = 40$. The comparison at $r = \frac{1}{2}$ concerns only the first-order term. Along the terminal row, $g(N, N; N)$ is a polynomial in the perturbation, since the recurrence for the linearization coefficients divides only by $a_0, \dots, a_{N-1}$. At $r = \frac{1}{2}$, computations for $N = 30, \dots, 77$, using maxima over successive six-size windows, give estimated envelope decay factors $0.503, 0.593, 0.640, 0.676$ for the absolute Taylor coefficients of orders $1, 2, 3, 4$, respectively. This is a finite-range computational observation, not an asymptotic statement. Exact terminal-row failures also occur for $r \le \frac{1}{2}$: at size $6.3 \cdot 10^{-6}$ for $r = \frac{1}{2}$, $N = 120$; at size $3.2 \cdot 10^{-3}$ for $r = \frac{2}{5}$, $N = 100$; and at size $1.6 \cdot 10^{-2}$ for $r = \frac{1}{3}$, $N = 120$.

**Computations away from $r = 1$.** The explicit fixed-size, one-kink, and log-concave-increment examples below are reproduced by `rigidity_probe.py` and `smallr_probe.py`. Exact rational computations certify the stated finite examples; the separately labelled high-precision calculations provide numerical evidence. No asymptotic conclusion is asserted. Here $g(N, N; N)$ is computed from column $N$ of the product table, an $O(N^2)$-operation recurrence, so sizes in the hundreds are cheap.

- For the rule $\beta = 1$, $\gamma = 0$, exact computation shows that property (P) fails at $r = 4/5$, with perturbation size $1/100$, for every $N = 6, \dots, 14$, and at $r = 19/20$, with perturbation size $1/1000$, for every $N = 4, \dots, 14$. The period-four sign pattern at $r = 1$ does not persist in general for $r < 1$: at $r = 4/5$ the first-order derivatives are $512/1875$ for $N = 4$ and $-71936/5859375$ for $N = 8$.

- Below $r = \frac{1}{2}$, the counterexamples we found to shape conditions occur at sizes that grow as $r$ decreases; we did not determine minimal sizes or prove a scaling. For constant-ratio arrays with convex profile, one-kink counterexamples ($m$ unit increments, then $1 + \delta$) have $g(N, N; N) < 0$ in exact arithmetic at

| $r$ | $2/5$ | $1/3$ | $3/10$ | $1/4$ | $2/7$ | $1/5$ | $1/10$ |
|---|---|---|---|---|---|---|---|
| $N$ | 20 | 40 | 40 | 60 | 80 | 100 | 500 |

- Convex profiles with log-concave increments (step ratios nonincreasing and at least 1) contain Krawtchouk, every geometric profile and the normalized quadratic profiles $d_\kappa(x) = (x + \kappa x^2)/(1 + \kappa)$, $\kappa \geq 0$, and they exclude the one-kink counterexamples. Yet $k$ steps of ratio $1 + 1/(2N)$ followed by flat increments give $g(N, N; N) < 0$ at $(r, N, k) = (\frac{1}{2}, 80, 40)$, $(\frac{2}{5}, 32, 22)$ and $(\frac{1}{3}, 90, 62)$ in exact arithmetic. In 80- and 150-digit floating-point arithmetic, which agree but are not exact sign certificates, the same happens at $(\frac{1}{4}, 250, 175)$ and $(\frac{1}{5}, 500, 400)$.

The exact examples above disprove the stated shape conditions at the listed parameters and sizes. The two larger high-precision examples provide additional numerical evidence. These findings establish neither minimal failure sizes nor general conclusions about smooth deformations.

## 10. Open problems

**Inequalities beyond geometric increments.** The geometric criterion imposes exact equalities, whereas the perturbation criterion provides small neighborhoods. Theorem 8.1 gives inequalities on the profile that retain the full Krawtchouk range with a constant ratio $b_i/c_i$. Theorem 8.6 and Corollary 8.7 allow the ratio $c_i/(b_i + c_i)$ to increase, provided the Newton coefficients of $b$ are those of $c$ with nonincreasing weights, Proposition 8.8 covers linearly increasing ratios with a linear profile, and Theorem 8.9 covers linearly increasing ratios with log-concave Newton profiles and nondecreasing $b$, and Theorem 8.11 extends this to increasing convex quadratic ratios. Find direct inequalities that cover other varying ratios, for instance nonlinear increasing ratios, which Theorems 8.6 and 8.9 do not cover in general. Log-concavity of the Newton coefficients of both $d$ and $c$ together with monotone $b$ does not suffice: $d = (0, 25, 55, 91)/91$, $c = (0, 24, 54, 90)/91$ gives $g(3, 3; 3) = -3/2002$. Increasing ratios with nondecreasing $b$ can fail as well: for the linear profile $d_i = i/3$ and the concave ratio $\rho = (1/2, 3/4, 5/6)$, for which $b_i = 1/6$ for $i \geq 1$, one has $g(3, 3; 3) = -1/24$, and the convex kinked ratio below fails at $N = 120$. Theorem 8.11 does not extend to cubic ratios: for $N = 4$, $d_i = i/4$ and $\rho(x) = \frac{19}{20} + \binom{x}{3}/320$, whose Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$ are nonnegative, $b = (0, \frac{1}{80}, \frac{1}{40}, \frac{9}{256}, \frac{3}{80})$ is nondecreasing, but $g(4, 4; 4) = -14011/30720000$. Even the bound $r \leq \frac{9}{10}$ does not restore (P) for cubic ratios: on linear profiles with $\kappa = 1/80$, which keeps $b$ nondecreasing, $\rho_i = \frac{9}{10} + \kappa\binom{i}{3}/\binom{N}{3}$ gives $g(48, 48; 48) < 0$ for $N = 48$, and $\rho_i = \frac{9}{10} + \kappa(i/N)^3$ gives $g(64, 64; 63) < 0$ for $N = 64$. Steeper ratios fail at smaller $r$, again with nondecreasing $b$ and nonnegative Newton coefficients of $\rho$: $\rho_i = \frac{1}{2} + 10^{i-N}/100$ gives $g(40, 40; 40) < 0$ for $N = 40$, and raising only the last ratio, $\rho_i = r$ for $i < N$ and $\rho_N = r + (1 - r)/N$, gives $g(40, 40; 38) < 0$ for $r = \frac{1}{2}$, $N = 40$ and $g(120, 120; 119) < 0$ for $r = \frac{2}{5}$, $N = 120$ (`runs/varratio/frontier_examples.py`). Write a cubic ratio as $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The cubic examples above violate $r\sigma \leq t\tau$, a consequence of log-concavity without internal zeros of $(r, t, \tau, \sigma)$: in the first two $t = \tau = 0$, and for $\kappa(i/N)^3$ one has $t\tau = 6\kappa^2/N^6 < r\sigma = 6r\kappa/N^3$. In sampled arrays with $t, \tau \geq 0$, $r\sigma =$

$u\,t\tau$ and nondecreasing $b$ on log-concave Newton profiles, no negative coefficients were found for the tested values $u \in \{\frac{1}{4}, 1, 4, 16, 64\}$, $r \in \{\frac{1}{10}, \frac{1}{3}, \frac{1}{2}, \frac{4}{5}, \frac{19}{20}, \frac{99}{100}\}$ and $4 \le N \le 18$ (full tables), and in the coefficients $g(m, N; k)$ for $u \in \{1, 4, 16\}$ and $N = 24, 32, 48, 64, 96$ (`runs/varratio/cubic_lc_test.py`); in these samples the operator $M$ of the proof of Theorem 8.9 was nonnegative for $u \le 4$ and $N \le 13$. For $t = \tau = 0$, by contrast, $M$ has negative entries; for instance $N = 6$, $d_i = i/6$, $r = \frac{1}{5}$ and $\sigma = \frac{1}{200}$ give a negative entry (`runs/varratio/cubic_diag.py`). Theorem 8.17 shows that $r\sigma \le u\,t\tau$ suffices for $u = 1$. Decide whether it suffices for larger $u$; the proof of Theorem 8.17 uses the condition only coefficientwise, and that argument stops at $u = 1$. Theorem 8.16 admits ratios with log-concave Newton coefficients, of any degree, on every log-concave Newton profile; its proof is computer-assisted. Find a proof of its key inequality (8.8) without computer certificates, and decide whether, for ratios of degree greater than three, log-concavity of the ratio's Newton coefficients can be weakened to conditions like the one in Theorem 8.17. The terminating row and products of degree sum greater than $N$ must remain part of the conclusion.

**Sharper stability radii.** For fixed $N$ and $0 < r < 1$, determine or estimate the largest coefficient radius for which every admissible stochastic perturbation has property (P). Corollary 9.11 shows that the order $(1-r)^N$ as $r \to 1$ is optimal, and Corollary 9.13 gives $\rho_N(r) = O((1-r^2)^N)$ for each fixed $r$; the gap to the lower bound $\frac{1}{2}(\min(r, 1-r)/(32N))^N$ of Theorem 5.3 remains open. The bound in Theorem 5.3 comes from a single-term lower margin and a worst-case recurrence amplification. Both estimates discard substantial information. A useful improvement would retain a transparent coefficient test and describe its dependence on both $N$ and the bias parameter.

**The symmetric endpoint.** At $r = 1$, the reference margin in Lemma 5.2 vanishes because parity forces many coefficients to be zero. Symmetric perturbations preserve the parity zeros, and Theorem 7.6 exploits this for Szwarc's quadratic system when $p_0 < 1/2$. At $p_0 = 1/2$, and for the symmetric Krawtchouk array, further coefficients on the parity support vanish. Determine perturbation directions and structural constraints that preserve positivity there; continuity alone does not control coefficients emerging from those zeros. For every $N \ge 1$, the symmetric Krawtchouk array is a limit of arrays whose product coefficients are strictly positive throughout the triangle support. For $N \ge 2$ this follows from Proposition 9.8; the two-point case is immediate. Theorem 9.4 and Corollary 9.6 impose two boundary equalities on admitted polynomial rules and on the analytic rules satisfying the holomorphic-extension hypothesis specified there. A description of the cone for all $N$ is open.

**Monotonicity-type criteria for finite systems.** Szwarc's Theorems 3.2 and 3.3 in [7] are open conditions of monotonicity type for infinite systems. For finite systems, property (P) at the boundary point is the hypergroup property of the corresponding birth–death chain, and Miclo [5, Theorem 31 and Proposition 6] adapts the wave-equation argument of Bakry and Huet to give inequality conditions for chains that are symmetric under $n \mapsto N - n$ and sufficiently convex. These conditions do not cover the full Krawtchouk or

Szwarc families. For symmetric Krawtchouk they fail when $N > 2$, but the cases $N = 1, 2$ are covered; the Szwarc instance $N = 1$, $p_0 = 1/2$ is the same covered three-point system. By Theorem 7.8, any criterion of this kind for nonsymmetric finite systems, valid including the terminal indices, immediately gives one for symmetric finite systems. Theorem 7.10 supplies such hypotheses for all triples except those whose pairwise sums all exceed $N$. Theorem 7.12 supplies such hypotheses for all triples under its strengthened conditions (i)–(iii), and Corollary 7.13 contains the arrays (2.4) with $q \geq 2$ but not the Krawtchouk arrays with $N \geq 3$. For constant-ratio arrays, Theorem 8.1 controls the deep triples by inequalities that contain the whole geometric criterion, in particular the Krawtchouk arrays. Theorems 8.6 and 8.9 and Proposition 8.8 extend this to ratios $c_i/(b_i + c_i)$ that increase in the ways described there; other nonconstant ratios remain open. Convexity of the profile does not suffice even with a constant ratio and $r$ bounded away from 1: for $d$ with increments 1 at the first seven steps and 6/5 at the remaining seven ($N = 14$), exact arithmetic gives $g(14, 14; 14) < 0$ at $r = 1/2$, and the analogous profile with $N = 20$, twelve unit increments and 21/20 afterwards gives $g(20, 20; 20) < 0$ at $r = 2/5$. Log-convex increments fail as well: fourteen unit increments followed by $(21/20)^t$, $t = 1, \ldots, 4$, give $g(18, 18; 18) < 0$ at $r = 1/2$. These examples show that the stated shape restrictions do not guarantee positivity at the specified biases and sizes. Each failure is witnessed by a negative terminal coefficient $g(N, N; N)$. Section 9 explains part of this: admission at the symmetric endpoint imposes two first-order boundary equalities for the polynomial rules of Theorem 9.4 and the analytic rules of Corollary 9.6, and the computational discussion there gives counterexamples to convex-profile conditions and to convex-profile conditions with log-concave increments at the stated parameters and sizes. Two questions remain. First, Corollary 9.13 gives an exponential upper bound on the admissible coefficient-ball radius around $K_{N,r}$ for every fixed $r \in (0, 1)$, witnessed by raising the top mass. Determine whether analogous upper bounds hold for terminal-row perturbations, and which uniform rules exhibit such thinness. The first-order comparison in the remark after that corollary changes at $r = \frac{1}{2}$. The exact terminal-row failures reported there for $r \leq \frac{1}{2}$ show that higher-order terms matter, so this comparison does not locate a threshold for finite perturbations. Second, decide whether any inequality criterion covers a range $0 < r \leq r_0$ for all $N$. A criterion given by linear inequalities in the coefficients admits a convex set of arrays, but the set of arrays satisfying property (P) need not be convex, even at fixed $(N, r)$ and with both endpoints admitted by Theorem 2.1. For $q \geq 1$ let $G_{N,q,r}$ be the array (2.4), so that $G_{N,1,r} = K_{N,r}$. For $q > 1$ and $0 < \lambda < 1$ the combination $(1 - \lambda)K_{N,r} + \lambda G_{N,q,r}$ is a stochastic array with constant ratio $b_i/c_i$ and increments proportional to $1 + t\, q^{i-N}$, where $t = \lambda N(q - 1)q^{N-1}/((1 - \lambda)(q^N - 1))$. Exact computation (`convexity_probe.py`) gives $g(N, N; N) < 0$ for this combination at $(r, N, q, t) = (4/5, 10, 10, 1)$, $(3/5, 20, 10, 2)$, $(1/2, 20, 10, 1)$, $(2/5, 60, 10, 2)$, $(1/3, 100, 10, 2)$, $(3/10, 200, 2, 5)$ and $(2/7, 200, 10, 5)$. Computations in 150- and 300-digit arithmetic agree on $g(N, N; N) < 0$ at $(1/4, 500, 3, 10)$; this is not an exact sign certificate. Hence, for each of the seven exactly certified parameter choices above, no sufficient criterion for property (P) whose admitted set is convex, in particular no sufficient criterion defined by a system of linear inequalities in the coefficients, can admit both $K_{N,r}$ and $G_{N,q,r}$. In our searches, the sizes of the examples

found tended to increase as $r$ decreased; at $r = 1/5$ no negative corner was found on the tested $(N, q, t)$ grid with $N \le 1000$. Sign conditions on the shape fail in the same way. Write $d_n = b_n + c_n$ and $\rho_n = c_n/d_n$ for $n \ge 1$. For $S_1, S_2 \in \{[0,\infty), (-\infty, 0], \{0\}, \mathbb{R}\}$ and $\rho_{\max} > 0$, consider the stochastic arrays with $b_0 = 0$, $a_n$ decreasing, $b_n$ and $c_n$ nondecreasing, $d_{n+2} - 2d_{n+1} + d_n \in S_1$ for $0 \le n \le N-2$, $\rho_{n+1} - \rho_n \in S_2$ for $1 \le n \le N-1$, and $\rho_n \le \rho_{\max}$. Each such class contains every $K_{N,r}$ with $r \le \rho_{\max}$, and for $S_1 = S_2 = \{0\}$ it consists of these arrays only. For every other choice of $(S_1, S_2)$ and every $\rho_{\max} \ge 83/400$, exact computation (`signpattern_probe.py`) gives an array in the class with $g(N, N; N) < 0$. In the first two examples, normalize the listed increments to sum to one. If $S_1$ contains a positive number, take constant ratio $1/5$ and eighty unit increments followed by twenty increments $11/10$ ($N = 100$). If $S_1$ contains a negative number, take constant ratio $1/10$ and seventeen unit increments followed by three increments $1/2$ ($N = 20$). If $S_1 = \{0\}$ and $S_2$ contains a positive number, take $d_n = n/N$ and $\rho_n = 1/5 + \max(n - 102, 0)/2400$ ($N = 120$). If $S_1 = \{0\}$ and $S_2$ contains a negative number, take $d_n = n/N$ and $\rho_n = 1/5 - \max(n - 68, 0)/1600$ ($N = 80$). Thus, for every $\rho_{\max} \ge 83/400$, among the classes just defined, only $S_1 = S_2 = \{0\}$ guarantees property (P) for all $N$. For $\rho_{\max} < 83/400$, these four examples do not settle all sign choices; the cases not excluded by them remain unresolved here. For instance, with $S_1 = \{0\}$ and increasing ratio starting from $\rho = 1/10$, our search found no failure for $N \le 300$. Criteria with conditions on higher differences, with mixed conditions, or with conditions that do not apply at every index are not covered. Szwarc-type certificates were also tested on the deep triples. Floating-point linear-programming searches returned feasible backward-cone certificates for every deep triple in the tested Krawtchouk arrays. For some triples, the joint search for one coefficient-independent weight at $r = 1/5$ and $r = 1/2$ was reported infeasible. These numerical findings concern the specified certificate construction; no exact infeasibility certificates or general exclusion of linear-inequality criteria are supplied. Checkerboard-restricted searches did not certify all deep triples at the tested $r < 1$. The exact marching calculations also produced examples with negative boundary flux. On the tested random perturbed arrays, the solver's certificate-feasibility results agreed with property (P).

**Other finite families.** Find analogous coefficient regions around other finite orthogonal systems and compare their geometry with the Krawtchouk neighborhoods. An algebra with a positive basis may supply the reference margin, but a recurrence criterion must express the resulting admissible deformations without requiring the unknown product coefficients as input.

## AI-use disclosure

OpenAI Codex was used to develop the proof arguments, search and check source passages, construct exact-arithmetic diagnostics, draft and revise this manuscript, and prepare the Lean translation. Separate AI-assisted reviews checked the algebraic proof, boundary cases, normalizations, and quantitative estimates; these reviews are not human peer review. Anthropic Claude was used to analyse Szwarc's Problem 2 against the survey text, to explore

discrete hyperbolic certificates numerically, and to develop and draft Sections 7, 8 and 9 together with their exact diagnostics. The kernel recurrence and the operator inequality of Section 8, the resummation in the proof of Theorem 8.9, its key inequality under nondecreasing $b$, Proposition 8.10, and the weighted detour and key inequality in the proof of Theorem 8.11 arose in research consultations with ChatGPT Pro; the closed form of the recurrence for every degree and the exact checks were completed with Claude, which also prepared the Lean formalizations of Theorem 8.1 and of the results of Sections 7 and 9 listed in the section on code and formalization. The review of Section 8 was therefore carried out by the same system that proposed its key steps. These sections were then reviewed in fourteen rounds by ChatGPT Pro (an AI review with independent exact checks, not human peer review); the first round failed, the corrections are recorded in `REVIEW.md`, and for each section the final confirmation round on the corrected text returned PASS. Theorem 8.9 was added afterwards; its first version, with the hypothesis $(2N-1)\kappa \le N(1-r)$ in place of nondecreasing $b$, passed a further review round with minor fixes and is machine-verified. The present form of Theorem 8.9 and Proposition 8.10 were added after that and passed a further review round with minor fixes; both are machine-verified. Theorem 8.11 was added last; its written proof passed a further review round with minor fixes, and it is machine-verified. Theorem 8.6, Corollary 8.7 and Proposition 8.8 were added afterwards and are machine-verified; their written proofs passed further review rounds. Theorem 8.12 and Corollary 8.13 were found and proved with Claude, without research consultation, and are machine-verified; their written proofs passed a further review round with minor fixes. Proposition 8.14, which is due to Walkup, was rederived with Claude together with the self-contained proof given here; that proof passed a further review round with minor fixes and is machine-verified. Theorem 8.16 and Lemma 8.15 were found and proved with Claude, without research consultation; the proof uses exact computer certificates, passed a further review round with minor fixes (including a rerun of the certificates), and is not machine-verified. Theorem 8.17 was found and proved with Claude, without research consultation; its proof uses exact computer certificates and is not machine-verified. The author set the research objective and retains responsibility for publication decisions and the final mathematical claims. No independent human verification is asserted here. “Machine-verified” refers only to the Lean theorems listed in the section on code and formalization, with the axioms stated there. Results and prose remarks not covered by those listed Lean theorems are supported by written proofs, AI review and exact numerical checks, but are not machine-verified.

## Code and formalization

The companion source tree contains `probe.py` and `verify.py`. The command

```
python3 -u verify.py --max-n 7
```

compares recurrence-matrix multiplication with direct multiplication modulo the terminal polynomial on exact rational inputs, checks the perturbation lower bounds, and replays two rejected coefficient-profile tests. It also checks Theorem 7.1, the margin of Lemma 7.4 and the negative coefficient for $p_0 > 1/2$ on exact Szwarc arrays, the identities of Lemma 7.3 in

the graded model, and the lower bound of Theorem 7.6 on rational symmetric perturbations. These computations are diagnostics and do not replace the proofs at arbitrary size. For Section 9, `rigidity_probe.py` checks Lemma 9.2 exactly for $N \leq 20$, the predicted failures of Theorem 9.4, the boundary layers of Proposition 9.9, the region of Lemma 9.5 and the fixed-$r$ computations, and `smallr_probe.py` replays the small-$r$ counterexamples.

The Lean development formalizes the following criteria:

| Paper result | Lean declaration |
|---|---|
| Theorem 2.1: geometric criterion | `Szkraw.geometric_criterion` |
| Lemma 2.2: coefficient parametrization | `Szkraw.geometric_parameters`, `Szkraw.geometric_coefficients_admissible`, `Szkraw.geometric_family_hypotheses`, `Szkraw.geometric_coefficients_N_one` |
| Lemma 3.1: basis and positivity | `Szkraw.squarefreeBasis`, `Szkraw.squarefree_basis_apply`, `Szkraw.squarefree_product_nonneg` |
| Lemma 3.2: positive associative inner product | `Szkraw.squarefree_trace_diagonal`, `Szkraw.tower_trace_square_pos`, `Szkraw.tower_trace_self_adjoint` |
| Lemma 3.3: three-term multiplication | `Szkraw.elementary_three_term`, `Szkraw.elementary_one` |
| Proposition 3.4: positive subalgebra | `Szkraw.elementary_products` |
| Lemma 4.1: positive normalization | `Szkraw.towerCharacter`, `Szkraw.tower_character_generator`, `Szkraw.tower_character_elementary`, `Szkraw.normalizer_pos`, `Szkraw.normalizer_one`, `Szkraw.character_normalizer_ratio` |
| Lemma 5.1: finite realization and triangle support | `Szkraw.finite_realization`, `Szkraw.product_triangle_zero` |
| Lemma 5.2: uniform positive margin | `Szkraw.kraw_triangle_margin` |
| Theorem 5.3: perturbation | `Szkraw.perturbation_criterion` |
| Corollary 5.4: systems outside the geometric family | `Szkraw.nongeometric_perturbation_example`, `Szkraw.relatively_open_coefficient_neighborhood` |
| Corollary 6.1: full positive Krawtchouk range | `Szkraw.full_krawtchouk_binomial_range`, `Szkraw.binomial_normalized_at_zero`, `Szkraw.binomial_node_probability_convention` |
| Corollary 6.2: dual $q$-Krawtchouk case | `Szkraw.additional_geometric_class` |
| Theorem 7.1: Szwarc's quadratic system | `Szkraw.szwarc_full`, `Szkraw.szwarc_sharp` |
| Lemma 7.2: edge idempotents | `Szkraw.edge_idempotents`, `Szkraw.cube_model` |
| Lemma 7.3: three-term action | `Szkraw.three_term_action`, `Szkraw.Uvec_top` |
| Lemma 7.4: margin for the quadratic system | `Szkraw.szwarc_margin` |
| Lemma 7.5: stability of a reference with a margin | `Szkraw.stability` |
| Theorem 7.6: symmetric criterion | `Szkraw.symmetric_criterion` |
| Corollary 7.7: symmetric systems outside both families | `Szkraw.symmetric_new` |
| Theorem 7.8: quadratic decomposition | `Szkraw.decomp_positivity`, `Szkraw.decomp_converse`, `Szkraw.even_block`, `Szkraw.odd_block`, `Szkraw.evenComp_admissible`, `Szkraw.oddComp_admissible` |
| Corollary 7.9: parity-split criterion | `Szkraw.parity_split`, `Szkraw.parity_split_sharp`, `Szkraw.ps_even_params`, `Szkraw.ps_odd_params` |
| Theorem 7.10: low products | `Szkraw.low_products` |

The entry for Theorem 2.1 includes a positive finite orthogonality measure, distinct support points, the terminal relation, and all finite product identities. The entry for Theorem 5.3 includes the explicit radius, positive triangle margin, and exact zeros outside the triangle. In Sections 3 and 4 the algebra $A$ is the iterated quadratic extension `Szkraw.tower`, whose generator adjoined at stage $N+1-j$ is $z_j$ with the square relation (3.1); subsets $S$ are encoded as $\{0,1\}$-vectors of length $N$, $\tau$ is `Szkraw.towerTrace`, $E_k$ is `Szkraw.elementary` and $t_k$ is `Szkraw.normalizer`. Lemma 2.2 is stated with $p = 1/r \geq 1$ in both directions. Lemma 3.3 is stated for $k \geq 1$; for $k = 0$ it reads $E_1 = E_1$, where $E_1$ is the sum of the generators (`Szkraw.elementary_one`). In Proposition 3.4 Lean states that every product $E_m E_n$ with $m, n \leq N$ is a combination of $E_0, \ldots, E_N$ with nonnegative coefficients; that $B$ is generated by $E_1$ and the independence of the $E_k$ are not stated separately. In Lemma 4.1 the character is an $\mathbb{R}$-algebra homomorphism with $z_j \mapsto pq^{N-j}$. In Corollary 5.4 Lean states, for the explicit arrays, (2.1), the failure of the constant-ratio hypothesis, nonnegative linearization and the margin of Theorem 5.3; for its last sentence it gives, for every $0 < r < 1$, an open set in the subspace of arrays satisfying the equalities in (2.1) that contains the Krawtchouk array at $r$ and consists of arrays with nonnegative linearization. In Corollary 6.1 Lean states nonnegative linearization of the array (2.4) with $q = 1$ and $r = p_0/(1-p_0)$, the orthogonality of its polynomials for the binomial weights at the nodes $1 - j/(N(1-p_0))$, and their value 1 at $j = 0$; the sharpness remark in its proof is not formalized. In Corollary 6.2, "outside the Krawtchouk recurrence class" is formalized as disagreement of $b_i + c_i$ with the profile of every Krawtchouk array. Proposition 6.3 is a comparison with the literature and is not formalized. The entries for Section 7 state nonnegativity of the coefficients $g(m, n; k)$: for Theorem 7.10 and Corollary 7.11 on all triples with two indices of sum at most $N$, and for Theorems 7.12, 7.15 and 7.16 and Corollaries 7.13 and 7.17 on all triples. Lemma 7.14 and the identities of Theorem 7.15 are stated for the matrices $R_m(J)$ and $R'_t(J) = \Pi R'_t(\Pi J \Pi)\Pi$. Theorem 7.1 is proved in Lean through Corollary 7.9 with $q = 1$, and Lemma 7.4 through the decomposition, whose components are Krawtchouk arrays; the symmetry of its measure holds for every finite realization of an array satisfying (2.1) with $b \equiv 0$ (`Szkraw.realization_symmetric`). In Corollary 7.7, "not of the form (2.4)" is formalized as the failure of the hypotheses of Theorem 2.1 and as disagreement with every array (2.4) in the Lean parametrization; its last sentence follows from the explicit coefficient box of positive radius in `Szkraw.symmetric_criterion`; openness and dimension are not stated separately in Lean. For Corollary 7.9, when $N \geq 2$, Lean proves that both normalized components satisfy the hypotheses of Theorem 2.1 and are the arrays (2.4) with parameters $(N, q, r)$ and $(N-1, q, r/q)$ (`Szkraw.ps_even_params`, `Szkraw.ps_odd_params`); `Szkraw.parity_split` also covers $N = 1$, where the odd component is the constant polynomial alone. In Theorem 7.8 the identity for $g(2a, 2b+1; 2c+1)$ is stated with the unnormalized odd component, which has the same product coefficients (`Szkraw.product_normalize`). The finite realization holds for every array satisfying (2.1) (`Szkraw.finite_realization`). The proofs formalize the cone argument and the identity $g(m, n; k) \int R_k^2 \, d\mu = g(m, k; n) \int R_n^2 \, d\mu$ in matrix form. The entry for Theorem 8.1 includes the same finite realization as the first and the nonnegativity of every coefficient $g(m, n; k)$; its proof formalizes Lemmas 8.3 and 8.4, the operator induction, and Lemma 8.2 in con-

verse form: the solution of (8.1) yields the linearization coefficients. The remark after the proof of Theorem 8.1 is formalized as identities for all real $u$ and $v$ (two-parameter form, `Szkraw.lc_twoParam`) and for all real $r$ (Bernstein form, `Szkraw.lc_bernstein`, whose coefficients $B_s$ are $\binom{D}{s}$ times the Bernstein coefficients); `Szkraw.constantRatio_twoParam` identifies the arrays of Theorem 8.1 with this family, and the proof formalizes the operator induction coefficientwise in $t$. All of them have compiled proofs without added axioms or admitted goals; the axiom audit reports only `propext`, `Classical.choice` and `Quot.sound`. Lemma 7.2 is stated for every commutative real algebra with elements $y_1, \dots, y_N$ satisfying $y_j^2 = (1-r)y_j + r$ whose squarefree monomials form a basis, which determines $A$ up to isomorphism, and for $r \neq -1$; `Szkraw.cube_model` shows that the real functions on $\{1, -r\}^N$ form such an algebra. Its basis assertion is stated as linear independence of the $f_l y_S$ together with equality of their span and the ideal $f_l A$. Lemma 7.3 is stated for $N \geq 1$, $p_0 < 1$ and $r = p_0/(1-p_0)$, in every commutative real algebra satisfying these relations, without a basis hypothesis; its recurrence is an equality in $A \times A^N$, which restricts to $V$, and `Szkraw.Uvec_top` states $U_{2N+1} = 0$ separately. Lemma 9.2 is stated for $N \geq 2$ as the derivative at $\varepsilon = 0$ of $g(N, N; N)$ along $K_{N,1} + \varepsilon v$. Its Lean proof replaces the beta integrals by an explicit matrix: the derivative $X$ of $R_N^{\varepsilon}(J_\varepsilon)$ satisfies $[J, X] = [\Pi, E]$ and $e_0^{\mathsf{T}} X = 0$, and $Z_{mn} = \binom{N}{n}(-1)^{n+s}/(2\binom{N-1}{s})$ for $m + n = 2s + 1$, $Z_{mn} = 0$ for $m + n$ even, satisfies $ZJ - JZ = e_N e_N^{\mathsf{T}}$ on the columns $n \geq 1$; hence the derivative equals $\operatorname{tr}((Z\Pi - \Pi Z)E)$, which is evaluated entry by entry. In Theorem 9.4 the set of arrays is a family of sets indexed by $N$ whose members satisfy $g(m, n; k) \geq 0$, admission is Definition 9.3, and the second statement is the vanishing of the derivative for every $N \geq 1$ above the degrees. Lemma 9.5 is stated with $U(x) = \int_0^1 \log|x - t|\, dt$ (`Szkraw.Ulog`) for $f$ complex differentiable on an open set containing $\Omega$, as a bound $C\sqrt{N}\, e^{-\eta N}$ for every $N \geq 1$; the cycle of its proof is the boundary of a finite union of squares of an axis-parallel grid with mesh $1/M$ for a positive integer $M$ and origin $\sqrt{2}(1 + i)$, so that no node $k/N$ lies on a grid line (`Szkraw.residue_identity`), and the Riemann-sum and Stirling estimates are `Szkraw.riemann_lower` and `Szkraw.factorial_le_stirling`. In Corollary 9.6 and Proposition 9.9 (2), a rule on $[0, 1]$ is analytic on an open set containing $\Omega$ if it is the restriction of a function complex differentiable there (`Szkraw.AnalyticOnOmega`); the second statement of Corollary 9.6 is the bound $|s_N|\, C\sqrt{N}\, e^{-\eta N}$ on the derivative for every $N \geq 2$, and the example is stated for every $N \geq 2$. In Proposition 9.9 (2) the boundary layers vanish beyond a fixed index $K$, the formulas for $\delta b_n$ and $\delta c_n$ are assumed for $1 \leq n \leq N - 1$ and $1 \leq n \leq N$ and all large $N$, and the hypothesis is that of the first statement; The first statement of Corollary 9.6 is obtained by specializing Proposition 9.9 (2) to zero boundary layers. In Proposition 9.10 the direction is given in the coordinates $b_1, \dots, b_{N-1}, c_1, \dots, c_N$ of Section 9, the hypothesis is that the derivative of $g(N, N; N)$ at $\varepsilon = 0$ is negative, and the conclusion is that $K_{N,r} + \varepsilon v$ satisfies (2.1) and has $g(N, N; N) < 0$; `Szkraw.productMatrix_contDiffAt` gives the smoothness used in its proof. In Corollary 9.11, $\rho_N(r)$ is the supremum of the radii with the stated property, and exact order means $c_1(1-r)^N \leq \rho_N(r) \leq c_2(1-r)^N$ for all $r$ in an interval $(r_0, 1)$. Lemma 9.7 is stated for the spectral measure of every array satisfying (2.1), whose node $x_0 = 1$ exists by `Szkraw.spectral_top_node`; the polynomials of $\mu_\varepsilon$ are identified by or-

thogonality and their value 1 at $x_0$ (`Szkraw.lower_measure`), and both parts are derivatives at $\varepsilon = 0$. In Proposition 9.8, $\Lambda_N$ is an explicit finite sum bounding the absolute ratios of the two first-order rates with nonzero $r$-rate (`Szkraw.lamN`), the lowered measure is identified by orthogonality and normalization, and the neighborhood is a coefficient box of some positive radius around the normalized array; the proof uses an explicit formula for the coefficients of $K_{N,r}$ (`Szkraw.kraw_g_formula`). In Proposition 9.12, Lean states that the normalized array satisfies (2.1), that its polynomials are orthogonal for the raised measure transported to $y = (1+\varepsilon)x - \varepsilon$ and equal 1 at the top node, and then the corner formula, its sign and the coefficient bound; the proof uses the polynomials $(1+\varepsilon S_n)R_n - \varepsilon \sum_{j<n} \pi_j R_j$ for any finite system (`TopMass.lean`). Theorem 8.9 is stated for arrays satisfying (2.1) whose profile $d_i = b_i + c_i$ has Newton coefficients log-concave without internal zeros, with $c_i = (r + \kappa i/N)(b_i + c_i)$ for $i \le N$, $0 < r \le 1$, $\kappa \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, using the same array-based encoding as in Theorem 8.1; the equivalence of the last condition with $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \le N(1-r)$ is `Szkraw.affine_bmono_iff`. Its proof formalizes the Leibniz formulas, the log-concavity of the Newton coefficients of $c$ through $\widehat{\kappa}$ (`Szkraw.affU_lcnz`), the key inequality through (8.3), the bound $\alpha u_1 \ge t u_0$ and the interpolation of $F$ (`Szkraw.AffSetting.key`, `Szkraw.Fz_nonneg`), and the resummation as an identity of matrices indexed by $\{0,\dots,N\}^2$ (`Szkraw.resum_nonneg`, `Szkraw.gkernel_resum`). The Lean proof uses the detour constant 4 for all four families, and for $K \ge 0$ the products with $(i,j) \in \{(1,n-1),(n-1,1)\}$ for $n \ge 3$ and $(2,n-2)$ for $n \ge 4$. Proposition 8.10 is stated for arrays satisfying (2.1) with $b_i + c_i = \nu_1 i + \nu_2 \binom{i}{2}$ and $c_i = \big(r + ti + \tau\binom{i}{2}\big)(b_i + c_i)$ for $i \le N$, $\nu_1 > 0$, $\nu_2 \ge 0$, $r > 0$, $t, \tau \ge 0$ and $b_0 \le \cdots \le b_N$; its proof identifies the iterated shift sums of the Newton coefficients with finite differences on $\Omega$ (`Szkraw.iterShift_padNewton`), uses the Leibniz rule for a quadratic factor (`Szkraw.fwdDiff_quad_mul`) and the inequality $\varphi_2(\ell)\varphi_2(\ell+1) \ge c_{\ell+1}\varphi_4(\ell)$ (`Szkraw.phi2_phi2`), and applies the resummation with $H_j = 0$ and $V_1 = -A_1^{\psi}$. The equivalent form of the monotonicity of $b$ stated after Proposition 8.10 is `Szkraw.quadprofile_bmono_iff`, stated for the same arrays without the monotonicity hypothesis; the Lean statement also omits the assumption $r > 0$. Theorem 8.11 is stated for arrays satisfying (2.1) whose profile has Newton coefficients log-concave without internal zeros, with $c_i = \big(r + ti + \tau\binom{i}{2}\big)(b_i + c_i)$ for $i \le N$, $r > 0$, $t, \tau \ge 0$ and $b_0 \le b_1 \le \cdots \le b_N$, using the array-based encoding of Theorem 8.1; the equivalent form of the monotonicity of $b$ stated after it is `Szkraw.quadratic_bmono_iff`, stated for the same arrays without the monotonicity hypothesis; the Lean statement also omits the assumption $r > 0$. Its proof formalizes the Leibniz rule for a quadratic factor in every order (`Szkraw.leibniz_quad`), the weighted detour (8.4) through the reduction to the configuration $u_k = q^{k-1}$ and the certificate (8.5) (`Szkraw.quadB_seq`, `Szkraw.cert_nonneg`), the key inequality (8.6) on $\Omega$ for $a \ge 3$ (`Szkraw.QuadSetting.key`), together with its reduced auxiliary inequality (`Szkraw.quad_key_red`), and the resummation with the third family $W_j$ (`Szkraw.resum_nonneg2`, `Szkraw.gkernel_resum2`). The Lean proof uses the detour constant 4 for $B$, $C'$, $H$ and $V$ and the constant 3 for $W$. For the remarks after Theorem 8.9, `Szkraw.profile_last_le` gives $d_{N-1} \le (N-1)/N$, with equality only for the linear profile, for arrays satisfying (2.1) whose profile has Newton coefficients log-concave without internal zeros (such profiles are convex), and `Szkraw.affine_range_remarks` states

the claims on the range of $\kappa$ for the condition $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \le N(1-r)$, which is equivalent to nondecreasing $b$ by `Szkraw.affine_bmono_iff`; that the linear-profile arrays are arrays of Proposition 8.8 is not stated separately. `Szkraw.affEx_remark`, `Szkraw.quadExample_remark` and `Szkraw.quadB_unit_fails` also show that the examples satisfy the hypotheses of Theorem 8.9, respectively Theorem 8.11. In Section 10 only the two counterexamples with $N = 3$ and the cubic counterexample with $N = 4$ are formalized; they are stated with all properties claimed for them, and the remaining computational statements there are not formalized. Theorem 8.6 is stated for arrays satisfying (2.1), with the two mixed inequalities written through iterated shift sums $\varphi_n(\ell) = \sum_h \binom{\ell}{h}\kappa_{n+h}$ of the padded Newton sequences (`Szkraw.MixedIneq`); its proof formalizes the kernel identity for the generator $z^{-1}c(\theta) + b(\theta) + z(1 - b(\theta) - c(\theta))$ for every array (`Szkraw.kernel_starV`), an operator induction with abstract move weights that need to be nonnegative and dominated only on $\Omega$ (`Szkraw.gkernel_nonneg`), and the mixed detour inequalities (`Szkraw.lemma_8_4_AV`, `Szkraw.lemma_8_4_CV`). In Corollary 8.7 the weights need to be nonincreasing on $1 \le k \le N$; its proof uses that the order $\psi_k\varphi_i \le \psi_i\varphi_k$ $(i \le k)$ is preserved by the shift sums (`Szkraw.mixedIneq_of_crossOrder`), and the statement includes the monotone ratio and, for weights constant on the support of $\kappa$, $b = w_0 c$. Proposition 8.8 is stated in the form given there. Every numbered result of Section 9 is formalized, and so are three consequences stated in its text: for every $d \in \mathbb{N}$, no set with property (P) admits all rules of a nonempty open set of polynomial rules of degree at most $d$, with rules given by coefficient vectors (`Szkraw.no_open_rule_set`); the single equality $\gamma(0) + \beta(1) + \gamma(1) = 0$ and the palindromic relations for the boundary layers for $k \ge 1$ and $N \ge k + K + 1$, where $K$ bounds the supports of $p, q, p', q'$, after Proposition 9.9 (`Szkraw.boundary_layers_palindromic`); and, for every fixed $r \in (0,1)$, the absence of coefficient balls of a positive radius independent of $N$ around all $K_{N,r}$, in the sense of $\rho_N(r)$, after Corollary 9.13 (`Szkraw.radius_no_uniform`). The description of $\Omega$ before Lemma 9.5 is formalized as well: $\Omega$ contains $[0,1]$ (`Szkraw.omega_contains`); there is $y_0$ with $0.26276 < y_0 < 0.26277$ such that $\frac{1}{2} + iy \in \Omega$ exactly when $|y| \le y_0$ (`Szkraw.omega_half_line`; its proof uses certified bounds for log, arctan, $\log 2$ and $\pi$); and near 0, $\Omega$ is the region $\operatorname{Re} x \ge a(\operatorname{Im} x)$ whose frontier is the curve $\operatorname{Re} x = a(\operatorname{Im} x)$, where $a(0) = 0$, $a(b)/b \to 0$ (vertical tangent), $a(b) > b/(2\log(1/b))$ for small $b > 0$, and $a$ is not real-analytic at 0 (`Szkraw.omega_corner_zero`); the analogous description at 1, with $\Omega$ to the left of the curve, is `Szkraw.omega_corner_one`, derived through the symmetry $x \mapsto 1 - \bar{x}$ of $U$ and of the frontier of $\Omega$ (`Szkraw.Ulog_one_sub`, `Szkraw.Ulog_conj`, `Szkraw.frontier_omega_sigma`). The remaining prose of Section 9, the computations and heuristics away from $r = 1$, is not formalized. Theorem 8.12 is stated with the Newton coefficients $\kappa_1, \dots, \kappa_N$ of $c$ log-concave without internal zeros; in Corollary 8.13 the ratio is $\rho(i) = \sum_{k \le i} \binom{i}{k}\rho_k$ with $(\rho_k)$ nonnegative, log-concave and without internal zeros on all $k \ge 0$, and $\rho_0 > 0$, and the Lean statement `Szkraw.lowdeg_newton_lc` gives the log-concavity of the Newton coefficients of $c$ as a separate conclusion. Proposition 8.14 is stated for real sequences on $\mathbb{N}$ with the predicate `Szkraw.LCNZ` (nonnegative, $u_i u_{i+2} \le u_{i+1}^2$, positive entries forming an interval) and for every sequence $\kappa$ with $f(x)g(x) = \sum_k \binom{x}{k}\kappa_k$ for all $x \in \mathbb{N}$ (`Szkraw.nval`); such $\kappa$ exists and is unique (`Szkraw.nval_mul`, `Szkraw.nval_inj`),

and the proof formalizes the layer cake, the box identity, (T1)–(T3) and the five cases.

The companion code is available at https://github.com/skypher/szkraw; the fixed revision cited in [10] identifies the reviewed proof and audit code. Run `make lean` to build the development and `make axioms` to build and audit the principal statements. The latter invokes `lean/CheckAxioms.lean`, which rejects dependencies outside `propext`, `Classical.choice`, and `Quot.sound` and tests rejection of unexpected axioms. The toolchain and dependencies are pinned in `lean/lean-toolchain` and `lean/lake-manifest.json`. The declaration-by-declaration correspondence, auxiliary formalization coverage, and verification commands are recorded in `RESEARCH.md` at that revision. Historical attribution and the open problems are prose claims, not machine-verified statements.

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